Quantitative Hamiltonian path counts and Wormald's angular Hessian

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Abstract. Divoux and Wang determined the leading constant $L = \cosh 1/\cos 1$ for the maximum number of Hamiltonian paths in a tournament. We prove the quantitative refinement

$$P(n) = \left(\frac{\cosh 1}{\cos 1} + O(n^{-1})\right)\frac{n!}{2^{n-1}}.$$

We also compute the full angular Hessian in Wormald's fixed-base formula and prove that it is positive definite for every regular base. Its determinant recovers the spectral ratio used by Divoux and Wang. Combining this identity with their counting and spectral results, the equality conditions of Fang and Huang, and Wormald's model comparisons yields the fixed-base expectation extrema. We give the additional equality analysis and nonregular-base estimates, and derive strict monotonicity of the carousel constants from the known spectrum. The original Hessian and carousel limit overlap with Bhattacharjee's work.

1. Introduction

A tournament is an orientation of a complete graph. Write $H(T)$ for the number of directed Hamiltonian paths in a tournament T, and $C(T)$ for the number of directed Hamiltonian cycles. Paths are counted as vertex orderings, whereas cyclic starting points are identified for cycles. Let

$$P(n) = \max_{|V(T)|=n} H(T), \qquad \mu_n = \frac{n!}{2^{n-1}}, \qquad \nu_n = \frac{(n-1)!}{2^n}.$$

Szele [10] observed that a uniform random tournament has μ_n Hamiltonian paths in expectation: each vertex ordering is directed with probability $2^{-(n-1)}$. Hence $P(n) \geq \mu_n$. The corresponding expected cycle count is ν_n for $n \geq 3$. These expectations provide the natural normalization for the extremal problem.

Adler, Alon and Ross [9] improved the random lower bound by choosing an almost complete packing of edge-disjoint triangles and orienting each triangle cyclically, independently of the others. A vertex ordering containing one of these triangles in three consecutive positions is more likely to form a directed path than it is in the independent-edge model. For a uniformly random vertex ordering, the number of such occurrences converges in distribution to Poisson(1), from which they obtained the lower bound $(e-o(1))\mu_n$. Wormald [6] used a different construction: the directions between large vertex classes are prescribed by a fixed tournament, while the tournaments within the classes are random. His calculation for carousel bases gives a cycle lower constant exceeding 2.85588 along an infinite sequence of orders, already close to the constant considered below.

The upper bounds developed through permanent estimates. If A is the adjacency matrix of a tournament, then per A counts its spanning unions of vertex-disjoint directed cycles. In particular, it bounds the number of Hamiltonian cycles. Alon [2] applied Brégman's inequality to this permanent and optimized the resulting expression over the outdegrees, obtaining $P(n) = O(n^{3/2}\mu_n)$. Friedgut and Kahn [7] used an entropy argument to improve the polynomial exponent; their cycle bound is $O(n^{\alpha}\nu_n)$ with $\alpha \approx 1.249$. These estimates identify the correct exponential scale, but leave a polynomial gap between the upper and lower bounds. At the scale of a leading constant, one must retain information that a bound in terms of the degrees alone discards.

Divoux and Wang [12] closed this gap. They proved that the maximum cycle count is $(L + o(1))\nu_n$ and the maximum path count is $(L + o(1))\mu_n$, where $L = \cosh 1/\cos 1$. Their proof still uses permanents,

but replaces a degree bound by a determinant approximation for nearly doubly stochastic matrices. After separating vertices with large degree imbalance, they classify a Hamiltonian cycle by the directed path segments that it induces on the exceptional set. These segments are encoded in modified adjacency matrices of the remaining tournament. Sinkhorn scaling and a determinant–permanent inclusion-exclusion formula then reduce the count to a ratio of determinants of a skew-symmetric matrix. A matrix comparison of Fang and Huang [11] bounds this ratio by its value for a transitive tournament. Odd carousels, and their one-vertex deletions at even orders, attain the bound asymptotically; a separate argument transfers the cycle result to paths.

Our first result refines the path asymptotic of Divoux and Wang [12, Theorem 1.4] by bounding the normalized error by $O(n^{-1})$ uniformly in the order. The leading constant and the carousel constructions are those of [12]. The proof below tracks the error through the permanent approximation, deletion, diagonal scaling, and the sum over deletion sets.

Theorem 1.1. There are absolute constants K', n_1 such that every integer $n \geq n_1$ satisfies

$$(L - K'/n)\mu_n \leq P(n) \leq (L + K'/n)\mu_n.$$

For odd n, the carousel Car_n has vertex set $\mathbb{Z}/n\mathbb{Z}$, with $i \to j$ precisely when the representative of $j - i$ modulo n belongs to $\{1, \ldots, (n-1)/2\}$. The lower bound is attained, up to an error of $O(\mu_n/n)$, by odd carousels and by the even tournaments obtained by deleting one vertex from Car_{n+1}.

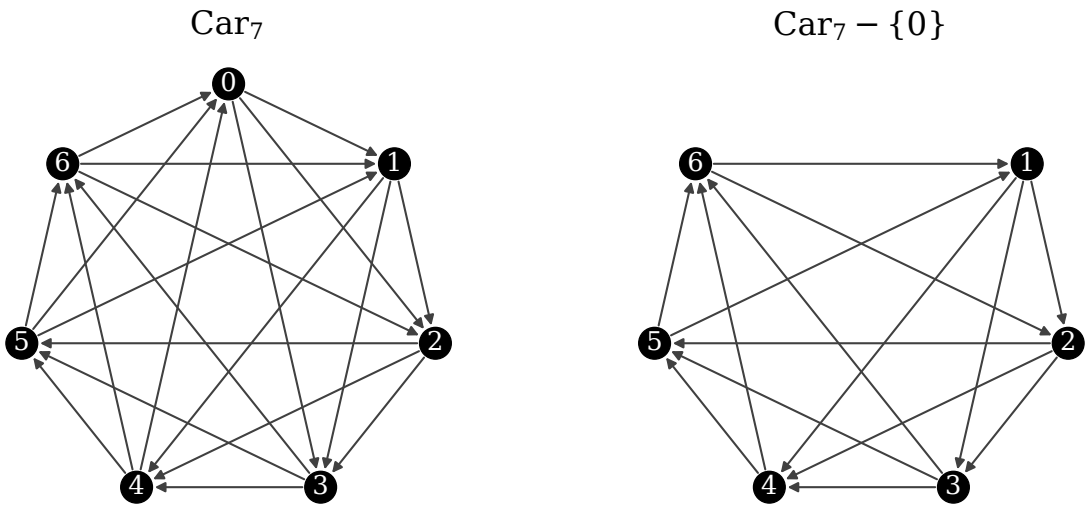


Figure 1: The carousel Car_7 and its even-order subtournament obtained by deleting vertex 0. Every vertex on the left has outdegree 3; those on the right have outdegree 2 or 3. All arcs are shown.

The second part of the paper concerns Wormald's random blow-up models. Fix an odd integer $d \geq 3$ and a tournament T^* on d vertices. Each vertex is replaced by a block of m vertices, and the directions between blocks are prescribed by T^*. In $\mathcal{T}(T^*, m)$ the internal tournaments are independent and uniform; in $\mathcal{R}(T^*, m)$ they are independent and uniform among regular tournaments, so m must be odd. Put $N = dm$. The expected cycle and path counts are normalized by ν_N and μ_N, respectively.

For a regular base, Theorem 4 of Wormald [6] expresses the limiting constant in terms of an angular Hessian, subject to its determinant being nonzero. He computes this determinant for several bases and conjectures that the condition always holds. We prove positive definiteness and evaluate the determinant for every regular base.

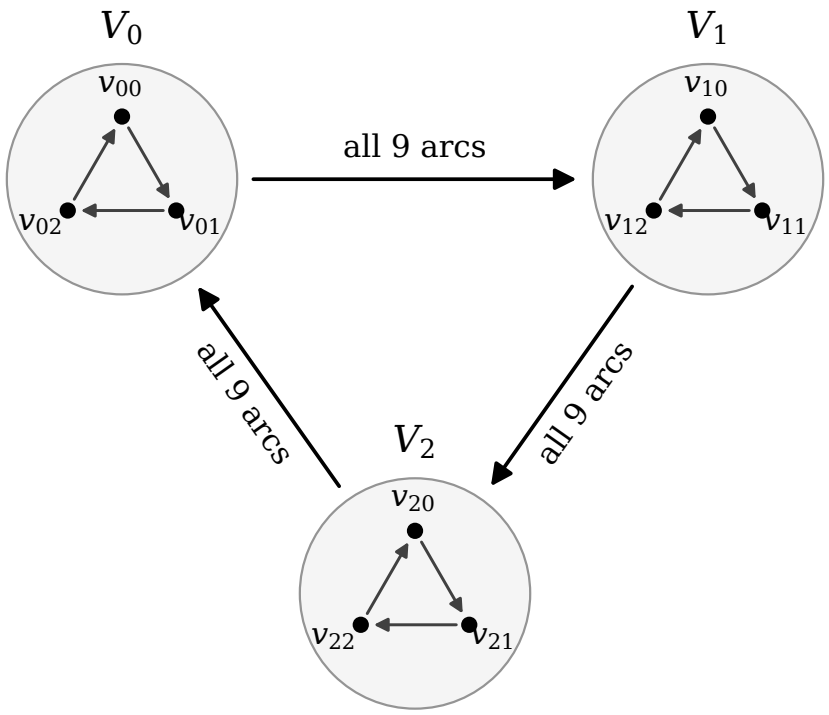


Figure 2: A blow-up of the directed triangle. Each internal block shown is a directed triangle; a thick arrow between blocks represents all 9 arcs in that direction. In the random models, the internal blocks are chosen independently from the specified distributions.

Theorem 1.2. Let T^* be regular, with adjacency matrix A. Set $S = A - A^T$, $C = S/d$, and $E = (e_1 - e_d, \dots, e_{d-1} - e_d)$. The Hessian $\mathbf{H}$ of Wormald's angular function at the origin is positive definite, and

$$\mathbf{H} = E^T(I + C^2)(I - C^2)^{-1}E, \qquad \mathcal{H} = d\,\frac{\det(I + C^2)}{\det(I - C^2)} > 0,$$

$$\sqrt{d/\mathcal{H}} = \rho_d(S) := \frac{\det(I + S/d)}{\det(I + iS/d)} \in \mathbb{Q}_{>0}.$$

Here $\mathcal{H} = \det \mathbf{H}$, and the square root is positive. The angular function is defined in Section 9. Together with Theorems 1, 4 and 5 of Wormald [6], this shows that, for fixed d, the normalized expected cycle and path counts in $\mathcal{T}(T^*, m)$ both tend to $\rho_d(S)$, while the corresponding limits in $\mathcal{R}(T^*, m)$ are $e^{1/d}\rho_d(S)$.

The model constants can also be obtained directly from Divoux and Wang [12, Proposition 5.1], without the Hessian calculation; Section 10.1 records that derivation. Their Proposition 7.1 and Lemma 9.1 already give the spectral maximum and the carousel value. The next corollary combines these results with the equality conditions of Fang and Huang [11], the zero-row-sum condition for a regular base, and a separate bound for nonregular bases.

Corollary 1.3. For each fixed odd $d \geq 3$, there exists M_d such that, for every $m \geq M_d$, the carousel Car_d is the unique base, up to isomorphism, maximizing the expected number of Hamiltonian cycles and paths in both models. For $\mathcal{R}(T^*, m)$, only odd m are considered.

The closed form below is the binomial evaluation in Divoux and Wang [12, Section 7 and Lemma 9.1], with the factor $e^{1/d}$ supplied by Wormald [6, Theorem 1]. The known frequencies in Fang and Huang [11, Lemma 4.2] give the following further consequence.

Corollary 1.4. Set

$$r_d = \frac{(1+1/d)^d + (1-1/d)^d}{(1+i/d)^d + (1-i/d)^d}, \qquad \mathfrak{r}_d = e^{1/d} r_d.$$

Along odd integers $d \geq 3$, the sequence $\mathfrak{r}_d$ increases strictly to L.

Theorem 1.2 directly proves Wormald's Hessian conjecture [6, Conjecture 6]. Corollary 1.3 combines existing counting and spectral results with the equality and nonregular-base arguments given here. The limit in Conjecture 2 follows from the closed form in [12] and Wormald's factor $e^{1/d}$; Corollary 1.4 adds strict monotonicity. Section 10 retains the expansion through d^{-5}, with remainder $O(d^{-6})$, as a calculation from that closed form.

Bhattacharjee [13, Theorems 1.1 and 1.3, Proposition 3.1, Lemma 3.2 and Corollary 3.3] also treats Wormald's Conjectures 2 and 6. His original angular Hessian, positivity, determinant/rational-square formulas, carousel value, limit and low-order expansion overlap with the corresponding results here. We retain direct derivations without claiming these as new relative to that work. We separately discuss monotonicity, equality, nonregular bases and eventual uniqueness at fixed base order. The global quantitative path problem in Theorem 1.1 is distinct.

1.1 The counting estimates and their relation to earlier methods

McCullagh [5] obtained the determinant approximation with an $O(n^{-1})$ relative error under moderate-deviation assumptions. Divoux and Wang [12, Proposition 3.1] prove a uniform $1 + o(1)$ version for nonnegative doubly stochastic matrices; their coefficient estimates and Gamma averaging can be refined to retain $O(n^{-1})$ in that setting. Section 3 gives the form needed here, with fixed entry and operator-norm bounds, and includes the detailed proof. The quantitative assertion of Theorem 1.1 concerns the final path count after deletion, scaling and summation; it does not assert a new determinant main term or a first $O(n^{-1})$ permanent approximation.

The proof expands the permanent by imposing distinct row and column indices separately. Double centering eliminates terms with a vertex of degree one. Components in which every vertex has degree two give Gaussian pairings and hence the determinant main term; the other components are estimated after contracting their degree-two chains. The resulting coefficient estimates hold over a range growing with the matrix order. For the tail, permanent polarization reduces the required complex-matrix bound to a positive semidefinite one. The complex Gaussian identity and arithmetic–geometric mean estimate used by Han and Niles-Weed [3, Lemmas 4.3–4.4] provide this bound; we then apply Cauchy's coefficient estimate. Together, the coefficient estimates and the tail bound give the uniform $O(n^{-1})$ error.

For paths, we use the identity

$$H(T) = \sum_{U \subseteq V(T)} \det(I + A[U]) \operatorname{per} A[U^c].$$

This is a specialization of the determinant–permanent identities discussed by Irving and Omar [1]. All its summands are nonnegative for a tournament. We can therefore estimate the permanent in each term without controlling cancellation in the final sum. The terms with large U have a sufficiently small factorial tail. For the remaining terms, the task is to apply the permanent approximation with an error uniform in both the tournament and the deletion set. We also retain the weights associated with each deletion-set size: their fixed moments are summable. Replacing every size by the largest permitted value would introduce unnecessary logarithmic losses.

The exceptional vertices require a further estimate. We split the adjacency matrix into four blocks according to the exceptional set and its complement, and expand the permanent over the matchings between these blocks. A cross-block matching removes core rows and core columns that need not have the same indices. Thus estimates restricted to principal submatrices do not suffice. Section 4 treats these nonprincipal

submatrices, performs the diagonal scaling, and records the change in the Gaussian determinant factor. This four-block expansion is the way exceptional vertices are counted here; the directed-segment classification in [12] concerns the cycle count.

It remains to ensure that these estimates give the claimed error for every tournament. We may assume $H(T) \geq \mu_n$. Brégman's inequality first bounds the sum of squared degree imbalances. Tournaments with a sufficiently large value of this sum are bounded directly; those with an exceptional vertex are handled by the cross-block expansion. For the remaining tournaments, the scaling calculation retains an exponential penalty for degree imbalance. Some intermediate errors have square-root size, which would be too large if estimated in isolation. Together with the penalty, they contribute only $O(n^{-1})$ to the upper bound. The main term is

$$\rho_n(S) = \frac{\det(I + S/n)}{\det(I + iS/n)}, \qquad S = A - A^T.$$

As in [12], the comparison of Fang and Huang [11] bounds this ratio by its value at the sign matrix of a transitive tournament. That value is $L + O(n^{-1})$. For the lower bound, the two-sided path estimate applies to odd carousels and to their even-order one-vertex deletions: the degree imbalances are respectively zero and of absolute value one. These arguments give Theorem 1.1. Using only the spectral-radius bound and the sum of squared frequencies in the last step gives Proposition 7.1; its role is to isolate what follows from the analytic counting estimates before the full moment comparison is used.

1.2 Wormald's models

Wormald [6] obtains the fixed-base counting formula by projecting a Hamiltonian cycle onto a closed walk in the base tournament. Inside each block, the cycle induces a collection of directed paths. Generating functions for these walks and paths lead to a multivariate coefficient problem, to which smooth-point asymptotics apply. For a regular base, the remaining constant involves the determinant of an angular Hessian defined through an implicit equation. This leaves both a nondegeneracy question and a comparison problem between different bases.

We differentiate the original implicit equation to obtain Theorem 1.2, including positive definiteness of the full Hessian. This supplies an unconditional justification for Wormald's nondegeneracy assumption. The identity $\sqrt{d/\det \mathbf{H}} = \rho_d(S)$ also links the Hessian to the existing spectral theory. The model limit $\rho_d(S)$ itself is already a direct application of [12, Proposition 5.1]; using Wormald's conditional Hessian formula to infer nondegeneracy would not be a valid substitute for the calculation.

For regular bases, we supplement the comparison in [12] with the equality classification from Fang and Huang [11]. Signed similarity together with zero row sums forces the carousel. Nonregular bases are bounded separately by Brégman's inequality. Since only finitely many bases occur at fixed order, the strict limiting gaps give Corollary 1.3. The short spectral proof of Corollary 1.4 and the detailed real-variable calculation are both included. The base order is fixed before the block order tends to infinity.

1.3 Further questions and organization

Theorem 1.1 does not identify the path maximizers at every finite order or determine the coefficient of the first correction term. It would also be useful to estimate M_d in terms of d. Allowing the base order and the block order to grow together would require error bounds uniform in the base order, beyond the fixed-base asymptotics used here.

Section 2 introduces notation. Sections 3 and 4 prove the permanent and scaling estimates, Section 5 establishes the path-counting identity, and Section 6 proves the spectral comparison. Sections 7 and 8 prove the auxiliary Proposition 7.1. Sections 9–11 concern the Hessian and the random blow-up models, and Section 12 completes the proof of Theorem 1.1. The appendices contain the auxiliary calculations and proofs of the standard tools used in the argument.

2. Preliminaries and notation

2.1 The random baseline and error conventions

Let $\mathfrak{S}_n$ be the symmetric group on $[n]$, and let I_π indicate that the consecutive edges of π all point forward. Each ordering uses $n-1$ distinct undirected edges, so

$$H(T) = \sum_{\pi \in \mathfrak{S}_n} I_\pi(T), \qquad \mathbb{E}I_\pi = \prod_{j=1}^{n-1} \frac{1}{2} = 2^{-(n-1)},$$

$$\mathbb{E}H(T) = \sum_{\pi \in \mathfrak{S}_n} \mathbb{E}I_\pi = n!2^{-(n-1)} = \mu_n.$$

The maximum is at least the mean, so $P(n) \geq \mu_n$. Let $\mathcal{T}_n$ denote all tournaments with vertex labels $[n]$. In this paper, a uniform $O(n^{-1})$ error means

$$\exists K < \infty \; \exists n_0 \; \forall n \geq n_0 \; \forall T \in \mathcal{T}_n : \quad |\text{error}(n,T)| \leq K/n.$$

The constants K, n_0 are uniform over tournaments and deletion sets. They may depend on additional fixed parameters b, A_0, B_0, when present.

Setting $p = \pi^2$, we have

$$\begin{aligned} C_* &= \frac{1+4/p}{1-4/p}\frac{3/2-4/p}{1/2+4/p} \\ &= \frac{p+4}{p-4}\frac{3p-8}{p+8} = \frac{3p^2+4p-32}{p^2+4p-32} \\ &= \frac{3\pi^4+4\pi^2-32}{\pi^4+4\pi^2-32}. \end{aligned}$$

The relative difference between the constants is

$$100\frac{C_*-L}{L} = 100\left(\frac{C_*\cos 1}{\cosh 1} - 1\right) = 0.0505359379058\ldots\%.$$

2.2 Sign matrices and spectral factors

Definition 2.1 (Sign matrices and scores).

The adjacency matrix A has zero diagonal and $A_{ij} = 1$ precisely when $i \to j$. Define

$$S = A - A^{\mathsf{T}}, \qquad 2A = J - I + S, \qquad s = S\mathbf{1}.$$

Here I is the identity matrix in the current dimension, $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}$, and $\mathbf{1}$ is the all-ones column vector. In $I_p, J_p, \mathbf{1}_p$, the subscript explicitly specifies the dimension p. The superscript T denotes transpose. If d_i^+ is the outdegree, then $s_i = 2d_i^+ - (n-1)$ and $\sum_i s_i = 0$. The Euclidean and maximum vector norms are

$$\|x\|_2 = \left(\sum_i |x_i|^2\right)^{1/2}, \quad \|x\|_\infty = \max_i |x_i|.$$

We use the notation

$$\mathcal{E} = \|s\|_2^2, \qquad d = \|s\|_\infty, \qquad a = \frac{s}{n-1}, \qquad \tau = \|a\|_2^2.$$

The normalized scores a and τ are defined for $n \geq 2$. An odd-order tournament is regular when every vertex has outdegree $(n-1)/2$, equivalently $s = 0$. An even-order tournament is balanced when every

outdegree is $n/2$ or $n/2-1$, equivalently $s_i \in \{-1,1\}$. Thus d measures the largest vertex imbalance, whereas τ measures the total squared imbalance after normalization.

Definition 2.2 (Submatrices and permanents).

For a vertex set U, the notation $A[U] = A[U,U]$ denotes the principal submatrix. For possibly different row and column sets R, C, the submatrix $A[R,C]$ selects those rows and columns, always in increasing original vertex-label order. The same convention applies to a general matrix $M[R,C]$. The complement U^c is taken in the current full vertex set. Empty sums equal zero and empty products equal one; both the determinant and the permanent of the empty matrix equal one. For a square matrix of order m, denoted by M, the permanent is

$$\operatorname{per} M = \sum_{\pi \in \mathfrak{S}_m} \prod_{i=1}^{m} M_{i,\pi(i)}.$$

Definition 2.3 (Frequencies and spectral factors).

The singular values of a real matrix M are the nonnegative square roots of the eigenvalues of $M^{\mathsf{T}}M$. The operator norm $\|M\|_{\mathrm{op}} = \sup_{\|x\|_2=1} \|Mx\|_2$ equals the largest singular value. The square of the Frobenius norm $\|M\|_F = (\sum_{i,j} |M_{ij}|^2)^{1/2}$ is the sum of the squared singular values. For the skew-symmetric matrix S, write $\pm i\lambda_j$ for its nonzero eigenvalue pairs, where $i^2 = -1$ and $\lambda_j > 0$, adding zero frequencies when needed. Each λ_j occurs twice among the singular values. Define

$$x_j = \frac{\lambda_j^2}{n^2}, \qquad D_n(S) = \det(I - S^{\mathsf{T}}S/n^2)^{-1/2},$$

$$\rho_n(S) = D_n(S)\det(I + S/n) = \prod_j \frac{1+x_j}{1-x_j}.$$

The skew-symmetric singular values occur in pairs, and $\|S\|_F^2 = n(n-1)$, so

$$\sum_j x_j = \frac{n-1}{2n}, \qquad \|S/n\|_{\mathrm{op}}^2 \le \frac{n-1}{2n} < \frac{1}{2}.$$

Therefore D_n and ρ_n are positive and bounded by absolute constants. In particular,

$$D_n(S) = \det(I + iS/n)^{-1}.$$

For every i,

$$s_i = \sum_{j \neq i} (A_{ij} - A_{ji}) = d_i^+ - [(n-1) - d_i^+] = 2d_i^+ - (n-1),$$

$$\sum_i s_i = \sum_{i<j} (S_{ij} + S_{ji}) = 0, \qquad \|S\|_F^2 = \sum_{i \neq j} 1 = n(n-1).$$

In a real orthonormal basis, a real skew-symmetric matrix becomes a direct sum of blocks $\left(\begin{smallmatrix} 0 & \lambda_j \\ -\lambda_j & 0 \end{smallmatrix}\right)$ and zero blocks. Consequently,

$$\det(I + S/n) = \prod_j (1 + \lambda_j^2/n^2) = \prod_j (1 + x_j),$$

$$\det(I - S^{\mathsf{T}}S/n^2) = \prod_j (1-x_j)^2,$$
$$D_n(S) = \prod_j (1-x_j)^{-1},$$
$$\det(I + iS/n) = \prod_j (1-\lambda_j/n)(1+\lambda_j/n) = \prod_j (1-x_j),$$
$$\rho_n(S) = \prod_j \frac{1+x_j}{1-x_j}.$$

Each frequency occurs twice in the squared Frobenius norm, so

$$2\sum_j \lambda_j^2 = n(n-1), \qquad \sum_j x_j = \frac{n-1}{2n}, \qquad \|S/n\|_{\mathrm{op}}^2 \le \sum_j x_j < \frac{1}{2}.$$

For $n = 1$, we have $S = [0]$, the frequency sum is empty, and $D_1(S) = \rho_1(S) = 1$.

Every log below is a natural logarithm. Using $-\log(1-x) = \int_0^x (1-u)^{-1}\,du \le 2x$ for $0 \le x \le 1/2$, we obtain

$$0 \le \log D_n(S) \le 2\sum_j x_j < 1, \qquad 1 \le D_n(S) < e.$$

Similarly, $\log(1+x) \le x$ gives $\log \rho_n(S) < 3/2$. Multiplication by $D_n(S)$ or $\rho_n(S)$ therefore preserves the order of the error. Lemma 6.1 gives a sharper bound.

3. A uniform permanent approximation

In this section we prove a uniform estimate for permanents. Write a matrix with unit row and column sums as $(J_n + E)/n$, where $E\mathbf{1} = E^{\mathsf{T}}\mathbf{1} = 0$. If the entries of E are bounded and $\|E/n\|_{\mathrm{op}} \le q < 1$, the main term is a determinant. Section 4 applies this estimate to scaled tournament submatrices.

McCullagh [5] obtained the determinant main term under moderate-deviation conditions. Here we give a self-contained version for a fixed entry bound C and singular-value spectral gap q, with an error uniform over the matrix class determined by these parameters. The proof controls the coefficients and the tail separately, so that the conclusion applies uniformly to the deletions and scalings in Section 4. A recent related approximation of Li [8, Theorem 2.1] assumes that the maximum absolute row or column sum of the centered perturbation is $o(n)$; this condition does not cover dense kernels whose corresponding sums have order n.

3.1 The theorem and the proof strategy

Theorem 3.1 (Uniform permanent approximation).

Fix $0 \le C < \infty$ and $0 \le q < 1$. Let $n \ge 1$, and suppose $E \in \mathbb{R}^{n\times n}$ satisfies

$$E\mathbf{1} = E^{\mathsf{T}}\mathbf{1} = 0, \qquad \max_{i,j} |E_{ij}| \le C, \qquad \|E/n\|_{\mathrm{op}} \le q.$$

Put $B = E/n$. Uniformly over all matrices satisfying these conditions,

$$\frac{\operatorname{per}(J_n + E)}{n!} = \det(I - BB^{\mathsf{T}})^{-1/2} + O_{C,q}(n^{-1}). \tag{3.1}$$

Equation (3.1) can also be written as the determinant factor times the relative error $1 + O_{C,q}(n^{-1})$. The subscript C, q means that the error constant and the threshold dimension may depend on these two parameters fixed in advance, but must apply simultaneously to every admissible matrix.

All determinant square roots in this section are the positive square roots on the relevant real intervals. The assumptions give $\|B\|_{\mathrm{F}}^2 \le C^2$. Thus the determinant factor is at least one and is uniformly bounded by a constant depending only on C, q.

Inclusion-exclusion for the row and column indices expresses each coefficient of the permanent as a sum over bipartite multigraphs. Centering eliminates degree-one vertices. The degree-two components correspond to Gaussian pairings and give the determinant main term. We classify the remaining components by their excess, the number of edges minus vertices; each unit of excess contributes a factor n^{-1}. We first estimate coefficients of degree at most αn, then use a complex-analytic bound for the remaining coefficients.

The proof uses the Gaussian series $G(t)$, which is a different object from the numerical bound $G = 100/19$ in Section 7.4. Put

$$f(t) = \frac{\operatorname{per}(J_n + tE)}{n!}, \qquad G(t) = \det(I - t^2 BB^{\mathsf{T}})^{-1/2}.$$

Choose $1 < \sigma < R < 1/q$; when $q = 0$, no upper restriction is imposed. For example, take

$$R = \frac{3+q}{2(1+q)}, \qquad \sigma = \frac{1+R}{2}.$$

In this section we define a separate control function $\Gamma(r)$, which is distinct from the score product Γ in Section 7:

$$\Gamma(r) = \exp\left(\frac{C^2 r^2}{2(1-q^2r^2)}\right), \qquad r \ge 0,\ rq < 1.$$

Throughout this section, when $q = 0$, upper restrictions involving $1/q$ are omitted. All constants depend only on C, q and the chosen radii. Section 3.6 combines them into an explicit error bound.

For these radii,

$$R - 1 = \frac{1-q}{2(1+q)} > 0, \qquad 1 - Rq = \frac{2-q-q^2}{2(1+q)} = \frac{(1-q)(q+2)}{2(1+q)} > 0,$$

$$\sigma - 1 = (R-1)/2 > 0, \qquad R - \sigma = (R-1)/2 > 0.$$

In this section, let s_i denote the singular values of B. Then

$$\sum_i s_i^2 = \sum_{i,j} B_{ij}^2 \le n^2 (C/n)^2 = C^2, \qquad 0 \le s_i \le q.$$

Using $-\log(1-u) = \int_0^u (1-v)^{-1}\,dv \le u/(1-u)$,

$$\begin{aligned}\log G(r) &= -\frac{1}{2}\sum_i \log(1 - r^2 s_i^2) \\ &\le \frac{1}{2}\sum_i \frac{r^2 s_i^2}{1 - r^2 s_i^2} \le \frac{r^2 \sum_i s_i^2}{2(1 - r^2 q^2)} \le \frac{C^2 r^2}{2(1 - r^2 q^2)}.\end{aligned}$$

Consequently $1 \le G(1) \le \Gamma(1)$. If $|f(1) - G(1)| \le K/n$, then

$$\left|\frac{f(1)}{G(1)} - 1\right| = \frac{|f(1) - G(1)|}{G(1)} \le K/n,$$

Thus the absolute error estimate also gives a relative error of the same order.

3.2 Coefficients and Gaussian pairing

Definition 3.1 (Permanent coefficients and rescaling).

Write $f(t) = \sum_{k=0}^{n} a_k t^k$. Expanding according to the entries taken from E gives

$$a_k = \frac{1}{(n)_k} \sum_{\substack{I,J \subseteq [n] \\ |I|=|J|=k}} \operatorname{per} E[I,J],$$

Here $0 \le k \le n$, so the denominator $(n)_k$ is nonzero. For $k > n$, define $a_k = 0$ directly, without using an expression containing $1/(n)_k$. Define

$$F_k = \frac{(n)_k}{n^k} a_k = \frac{1}{k!} \sum_{\substack{i_1,\dots,i_k \text{ distinct} \\ j_1,\dots,j_k \text{ distinct}}} \prod_{\ell=1}^{k} B_{i_\ell j_\ell}, \qquad F(t) = \sum_{k=0}^{n} F_k t^k.$$

The sum on the right remains meaningful when $k > n$: its indexing set is empty, so its value is zero. The formal identities below use this convention.

Definition 3.2 (Set partitions and weights).

For a finite set E, let $\mathfrak{P}(E)$ be the set of all partitions of E into disjoint nonempty blocks, and put $\mathfrak{P}_k = \mathfrak{P}([k])$. The indicator that the k coordinates are distinct has a standard inclusion-exclusion expansion on the partition lattice. The permanent calculation in [5, Section 8] also uses an expansion over pairs of partitions. A block of size d has Möbius weight $(-1)^{d-1}(d-1)!$; here the weight of an entire partition is defined as $\mu(\pi) = \prod_{D \in \pi} (-1)^{|D|-1}(|D|-1)!$. Apply this expansion separately to the row and column coordinates of F_k. Each partition pair determines a bipartite multigraph: its edges have distinct labels $1, \dots, k$, the vertices on the two sides are the blocks of the row and column partitions, and each block of size d receives the weight above.

The numerical label of each vertex is summed independently over $[n]$; different vertices may have the same numerical label. If a row vertex has degree one, summing its label gives a column sum of B, which is zero. A column vertex of degree one similarly gives a zero row sum. Thus only graphs in which every vertex has degree at least two contribute.

For example, in degrees two and three, each surviving row or column partition has a single block. The products of the Möbius weights on the two sides are 1 and 4, respectively. Dividing by $k!$ gives

$$F_2 = \frac{1}{2} \sum_{i,j} B_{ij}^2, \qquad F_3 = \frac{2}{3} \sum_{i,j} B_{ij}^3, \qquad |F_3| \le \frac{2C^3}{3n}.$$

The quadratic term is part of the Gaussian main term, whereas the cubic term is $O(n^{-1})$.

A connected component whose vertices all have degree two is called a pure degree-two component. Their total exponential generating function is $G(t)$. This can be checked directly with two independent standard real Gaussian vectors X, Y. By Wick's formula,

$$\mathbb{E} \exp(t X^{\mathsf{T}} B Y)$$

the coefficient of t^k in this expression is the sum over pairing partitions on the two sides of the edge labels, precisely the graphs whose vertices all have degree two. Integrating first with respect to the conditional Gaussian distribution and then diagonalizing BB^{T} gives

$$\mathbb{E} \exp(t X^{\mathsf{T}} B Y) = \mathbb{E} \exp\left(\frac{t^2}{2} X^{\mathsf{T}} B B^{\mathsf{T}} X \right) = \det(I - t^2 B B^{\mathsf{T}})^{-1/2}.$$

This analytic identity holds when $|t| \|B\|_{\mathrm{op}} < 1$ and determines all formal coefficients. In particular, the coefficients of G are nonnegative, and

$$G(r) \le \Gamma(r) \qquad (r \ge 0,\ rq < 1),$$

because $-\log(1-x) \le x/(1-x)$ and $\sum_i s_i^2 = \|B\|_{\mathrm{F}}^2 \le C^2$; here s_i are the singular values of B.

Remove all pure degree-two components from every contributing graph, and call what remains its core. A core need not be connected. If it has h edges and v vertices, define its excess as $j = h - v$. Every nonempty core satisfies $j \ge 1$. Decomposition into components with labeled edges gives the coefficientwise identity

$$F(t) = G(t)\left(1 + \sum_{j \ge 1} C_j(t)\right),$$

Here C_j is the generating series for cores of excess j, with the Möbius signs and matrix contractions included in its coefficients. The identity is formal: a fixed degree k involves only finitely many $j < k$, so no convergence in j at nonzero t is required. The distinct-index definition of F_k makes all coefficients above degree n vanish.

Definition 3.3 (Core series).

For $\pi, \sigma \in \mathfrak{P}_h$, let $D_\pi(a)$ and $D_\sigma(a)$ denote the two partition blocks containing edge label a. Give the row blocks labels $x_D \in [n]$ and the column blocks labels $y_E \in [n]$, and define the contraction

$$\mathcal{W}_B(\pi, \sigma) = \sum_{x:\pi \to [n]} \sum_{y:\sigma \to [n]} \prod_{a=1}^{h} B_{x_{D_\pi(a)}, y_{D_\sigma(a)}}.$$

Both sums run over all functions, with no injectivity requirement. Define $\mathcal{K}_{j,h}$ to be the set of partition pairs satisfying the following conditions: every block has size at least two; every connected component of the associated bipartite multigraph contains a block of size at least three; and $h - |\pi| - |\sigma| = j$. Here $|\pi|$ counts blocks, not elements of the original set. A connected component is a maximal set of vertices connected by successive edges. Degree counts incident half-edges, with parallel edges counted separately; after compression, a loop contributes two to the degree of its vertex. Define the core series by

$$C_j(t) = \sum_{h=0}^{\infty} c_{j,h} t^h, \qquad c_{j,h} = \frac{1}{h!} \sum_{(\pi,\sigma) \in \mathcal{K}_{j,h}} \mu(\pi)\mu(\sigma)\mathcal{W}_B(\pi, \sigma).$$

For fixed h, the sum is finite. A nonempty core has at least one block on either side, so $j \le h - 2$ and $c_{j,h} = 0$ for $h \le j + 1$. We shall also use the weaker condition $j < h$. The empty graph contributes $F_0 = G(0) = 1$ and is not included in C_j.

Write $g_k = [t^k]G(t)$ for $k \ge 0$, where $[t^k]$ denotes coefficient extraction. Thus $g_0 = 1$ and $g_k = 0$ for odd k. Lemma 3.2 will show that the weighted norm $\|C_j\|_R = \sum_h |c_{j,h}| R^h$ is finite.

Fix the entries selected from E by specifying the row set I and image set J, both of size k. The total weight of bijections at these positions is per $E[I, J]$. The remaining $n - k$ rows can be bijected arbitrarily to the remaining columns in $(n - k)!$ ways. Hence

$$[t^k] \operatorname{per}(J + tE) = (n-k)! \sum_{|I|=|J|=k} \operatorname{per} E[I, J],$$

$$a_k = \frac{(n-k)!}{n!} \sum_{|I|=|J|=k} \operatorname{per} E[I, J] = \frac{1}{(n)_k} \sum_{|I|=|J|=k} \operatorname{per} E[I, J].$$

Given the row set, column set, and a bijection, labeling its k matching edges successively by $1, \ldots, k$ can be done in $k!$ ways. Thus

$$\sum_{\boldsymbol{i},\boldsymbol{j}\ \text{distinct}} \prod_{h=1}^{k} B_{i_h j_h} = k! \sum_{|I|=|J|=k} \operatorname{per} B[I,J] = \frac{k!}{n^k} \sum_{|I|=|J|=k} \operatorname{per} E[I,J].$$

For a condition $\mathcal{A}$, $\mathbf{1}\{\mathcal{A}\}$ equals one when the condition holds and zero otherwise. It is a scalar indicator, distinct from the all-ones vector $\mathbf{1}_p$. For a partition π, define

$$\mu(\pi) = \prod_{D\in\pi} (-1)^{|D|-1}(|D|-1)!, \quad \mathbf{1}_\pi(\boldsymbol{i}) = \prod_{D\in\pi} \mathbf{1}\{i_h \text{ constant on } D\}.$$

Then

$$\mathbf{1}\{i_1,\dots,i_k \text{ distinct}\} = \sum_{\pi\in\mathfrak{P}_k} \mu(\pi)\mathbf{1}_\pi(\boldsymbol{i}).$$

To verify this, let $D_1,\dots,D_r$ be the equality classes of the given coordinates. The right-hand side factors as

$$\prod_{a=1}^{r} \left(\sum_{\pi\in\mathfrak{P}(D_a)} \mu(\pi) \right).$$

The permutation sign is $\operatorname{sgn}(\pi) = (-1)^{\#\{(i,j):i<j,\ \pi(i)>\pi(j)\}}$: it is one for an even number of inversions and minus one for an odd number. A block of size d can be arranged into a directed cycle in $(d-1)!$ ways, and such a cycle has sign $(-1)^{d-1}$. Thus each parenthesis is the sum of the signs of all permutations of that set. This equals one for a singleton. For size at least two, multiplication by a fixed transposition bijects positive and negative terms, so they cancel and the sum is zero.

Substituting on both sides gives

$$F_k = \frac{1}{k!} \sum_{\pi,\sigma\in\mathfrak{P}_k} \mu(\pi)\mu(\sigma) \sum_{\boldsymbol{i},\boldsymbol{j}} \mathbf{1}_\pi(\boldsymbol{i})\mathbf{1}_\sigma(\boldsymbol{j}) \prod_{h=1}^{k} B_{i_h j_h}.$$

If π has a singleton block $\{h\}$, summing the label i_h, which appears only once, gives $\sum_i B_{ij_h} = 0$. The column side is identical. For $k=2$ only one pair of two-element blocks survives, with weight $(-1)(-1)=1$; for $k=3$ only one pair of three-element blocks survives, with weight $2\cdot 2 = 4$. Hence

$$F_2 = \frac{1}{2!}\sum_{i,j} B_{ij}^2, \quad F_3 = \frac{4}{3!}\sum_{i,j} B_{ij}^3, \quad |F_3| \le \frac{2}{3}n^2(C/n)^3 = \frac{2C^3}{3n}.$$

Put $K = BB^{\mathsf{T}}$ and $Q = B^{\mathsf{T}}B$. The fourth-order term is

$$F_4 = \frac{1}{8}(\operatorname{tr} K)^2 + \frac{1}{4}\operatorname{tr}(K^2) - \frac{3}{4}\sum_i \left(\sum_j B_{ij}^2\right)^2 - \frac{3}{4}\sum_j \left(\sum_i B_{ij}^2\right)^2 + \frac{3}{2}\sum_{i,j} B_{ij}^4.$$

The first three classes of partition pairs are: both sides have type $2+2$ ($3\cdot 3 = 9$ pairs); one side has type 4 and the other type $2+2$ (three pairs in each orientation, with block weight -6); and both sides have type 4 (weight 36). Dividing by $4! = 24$ gives the stated coefficients. The three identical pairing pairs give $3/24 = 1/8$, and the six distinct pairing pairs give $6/24 = 1/4$. This also directly confirms the separation of the main term g_4 from the non-Gaussian correction.

A standard real Gaussian variable X has density $(2\pi)^{-1/2}e^{-x^2/2}$. For $u \in \mathbb{R}$, completing the square gives

$$\mathbb{E}e^{uX} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2/2+ux}\,dx = e^{u^2/2}\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-u)^2/2}\,dx = e^{u^2/2}.$$

For $v \in \mathbb{R}^n$, independence gives the multivariate generating function $\mathbb{E}e^{v^{\mathsf{T}}X} = e^{\|v\|^2/2}$. Differentiating with respect to the coordinates gives

$$\mathbb{E}\prod_{h=1}^{2r} X_{i_h} = \sum_{\mathcal{P}\text{ a pairing}} \prod_{\{a,b\}\in\mathcal{P}} \mathbf{1}\{i_a = i_b\}, \qquad \mathbb{E}\prod_{h=1}^{2r+1} X_{i_h} = 0.$$

We record the real Gaussian moment calculation. A standard real Gaussian vector has independent coordinates with density $(2\pi)^{-1/2}e^{-x^2/2}$. Put

$$m_r = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^r e^{-x^2/2}\,dx.$$

Normalization gives $m_0 = 1$, and symmetry of the integral of an odd function gives $m_{2r+1} = 0$. Integration by parts for $r \geq 1$ gives

$$\begin{aligned} m_{2r} &= -\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^{2r-1}\,d(e^{-x^2/2}) \\ &= \frac{2r-1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^{2r-2}e^{-x^2/2}\,dx \\ &= (2r-1)m_{2r-2} = (2r-1)(2r-3)\cdots 1. \end{aligned}$$

The boundary term vanishes because $|x|^a e^{-x^2/2} \to 0$ for every fixed integer $a \geq 0$. With $(-1)!! = 1$, this can be written uniformly as $m_{2r} = (2r-1)!!$, including $r = 0$.

For an index list $i_1, \ldots, i_k$, define the multiplicity $c_a = \#\{h : i_h = a\}$. Independence gives the finite product

$$\mathbb{E}\prod_{h=1}^{k} X_{i_h} = \prod_{a=1}^{n} m_{c_a}.$$

If any c_a is odd, the right-hand side is zero. If all c_a are even, every admissible pairing must pair positions with equal indices. The number of pairings of c_a positions satisfies

$$p_0 = 1, \qquad p_{2r} = (2r-1)p_{2r-2} = (2r-1)!!.$$

Thus the total number of admissible pairings is $\prod_a (c_a - 1)!!$, exactly the moment computed above. Defining the Kronecker symbol $\delta_{ab} = \mathbf{1}\{a = b\}$ gives

$$\mathbb{E}\prod_{h=1}^{2r} X_{i_h} = \sum_{\mathcal{P}} \prod_{\{a,b\}\in\mathcal{P}} \delta_{i_a i_b}.$$

Here $\mathcal{P}$ ranges over pairings of $[2r]$. The empty pairing contributes one when $r = 0$, and odd moments vanish.

First take a real t satisfying $|t|q < 1$. Integrate with respect to Y, and then orthogonally diagonalize BB^{T}. The standard-vector density is $(2\pi)^{-n/2}e^{-\|x\|_2^2/2}$. An orthogonal transformation preserves $\|x\|_2$ and has absolute Jacobian one, so the transformed density is unchanged and still factors into the coordinate densities. Therefore,

$$\begin{aligned}\mathbb{E}e^{tX^{\mathsf{T}}BY} &= \mathbb{E}_X e^{t^2\|B^{\mathsf{T}}X\|^2/2} \\ &= \prod_i \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(1-t^2s_i^2)x^2/2}\,dx \\ &= \prod_i (1-t^2s_i^2)^{-1/2} = G(t).\end{aligned}$$

To extend the identity to the complex disk, put $U = X^{\mathsf{T}}BY$ and choose a real r with $|t| < r < 1/q$. If $q = 0$, then $B = 0$ and the result is immediate. For real u,

$$e^{r|u|} \le e^{ru} + e^{-ru}.$$

Since Y has the same distribution as $-Y$, U has the same distribution as $-U$. Hence

$$\mathbb{E}e^{r|U|} \le \mathbb{E}e^{rU} + \mathbb{E}e^{-rU} = 2G(r) < \infty.$$

Nonnegativity of the terms in the exponential series gives, for $k \ge 0$,

$$\frac{r^k|U|^k}{k!} \le e^{r|U|} \quad \Longrightarrow \quad \mathbb{E}|U|^k \le \frac{k!}{r^k}\mathbb{E}e^{r|U|} \le \frac{2k!}{r^k}G(r) < \infty.$$

Consequently, for every integer $N \ge 0$, expectation can be taken term by term in the finite sum:

$$\sum_{k=0}^{N} \frac{r^k}{k!}\mathbb{E}|U|^k = \mathbb{E}\sum_{k=0}^{N} \frac{r^k|U|^k}{k!} \le \mathbb{E}e^{r|U|} \le 2G(r).$$

Next let $s = |t|/r < 1$. Bound the remainder of the exponential series pointwise:

$$\left| e^{tU} - \sum_{k=0}^{N} \frac{t^kU^k}{k!} \right| \le \sum_{k>N} s^k \frac{r^k|U|^k}{k!} \le s^{N+1}e^{r|U|}.$$

Taking expectations yields the explicit error

$$\left| \mathbb{E}e^{tU} - \sum_{k=0}^{N} \frac{t^k}{k!}\mathbb{E}U^k \right| \le 2G(r)s^{N+1} \longrightarrow 0.$$

This proves that, for every complex t with $|t| < 1/q$,

$$\mathbb{E}e^{tU} = \sum_{k=0}^{\infty} \frac{t^k}{k!}\mathbb{E}U^k, \qquad \sum_{k=0}^{\infty} \frac{|t|^k}{k!}|\mathbb{E}U^k| \le 2G(r) < \infty.$$

We compare these coefficients with those of $G(t)$. Let s_i be the singular values of B. The series for each factor is determined by $c_0 = 1$ and the recurrence

$$c_{m+1} = \frac{2m+1}{2m+2}c_m, \qquad c_m = \frac{(2m)!}{4^m(m!)^2}, \qquad (1-z)^{-1/2} = \sum_{m\ge0} c_m z^m \quad (|z| < 1).$$

The recurrence follows by coefficientwise comparison in $(1-z)f'(z) = f(z)/2$; since $c_{m+1}/c_m < 1$, the series on the right converges absolutely for $|z| < 1$. Further,

$$\frac{d}{dz}\left[(1-z)f(z)^2\right] = -f(z)^2 + 2(1-z)f(z)f'(z) = -f(z)^2 + f(z)^2 = 0.$$

Its value at zero is one, so $(1-z)f(z)^2 = 1$, which selects the branch taking value one at zero. Multiplying the finitely many factors gives

$$G(t) = \prod_i \left(\sum_{m \geq 0} c_m s_i^{2m} t^{2m} \right), \qquad |t|q < 1.$$

On the real interval this agrees with the Gaussian integral already computed. If one coefficient differed from that of $\mathbb{E}U^k/k!$, choose the least differing degree k_0, divide the difference of the two series by t^{k_0}, and let $t \to 0$ along the real axis. This would give

$$0 = \lim_{t \to 0} \frac{\mathbb{E}e^{tU} - G(t)}{t^{k_0}} = \frac{\mathbb{E}U^{k_0}}{k_0!} - [t^{k_0}]G(t) \neq 0,$$

a contradiction. Hence all coefficients agree. Both sides are represented by absolutely convergent series on the stated complex disk, so the identity also holds there. The exponential factor $1/k!$ matches the factor $1/k!$ in the exponential generating function for labeled edges; Wick pairings correspond to partitions whose blocks on both sides all have size two. We interchange convergent series only within the fixed spectral-gap disk. The infinite sum over core excess j is still used only as a formal coefficient identity.

In any contributing graph every degree is at least two, so

$$2(h - v) = \sum_u (\deg u - 2) \geq 0.$$

If $h - v = 0$, all vertices have degree exactly two and every connected component is pure degree two. After removing those components, any nonempty core has $j = h - v \geq 1$. Splitting the edge-label set between the pure degree-two part and the core gives a factor $\binom{k}{h}$; multiplying by $1/k!$ gives $1/[h!(k-h)!]$, so the generating functions multiply exactly.

Put $K = BB^{\mathsf{T}}$. The matrix eigendecomposition and $-\log(1-u) = \sum_{r \geq 1} u^r/r$ give

$$\log G(t) = \frac{1}{2} \sum_{r=1}^{\infty} \frac{t^{2r}}{r} \operatorname{tr}(K^r) = A_s t^2 + B_s t^4 + C_s t^6 + O(t^8),$$

Within this paragraph define the scalars $A_s = \operatorname{tr} K/2$, $B_s = \operatorname{tr}(K^2)/4$, and $C_s = \operatorname{tr}(K^3)/6$. Exponentiating coefficient by coefficient gives

$$\begin{aligned} G(t) &= 1 + (A_s t^2 + B_s t^4 + C_s t^6) + \frac{1}{2}(A_s t^2 + B_s t^4)^2 + \frac{1}{6}(A_s t^2)^3 + O(t^8) \\ &= 1 + A_s t^2 + (B_s + A_s^2/2)t^4 + (C_s + A_s B_s + A_s^3/6)t^6 + O(t^8). \end{aligned}$$

Therefore

$$\begin{aligned} g_2 &= \tfrac{1}{2} \operatorname{tr} K, \\ g_4 &= \tfrac{1}{4} \operatorname{tr}(K^2) + \tfrac{1}{8} (\operatorname{tr} K)^2, \\ g_6 &= \tfrac{1}{6} \operatorname{tr}(K^3) + \tfrac{1}{8} \operatorname{tr} K \operatorname{tr}(K^2) + \tfrac{1}{48} (\operatorname{tr} K)^3. \end{aligned}$$

All odd coefficients are zero because $G(t)$ contains only t^2. The scalars A_s, B_s, C_s are used only in this exponential expansion.

There are three two-element partitions $12|34, 13|24, 14|23$. Choosing one on each side gives nine pairs: three identical and six distinct. Identical pairings give the matrix-label sum $(\sum_{ij} B_{ij}^2)^2 = (\operatorname{tr} K)^2$; distinct pairings give $\operatorname{tr}(K^2)$. Thus the pure degree-two contribution is

$$\frac{3}{4!}(\operatorname{tr} K)^2 + \frac{6}{4!} \operatorname{tr}(K^2) = g_4.$$

If one side is a four-element block, its Möbius weight is -6. The other side has three possible pairings into two-element blocks, each with weight $(-1)^2 = 1$. The row-colored degree-four term is therefore

$$\frac{3(-6)}{24} \sum_i \sum_{j,k} B_{ij}^2 B_{ik}^2 = -\frac{3}{4} \sum_i \left(\sum_j B_{ij}^2 \right)^2 .$$

Exchanging row and column colors gives the column-colored degree-four term

$$-\frac{3}{4} \sum_j \sum_{i,k} B_{ij}^2 B_{kj}^2 = -\frac{3}{4} \sum_j \left(\sum_i B_{ij}^2 \right)^2 .$$

If both sides are four-element blocks, the product of the weights is $(-6)^2 = 36$. The last term is therefore

$$\frac{36}{24} \sum_{i,j} B_{ij}^4 = \frac{3}{2} \sum_{i,j} B_{ij}^4.$$

The four classes sum to the expression for F_4 above. Partitions containing degree-one vertices contribute zero by centering.

3.3 Estimating non-Gaussian cores

We first sum over the internal indices of each degree-two chain and bound the resulting matrix product by its operator norm. After these chains are compressed, a graph of excess j has at most $2j$ vertices. Put

$$W = \max\left\{1, CR + \frac{C^2R^2}{1-Rq}\right\}, \qquad D = 710W^3, \qquad T_1 = \frac{3}{2}W^2 + \frac{10}{3}W^3.$$

Lemma 3.2 (Compressed core activity).
With the notation above,

$$\|C_j\|_R \le (Dj/n)^j \qquad (j \ge 1), \tag{3.2}$$

and the first excess satisfies the sharper bound

$$\|C_1\|_R \le T_1/n.$$

Here $\|H\|_R = \sum_{k\ge 0} |[t^k]H| R^k$.

Proof.

Compress all maximal paths whose internal vertices have degree two. Each compressed edge represents a chain of positive length whose endpoints have degree at least three. Loops, parallel edges, and disconnected compressed graphs are allowed. If b is the number of vertices of degree at least three and e is the number of compressed edges, compression preserves excess and gives

$$e = b + j, \qquad 2j = \sum_{i=1}^{b} (d_i - 2), \qquad 1 \le b \le 2j, \qquad e \le 3j.$$

The decoration counting identity. Fix the excess and the original number of edges. Sum over all ordered degree sequences $(d_1, \dots, d_b)$, assignments of the two vertex colors, pairings of labeled half-edges, and positive chain lengths compatible with the colors. After removing the temporary labels, the absolute Möbius compensation is $1/(b! \prod_i d_i)$. For a chain contraction, its internal numerical indices must be summed before taking the absolute value.

The structural vertices are the blocks of the two partitions of the edge labels, and their numerical indices lie in $[n]$. Distinct blocks remain distinct vertices even when their numerical indices coincide. Möbius inversion has already accounted for the original distinctness constraints, so we count partition pairs directly. Temporarily label the b vertices of high degree and the d_i half-edges at each vertex, and use these labels to choose a unique reading direction for every chain. After summing over all ordered degree sequences, every original core with labeled edges has $b!\prod_i d_i!$ decorations. For any fixed degree sequence, only compatible vertex labelings occur; summing over all sequences produces exactly the full $b!$ factor. Every structural vertex is identifiable by its color and incident original edge labels, and every incident half-edge by its original edge label. There is therefore no additional automorphism stabilizer. Conversely, a decorated half-edge pairing and all positive chain lengths determine every chain position. The original h edge labels can be assigned to these positions in $h!$ ways. When the bipartite color-parity conditions hold, the internal vertices and their colors are then determined as well.

Removing the $h!$ factor from the original exponential generating function and the temporary decorations, and restoring the absolute Möbius weights $\prod_i (d_i-1)!$ of the high-degree vertices, gives the compensation

$$\frac{1}{b!\prod_i d_i}.$$

This step includes loops and chain reversal. Order the two distinct labeled half-edges in each pair, and read the chain starting from the smaller one. Reversal changes the list of original edge labels occupying these positions; it does not supply an additional free factor of two. The two half-edges of a loop follow the same convention. Internal degree-two vertices have absolute Möbius weight one, and there is no additional factorial for chain length. Chains with endpoints of different colors have odd length; those with endpoints of the same color have even length. In particular, every loop has even length at least two. These restrictions are part of the exact identity.

For example, take one row-colored vertex of degree four and two loops of length two, each passing through its own column-colored vertex of degree two. Then $j=b=1$ and $h=4$. There are three ways to split the four original edge labels into two pairs. The degree-four vertex has Möbius weight $(-1)^3 3! = -6$, whereas the product of the two degree-two weights is $+1$. After division by $4!$, the signed contribution is

$$-\frac{3}{4}\sum_i \left(\sum_k B_{ik}^2\right)^2.$$

Even if numerical indices coincide in some summands, the two structural column vertices remain distinct. Exchanging colors gives $-\frac{3}{4}\sum_k(\sum_i B_{ik}^2)^2$. The absolute compensation bound uses the absolute values of these terms. As an example with parallel edges, take three chains of length one between a row vertex and a column vertex. Their contribution is $\frac{(2!)^2}{3!}\sum_{i,k} B_{ik}^3 = \frac{2}{3}\sum_{i,k} B_{ik}^3$, exactly the cubic term above. These examples include loops, parallel edges, and the reversal convention, while the original graph remains bipartite.

We discard color restrictions only when taking an upper bound. For fixed j, b, the resulting total compensation is

$$U_{j,b} = \frac{2^b}{b!}\frac{(2e)!}{2^e e!}\sum_{\substack{d_1+\cdots+d_b=2e\\ d_i\geq 3}} \frac{1}{d_1\cdots d_b}, \qquad e=b+j.$$

In the following estimate, each chain's internal numerical indices must be summed before taking absolute values. Replacing B by $|B|$ entry by entry at the outset would lose the operator spectral-gap control. A chain of length one is a matrix entry, whose absolute value is at most C/n. For a chain of length $\ell \geq 2$, first sum its internal labels. Its contraction is an entry of an alternating product of B, B^{T}. The Euclidean norms of the row or column vectors at the ends are at most $C/\sqrt{n}$, and the operator norms of the intervening factors are at most q. Thus the absolute value of the contraction is at most

$$\frac{C^2}{n}q^{\ell-2}.$$

Summing over lengths with weight R bounds the total activity of each chain by W/n. Summing the labels of the b high-degree vertices contributes at most n^b. Hence

$$\|C_j\|_R \le n^{-j}\sum_{b=1}^{2j} U_{j,b}W^{b+j}.$$

The number of degree compositions is $\binom{2j-1}{b-1} \le 4^j$: write $d_i = 3 + x_i$, so that $\sum_i x_i = 2j - b$. In addition, $\prod_i d_i^{-1} \le 3^{-b}$, and the number of half-edge pairings is at most $(6j)^{b+j}$. Therefore

$$\sum_{b=1}^{2j} U_{j,b}W^{b+j} \le (24jW)^j \sum_{b=1}^{2j} \frac{(4jW)^b}{b!}.$$

Taking $\theta = 1/(2W) \le 1/2$ gives

$$\sum_{b=0}^{2j} \frac{(4jW)^b}{b!} \le \theta^{-2j}\exp(4jW\theta) = (2\mathrm{e}W)^{2j}.$$

Thus the activity is at most $(96\mathrm{e}^2W^3j)^j \le (710W^3j)^j$. For the last inequality, use $\mathrm{e} < 87/32$, obtained by summing the exponential series through degree six and bounding its tail by $8/(7 \cdot 7!)$.

For $j = 1$ only $b = 1, 2$ are possible. The compensation formula gives, respectively,

$$U_{1,1} = \frac{3}{2}, \qquad U_{1,2} = \frac{10}{3}.$$

Combining this with the chain bound gives $\|C_1\|_R \le T_1/n$.

Compressing one degree-two vertex removes one vertex and one edge, so $h - v$ is unchanged. The compressed graph satisfies

$$2e = \sum_{i=1}^{b} d_i, \quad e - b = j, \quad 2j = \sum_{i=1}^{b}(d_i - 2) \ge b,$$

Consequently $b \le 2j$ and $e = b + j \le 3j$. The numbers of temporary decorations and original edge labels cancel against the Möbius weights exactly as follows:

$$\frac{1}{h!} \cdot \frac{h!}{b!\prod_i d_i!} \cdot \prod_i (d_i - 1)! = \frac{1}{b!}\prod_i \frac{(d_i - 1)!}{d_i!} = \frac{1}{b!\prod_i d_i}.$$

For $2e$ half-edges, the number of pairings follows from $p_{2e} = (2e - 1)p_{2e-2}$ and $p_0 = 1$:

$$p_{2e} = (2e - 1)(2e - 3)\cdots 1 = \frac{(2e)!}{2^e e!}.$$

There are 2^b color assignments. Ignoring compatibility restrictions on colors and chain lengths therefore gives $U_{j,b}$. A chain of length one has contraction B_{ij}. For length $\ell \ge 2$, write the contraction as $r^{\mathsf{T}}M_1 \cdots M_{\ell-2}c$, where r, c are rows or columns of B. Then

$$\|r\|_2^2, \|c\|_2^2 \le n(C/n)^2 = C^2/n,$$

$$|r^{\mathsf{T}}M_1 \cdots M_{\ell-2}c| \le \|r\|_2 \prod_a \|M_a\|_{\mathrm{op}}\|c\|_2 \le (C^2/n)q^{\ell-2}.$$

Weighting by length gives

$$R\frac{C}{n}+\sum_{\ell=2}^{\infty}R^{\ell}\frac{C^2}{n}q^{\ell-2}=\frac{1}{n}\left(CR+\frac{C^2R^2}{1-Rq}\right)\le W/n.$$

The b remaining vertices have at most n^b numerical labelings, giving $n^b(W/n)^e=n^{-j}W^{b+j}$. Summing the internal chain indices preserves the factor $q^{\ell-2}$.

To count degree compositions, put $x_i=d_i-3\ge 0$. Then $\sum_i x_i=2j-b$, so

$$\#\{(d_1,\dots,d_b)\}=\binom{(2j-b)+b-1}{b-1}=\binom{2j-1}{b-1}\le 2^{2j-1}\le 4^j.$$

Moreover, $\prod_i 1/d_i\le 3^{-b}$ and the number of pairings is at most $(2e)^e\le(6j)^{b+j}$. Thus

$$\begin{aligned}U_{j,b}W^{b+j}&\le\frac{2^b}{b!}(6j)^{b+j}4^j3^{-b}W^{b+j}\\&=(24jW)^j\frac{(4jW)^b}{b!}.\end{aligned}$$

If $0<\theta\le 1$ and $b\le 2j$, then $\theta^b\ge\theta^{2j}$. Hence

$$\sum_{b=0}^{2j}\frac{(4jW)^b}{b!}\le\theta^{-2j}\sum_{b=0}^{\infty}\frac{(4jW\theta)^b}{b!}=\theta^{-2j}e^{4jW\theta}.$$

Taking $\theta=1/(2W)$ gives $(2W)^{2j}e^{2j}$. Multiplication by $(24jW)^j$ yields $(96e^2W^3j)^j$. The numerical estimate follows from

$$e<\sum_{h=0}^{6}\frac{1}{h!}+\frac{1}{7!}\sum_{r=0}^{\infty}8^{-r}=\frac{1957}{720}+\frac{1}{4410}<\frac{87}{32},$$

$$96(87/32)^2=\frac{22707}{32}=709.59375<710.$$

When $j=1$, $b=1,e=2,d_1=4$. Thus

$$U_{1,1}=\frac{2}{1!}\frac{4!}{2^2\,2!}\frac{1}{4}=\frac{3}{2}.$$

When $b=2,e=3$, the only degree sequence is $(3,3)$. Hence

$$U_{1,2}=\frac{2^2}{2!}\frac{6!}{2^3\,3!}\frac{1}{3\cdot 3}=\frac{10}{3}.$$

This gives $T_1=(3/2)W^2+(10/3)W^3$ term by term.

Fix a compressed core and assign numerical labels to its b remaining high-degree vertices. Its e compressed edges have internal structural vertex sets $V_1,\dots,V_e$, respectively; these sets are pairwise disjoint. Numerical labels may coincide, but each structural vertex has its own independent summation variable. The distributive law for finite sums therefore gives

$$\sum_{\text{all internal labels}}\prod_{r=1}^{e}\left(\prod_{a\in\text{chain}_r}B_{i_aj_a}\right)=\prod_{r=1}^{e}\left[\sum_{\text{internal labels of chain }r}\prod_{a\in\text{chain}_r}B_{i_aj_a}\right].$$

Each bracket is a single entry of an alternating matrix product; only now do we take its absolute value. Summation over positive chain lengths also separates into e geometric series. With the high-degree labels

fixed, the total absolute activity is thus at most $(W/n)^e$. Summing over the n^b choices of high-degree labels then gives

$$n^b(W/n)^e = W^e n^{b-e} = W^{b+j}n^{-j}.$$

The internal vertex sets of the chains are disjoint, so their summation variables are independent; numerical labels may still coincide.

When $j = 1, b = 1$, there is one degree-four vertex with four half-edges. The only pairings are

$$(12)(34), \quad (13)(24), \quad (14)(23).$$

Each pairing gives two loops. There are two colors and the compensation denominator is $b!d_1 = 4$, giving a total of $2 \cdot 3/4 = 3/2$.

When $j = 1, b = 2$, both vertices have degree three. Half-edge pairings fall into two classes. If all three edges join the vertices, there are $3! = 6$ pairings. If each vertex has one loop and the remaining two half-edges form an edge, there are $3 \cdot 3 = 9$ pairings. The two counts sum to fifteen. There are four vertex-color assignments and the compensation denominator is $2! \cdot 3 \cdot 3 = 18$. Therefore

$$\frac{4}{18}(6 + 9) = \frac{10}{3}.$$

Dropping the color and length compatibility conditions gives an upper bound. The two cases give the two terms of T_1. □

3.4 A linear coefficient window and factorial restoration

The activity bound in (3.2) is useful when j/n is small. Take

$$\alpha = \min\left\{\frac{1}{4}, \frac{1}{16D}, \frac{\log\sigma}{2}\right\}, \qquad M = \lfloor \alpha n \rfloor.$$

and put $v_j = (Dj/n)^j$. When $j + 1 \le M$,

$$\frac{v_{j+1}}{v_j} = \frac{D(j+1)}{n}\left(1 + \frac{1}{j}\right)^j \le \frac{\mathrm{e}D(j+1)}{n} \le 3D\alpha < \frac{1}{2}.$$

Hence, for $n \ge 4/\alpha$,

$$\sum_{j=2}^{M} v_j \le 2v_2 = \frac{8D^2}{n^2}.$$

In the formal core identity, contributions of degree at most M can come only from excess $j \le M$. Combining nonnegativity of the coefficients of G, the activity lemma, and the convolution inequality for weighted coefficient sums gives

$$\sum_{k=0}^{M} |F_k - [t^k]G|R^k \le \Gamma(R)\left(\frac{T_1}{n} + \frac{8D^2}{n^2}\right).$$

This estimate concerns only coefficients of degree at most M.

Write $g_k = [t^k]G$ and, for $0 \le k \le M$, put $r_k = n^k/(n)_k$. Since $\alpha \le 1/4$,

$$\log r_k = \sum_{\ell=0}^{k-1} -\log(1 - \ell/n) \le \frac{k(k-1)}{2n(1-\alpha)} \le \frac{k^2}{n} \le \alpha k.$$

The condition $\alpha \le (\log \sigma)/2$ implies $r_k \le \sigma^k$. Moreover,

$$0 \le r_k - 1 \le \frac{k(k-1)}{2n(1-\alpha)} r_k.$$

Since $a_k = r_k F_k$, we can decompose

$$a_k - g_k = r_k(F_k - g_k) + (r_k - 1)g_k.$$

The preceding R-weighted estimate bounds the first part. For the second part we use the second moment of the Gaussian coefficients:

$$\sum_{k\ge 0} k(k-1) g_k \sigma^k = \left((t\partial_t)^2 - t\partial_t\right) G(t)\big|_{t=\sigma}.$$

Put $a_i = \sigma^2 s_i^2$. Differentiating $G(t) = \prod_i (1 - t^2 s_i^2)^{-1/2}$ gives

$$\frac{\left((t\partial_t)^2 - t\partial_t\right) G(t)}{G(t)}\bigg|_{t=\sigma} = \left(\sum_i \frac{a_i}{1-a_i}\right)^2 + \sum_i \frac{a_i(1+a_i)}{(1-a_i)^2}.$$

Define

$$u_\sigma = \frac{\sigma^2 C^2}{1-\sigma^2 q^2}, \qquad v_\sigma = \frac{\sigma^2 C^2 (1+\sigma^2 q^2)}{(1-\sigma^2 q^2)^2},$$

$$K_G = \frac{\Gamma(\sigma)}{2(1-\alpha)} \left(u_\sigma^2 + v_\sigma\right).$$

The two terms in the derivative formula are at most u_σ^2 and v_σ, respectively. Therefore,

$$\sum_{k=0}^{M} (r_k - 1) g_k \le K_G / n.$$

This gives the first two terms in the error bound.

Since $e < 3$ and $\alpha \le 1/(16D)$,

$$\frac{v_{j+1}}{v_j} \le 3D\alpha \le 3/16 < 1/2, \quad \sum_{j=2}^{M} v_j \le v_2 \sum_{r=0}^{\infty} 2^{-r} = 2(2D/n)^2 = 8D^2/n^2.$$

If $H(t) = \sum h_k t^k$ and $K(t) = \sum k_k t^k$, then

$$\sum_{r=0}^{M} |[t^r](HK)| R^r \le \sum_{a,b\ge 0} |h_a||k_b| R^{a+b} = \|H\|_R \|K\|_R.$$

In coefficients with $r \le M$, only $j \le M$ are needed. We therefore truncate to a finite excess sum before applying this inequality; we do not evaluate $\sum_{j\ge 1} C_j(R)$.

The integral estimate for factorial restoration is

$$-\log(1-u) = \int_0^u \frac{dv}{1-v} \le \frac{u}{1-\alpha} \quad (0 \le u \le \alpha).$$

Summing over $u = 0, 1/n, \ldots, (k-1)/n$ gives

$$\log r_k \le \frac{k(k-1)}{2n(1-\alpha)} \le k^2/n \le \alpha k, \quad r_k \le e^{\alpha k} \le \sigma^k.$$

The inequality $e^x - 1 \le xe^x$ gives $r_k - 1 \le [k(k-1)/(2n(1-\alpha))]r_k$. Consequently,

$$\begin{aligned}\sum_{k=0}^{M} |a_k - g_k| &\le \sum_{k=0}^{M} r_k|F_k - g_k| + \sum_{k=0}^{M}(r_k - 1)g_k \\ &\le \sum_{k=0}^{M} R^k|F_k - g_k| + \frac{1}{2n(1-\alpha)} \sum_{k\ge 0} k(k-1)g_k\sigma^k.\end{aligned}$$

Put $D_t = t\partial_t$ and $a_i(t) = t^2 s_i^2$. Successive differentiation gives

$$\begin{aligned}D_t \log G &= \sum_i \frac{a_i}{1-a_i} =: A(t), \\ D_t A &= \sum_i \frac{2a_i}{(1-a_i)^2}, \\ D_t G &= GA, \\ D_t^2 G &= G\left[A^2 + \sum_i \frac{2a_i}{(1-a_i)^2}\right], \\ \frac{(D_t^2 - D_t)G}{G} &= A^2 + \sum_i \left[\frac{2a_i}{(1-a_i)^2} - \frac{a_i}{1-a_i}\right] \\ &= A^2 + \sum_i \frac{a_i(1+a_i)}{(1-a_i)^2}.\end{aligned}$$

At $t = \sigma$ we have $a_i \le \sigma^2 q^2$, so

$$A(\sigma) \le \frac{\sigma^2 C^2}{1-\sigma^2 q^2} = u_\sigma, \quad \sum_i \frac{a_i(1+a_i)}{(1-a_i)^2} \le \frac{\sigma^2 C^2(1+\sigma^2 q^2)}{(1-\sigma^2 q^2)^2} = v_\sigma.$$

Together with $G(\sigma) \le \Gamma(\sigma)$, this gives exactly K_G/n.

Define the projection $\mathcal{P}_M \sum_k h_k t^k = \sum_{k=0}^{M} h_k t^k$. A nonempty core of excess j has at least $j+1$ original edges, so in degrees at most M no $j > M$ occurs. Thus, exactly,

$$\mathcal{P}_M(F - G) = \mathcal{P}_M\left(G \sum_{j=1}^{M} C_j\right).$$

The right-hand side involves only finitely many core series, so its weighted coefficient norm is legitimate. The triangle inequality and the convolution bound give

$$\begin{aligned}\sum_{k=0}^{M} |F_k - g_k| R^k &= \|\mathcal{P}_M(F-G)\|_R \\ &\le \left\| G \sum_{j=1}^{M} C_j \right\|_R \\ &\le \|G\|_R \sum_{j=1}^{M} \|C_j\|_R \\ &\le \Gamma(R) \left[T_1/n + \sum_{j=2}^{M} (Dj/n)^j \right] \\ &\le \Gamma(R)(T_1/n + 8D^2/n^2).\end{aligned}$$

Only finitely many weighted norms of C_j occur. On restoring a_k, the first part satisfies

$$\sum_{k=0}^{M} r_k |F_k - g_k| \le \sum_{k=0}^{M} \sigma^k |F_k - g_k| \le \sum_{k=0}^{M} R^k |F_k - g_k|.$$

The second part, $(r_k - 1)g_k$, is bounded by the Gaussian derivative moment above.

3.5 Controlling the remaining coefficients

We have controlled coefficients through degree M, where M is proportional to n. To estimate higher coefficients by Cauchy's inequality, it suffices, on the fixed circle $|t| = R > 1$, to bound f by $\exp(O(\sqrt{n}))$. The next two lemmas give this bound while retaining the normalization factor $n!/n^n$.

Lemma 3.3 (Permanent polarization).

For a complex square matrix Z, its singular values are the nonnegative square roots of the eigenvalues of Z^*Z, and the corresponding norms are computed as in Definition 2.3. Write $|Z| = (Z^*Z)^{1/2}$ for its operator absolute value, taking the Hermitian positive semidefinite square root, rather than the absolute values of individual entries. For a Hermitian matrix, $H \ge 0$ means that $x^*Hx \ge 0$ holds for every vector. Then

$$|\operatorname{per} Z| \le \sqrt{\operatorname{per}|Z| \ \operatorname{per}|Z^*|}.$$

Proof.

On complex vector spaces we use the inner product $\langle x, y\rangle = x^*y$. In $(\mathbb{C}^n)^{\otimes n}$, put

$$\xi = (n!)^{-1/2} \sum_{\pi \in \mathfrak{S}_n} e_{\pi(1)} \otimes \cdots \otimes e_{\pi(n)}.$$

Then $\|\xi\| = 1$ and $\langle \xi, Z^{\otimes n}\xi\rangle = \operatorname{per} Z$. Take the polar decomposition $Z = U|Z|$. If Z is singular, extend the partial isometry in its polar decomposition to a unitary matrix U. Write $T = |Z|^{\otimes n}$ and $V = U^{\otimes n}$. Applying Cauchy–Schwarz in the semi-inner product induced by the positive semidefinite operator T gives

$$|\langle \xi, VT\xi\rangle| \le \sqrt{\langle V^*\xi, TV^*\xi\rangle \langle \xi, T\xi\rangle}.$$

The two factors are $\operatorname{per}(U|Z|U^*) = \operatorname{per}|Z^*|$ and $\operatorname{per}|Z|$, respectively. Both are nonnegative because they are diagonal matrix elements of positive semidefinite tensor powers. □

Lemma 3.4 (A positive semidefinite permanent bound).

Let $n \geq 1$, and let H be an $n \times n$ Hermitian positive semidefinite matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then

$$\operatorname{per} H \leq \frac{n!}{n^n} h_n(\lambda_1, \ldots, \lambda_n),$$

Here h_n is the complete homogeneous symmetric polynomial of degree n, defined by

$$h_n(\lambda_1, \ldots, \lambda_n) = \sum_{\substack{k_1, \ldots, k_n \geq 0 \\ k_1 + \cdots + k_n = n}} \prod_{i=1}^{n} \lambda_i^{k_i}.$$

All k_i are integers. Every zeroth power in this polynomial equals one, including, when $\lambda_i = 0$, the expression λ_i^0. This polynomial is a different object from the scalar weight $h_k = ((k+1)/2)^{k/2}$ in Section 2. If $\lambda_1 = 1$ and $0 \leq \lambda_i < 1$ for $i \geq 2$, then

$$\operatorname{per} H \leq \frac{n!}{n^n} \prod_{i=2}^{n} (1 - \lambda_i)^{-1}.$$

Proof.

Let Z be a circularly symmetric complex Gaussian vector with covariance matrix H. The complex Wick formula gives $\operatorname{per} H = \mathbb{E} \prod_i |Z_i|^2$. Pointwise AM–GM bounds this product by $n^{-n} \|Z\|^{2n}$. After unitary diagonalization, $\|Z\|^2$ has the same distribution as $\sum_i \lambda_i |\zeta_i|^2$, where ζ_i are independent standard circularly symmetric complex Gaussian variables satisfying $\mathbb{E}|\zeta_i|^{2k} = k!$. Expanding the polynomial therefore gives

$$\mathbb{E} \left(\sum_i \lambda_i |\zeta_i|^2 \right)^n = n! h_n(\lambda_1, \ldots, \lambda_n).$$

Finally, $h_n(1, \lambda_2, \ldots, \lambda_n)$ is the sum of all monomials in $\lambda_2, \ldots, \lambda_n$ of total degree at most n. It is bounded by the product of the corresponding infinite geometric series. □

The Gaussian identity and AM–GM comparison in Lemma 3.4 are the tools used by Han–Niles-Weed [3, arXiv v2, Lemmas 4.3–4.4] for positive semidefinite inputs. Lemma 3.3 connects them to the general matrices needed here.

Put $\beta = RC/(1 - Rq)$ and write $P_n = J_n/n$ for the orthogonal projection onto the all-ones direction. In general, $u^\perp = \{v : u^* v = 0\}$ denotes the orthogonal complement of the vector u. Double centering gives $P_n B = B P_n = 0$. Consequently, on $|t| = R$,

$$|P_n + tB| = P_n + R(B^{\mathsf{T}} B)^{1/2}, \qquad |(P_n + tB)^*| = P_n + R(B B^{\mathsf{T}})^{1/2}.$$

Both have eigenvalue 1 in the all-ones direction and, on $\mathbf{1}^\perp$, eigenvalues Rs_i, including zero singular values when needed. Since $Rq < 1$, the two lemmas and the exact identity $f(t) = (n^n/n!) \operatorname{per}(P_n + tB)$ give

$$|f(t)| \leq \prod_i (1 - Rs_i)^{-1} \leq \exp\left(\frac{R}{1 - Rq} \sum_i s_i \right) \leq \exp(\beta \sqrt{n}).$$

The last step uses $\sum_i s_i \leq \sqrt{n}\, \|B\|_{\mathrm{F}} \leq C\sqrt{n}$. Cauchy's coefficient estimate therefore gives

$$\sum_{k > M} |a_k| \leq \frac{R}{R - 1} \exp\bigl(\beta \sqrt{n} - \alpha n \log R \bigr).$$

For the Gaussian tail, nonnegativity of the coefficients gives the slightly weaker but convenient bound

$$\sum_{k>M} g_k \le \Gamma(\sigma)\sigma^{-M} \le \sigma\Gamma(\sigma)\exp(-\alpha n\log\sigma)\,.$$

Both tails decay exponentially in n. Combining them with the $O(n^{-1})$ estimate in the finite window proves Theorem 3.1.

For any M,

$$\begin{aligned}\langle \xi, M^{\otimes n}\xi\rangle &= \frac{1}{n!}\sum_{\pi,\sigma\in\mathfrak{S}_n}\prod_{h=1}^{n} M_{\pi(h),\sigma(h)}\\ &= \frac{1}{n!}\sum_{\pi\in\mathfrak{S}_n}\sum_{\tau\in\mathfrak{S}_n}\prod_{i=1}^{n} M_{i,\tau(i)} = \operatorname{per} M.\end{aligned}$$

Here $\tau = \sigma\circ\pi^{-1}$; for fixed π, $\sigma\mapsto\tau$ is a bijection. The $n!$ basis tensors are mutually orthogonal, so $\|\xi\|^2 = (n!)^{-1}n! = 1$. If $T\ge 0$,

$$|\langle a, Tb\rangle|^2 = |\langle T^{1/2}a, T^{1/2}b\rangle|^2 \le \langle a, Ta\rangle\langle b, Tb\rangle.$$

Taking $a = V^*\xi, b = \xi$ and using $VTV^* = (U|Z|U^*)^{\otimes n} = |Z^*|^{\otimes n}$ gives Lemma 3.3.

A standard circularly symmetric complex variable $\zeta = x + iy$ has density $\pi^{-1}e^{-|z|^2}$. Integration in polar coordinates gives

$$\begin{aligned}\mathbb{E}|\zeta|^{2k} &= \pi^{-1}\int_0^{2\pi} d\theta \int_0^{\infty} r^{2k}e^{-r^2} r\,dr\\ &= \int_0^{\infty} u^k e^{-u}\,du = k!.\end{aligned}$$

Writing $I_k = \int_0^{\infty} u^k e^{-u}\,du$, integration by parts gives $I_k = kI_{k-1}$ and $I_0 = 1$.

For integers $a, b \ge 0$, polar integration gives the one-dimensional mixed moment

$$\begin{aligned}\mathbb{E}[\zeta^a(\overline{\zeta})^b] &= \frac{1}{\pi}\int_0^{\infty} r^{a+b+1}e^{-r^2}\,dr\int_0^{2\pi} e^{i(a-b)\theta}\,d\theta\\ &= \begin{cases} a!, & a = b,\\ 0, & a\ne b.\end{cases}\end{aligned}$$

If $a\ne b$, the angular integral is $[e^{i(a-b)\theta}/(i(a-b))]_0^{2\pi} = 0$. If $a = b$, the angular integral is 2π; the substitution $u = r^2$ then gives $\int_0^{\infty} u^a e^{-u}\,du = a!$.

Let $\zeta_1,\dots,\zeta_n$ be independent, and put $Z = L\zeta$ and $H = LL^*$. For index lists $\boldsymbol{a}, \boldsymbol{b}\in[n]^n$, the preceding identity and independence imply that the mixed moment is zero if some numerical index has different multiplicities in the two lists. If every multiplicity agrees and is c_j, the mixed moment is $\prod_j c_j!$. This is exactly the number of bijective pairings that match every position in the first list to a position of equal value in the second. Thus

$$\mathbb{E}\prod_{i=1}^{n}\zeta_{a_i}\prod_{j=1}^{n}\overline{\zeta}_{b_j} = \sum_{\sigma\in\mathfrak{S}_n}\prod_{i=1}^{n}\delta_{a_i,b_{\sigma(i)}}.$$

Expanding the product and interchanging the finite sums gives

$$\begin{aligned}\mathbb{E}\prod_{i=1}^{n}|Z_i|^2 &= \sum_{\boldsymbol{a},\boldsymbol{b}}\prod_i L_{i,a_i}\prod_j \overline{L}_{j,b_j}\sum_{\sigma\in\mathfrak{S}_n}\prod_i \delta_{a_i,b_{\sigma(i)}}\\ &= \sum_{\sigma\in\mathfrak{S}_n}\sum_{\boldsymbol{a}}\prod_i L_{i,a_i}\overline{L}_{\sigma(i),a_i}\\ &= \sum_{\sigma\in\mathfrak{S}_n}\prod_i\left(\sum_{a=1}^{n} L_{i,a}\overline{L}_{\sigma(i),a}\right)\\ &= \sum_{\sigma\in\mathfrak{S}_n}\prod_i H_{i,\sigma(i)} = \operatorname{per} H.\end{aligned}$$

The second line uses $b_{\sigma(i)} = a_i$ to sum out all b indices. The third line uses the independent summation of each a_i. The matrix L may be singular: take $L = H^{1/2}$. Thus the proof includes positive semidefinite matrices with zero eigenvalues. The same second-moment calculation also determines the meaning of covariance: $\mathbb{E}Z_i\overline{Z}_j = H_{ij}$ and $\mathbb{E}Z_iZ_j = 0$.

For every sample $|Z_i|^2 \geq 0$. AM–GM gives

$$\prod_i |Z_i|^2 \leq \left(\frac{\sum_i |Z_i|^2}{n}\right)^n, \qquad \operatorname{per} H \leq n^{-n}\mathbb{E}\|Z\|_2^{2n}.$$

A unitary transformation preserves $\|Z\|$. Applying the multinomial theorem term by term gives

$$\begin{aligned}\mathbb{E}\left(\sum_i \lambda_i|\zeta_i|^2\right)^n &= \sum_{k_1+\cdots+k_n=n}\frac{n!}{\prod_i k_i!}\prod_i \lambda_i^{k_i}\mathbb{E}|\zeta_i|^{2k_i}\\ &= n!\sum_{k_1+\cdots+k_n=n}\prod_i \lambda_i^{k_i} = n!h_n(\lambda).\end{aligned}$$

If $\lambda_1 = 1$, eliminate the first variable using $k_1 = n - \sum_{i\geq 2} k_i$:

$$h_n(1,\lambda_2,\ldots,\lambda_n) = \sum_{\substack{k_2,\ldots,k_n\geq 0\\ \sum k_i\leq n}}\prod_{i\geq 2}\lambda_i^{k_i} \leq \prod_{i=2}^{n}\sum_{k_i=0}^{\infty}\lambda_i^{k_i} = \prod_{i=2}^{n}(1-\lambda_i)^{-1}.$$

Write $P = P_n$. Since $P^2 = P$ and $PB = B^{\mathsf{T}}P = BP = PB^{\mathsf{T}} = 0$,

$$(P+tB)^*(P+tB) = P + R^2B^{\mathsf{T}}B \quad (|t| = R).$$

These two parts act on orthogonal subspaces, so the square root is $P + R(B^{\mathsf{T}}B)^{1/2}$. The two polarization factors have the same eigenvalues $1, Rs_1, \ldots$. The zero singular value corresponding to the all-ones direction may be omitted from the product because its factor is one. Define $A_R = \prod_i(1-Rs_i)^{-1} > 0$. Take logarithms only of this positive upper bound, avoiding, when $f(t) = 0$, the undefined expression $\log|f(t)|$. Retaining the normalization exactly gives

$$\begin{aligned}|f(t)| &= \frac{n^n}{n!}|\operatorname{per}(P+tB)| \leq \prod_i(1-Rs_i)^{-1},\\ \log A_R &= -\sum_i \log(1-Rs_i) \leq \sum_i \frac{Rs_i}{1-Rs_i} \leq \frac{R}{1-Rq}\sum_i s_i \leq \frac{RC}{1-Rq}\sqrt{n} = \beta\sqrt{n}.\end{aligned}$$

Hence $|f(t)| \leq A_R \leq e^{\beta\sqrt{n}}$, including the case $f(t) = 0$. Zeros are possible for admissible inputs: take $n = 3$ and

$$E = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \qquad B = E/3.$$

This matrix is doubly centered, and

$$B^{\mathsf{T}}B = \frac{1}{9}(3I_3 - J_3).$$

The eigenvalue of J_3 is three in the all-ones direction and zero on the zero-sum subspace. The eigenvalues of $B^{\mathsf{T}}B$ are therefore $0, 1/3, 1/3$, giving $\|B\|_{\text{op}} = 1/\sqrt{3} \le 3/5 =: q < 1$. The six permanent terms come from the identity permutation, three transpositions, and two three-cycles. Hence

$$\begin{aligned} 6f(t) &= 1 + 3(1-t^2) + (1+t)^3 + (1-t)^3 \\ &= 1 + 3 - 3t^2 + 2 + 6t^2 = 6 + 3t^2, \\ f(i\sqrt{2}) &= 1 + (i\sqrt{2})^2/2 = 0. \end{aligned}$$

Taking $R = \sqrt{2} < 5/3 = 1/q$ and $\sigma = (1+\sqrt{2})/2$ gives a zero of f on the chosen circle $|t| = R$. Thus $\log|f(t)|$ is not defined everywhere on this circle.

Cauchy's coefficient formula on the circle $t = Re^{i\theta}$ is

$$a_k = \frac{1}{2\pi R^k}\int_0^{2\pi} f(Re^{i\theta})e^{-ik\theta}\,d\theta, \qquad |a_k| \le e^{\beta\sqrt{n}}R^{-k}.$$

Thus

$$\sum_{k>M}|a_k| \le e^{\beta\sqrt{n}}\frac{R^{-M}}{R-1} \le \frac{R}{R-1}e^{\beta\sqrt{n}-\alpha n\log R}.$$

Here $M \ge \alpha n - 1$. The coefficients on the Gaussian side are nonnegative, so

$$\sum_{k>M} g_k \le \sigma^{-M}\sum_{k>M} g_k\sigma^k \le \sigma\Gamma(\sigma)e^{-\alpha n\log\sigma}.$$

Put $c = \alpha\log R > 0$. For $n \ge (2\beta/c)^2$, we have $\beta\sqrt{n} - cn \le -cn/2$. Since $ne^{-cn/2} \to 0$, the first tail is $o(n^{-1})$, and the same argument applies to the second. Here we use the linear growth of $M = \lfloor \alpha n \rfloor$.

3.6 Combining the error bounds

Proposition 3.5 (An explicit error bound).

For every integer $n \ge 4/\alpha$,

$$|f(1) - G(1)| \le \frac{\Gamma(R)T_1 + K_G}{n} + \frac{8\Gamma(R)D^2}{n^2} + \frac{R}{R-1}\exp\bigl(\beta\sqrt{n} - \alpha n\log R\bigr) + \sigma\Gamma(\sigma)\exp(-\alpha n\log\sigma)\,.$$

The first two terms come from the coefficient window and factorial-restoration errors in Section 3.4. The last two come from the tail bounds in Section 3.5. Both exponential terms are $o(n^{-1})$, giving the uniform error rate in Theorem 3.1.

Put $c_1 = \alpha\log R > 0$, $c_2 = \alpha\log\sigma > 0$, and $\beta = RC/(1-Rq) \ge 0$. For

$$n \ge N_1 := \max\{1, (2\beta/c_1)^2\},$$

we have

$$\sqrt{n} \geq 2\beta/c_1 \implies \beta\sqrt{n} \leq (c_1/2)n \implies \beta\sqrt{n} - c_1 n \leq -c_1 n/2.$$

For any fixed $c > 0$, the function $u(x) = xe^{-cx}$ satisfies, on $x \geq 0$,

$$u'(x) = e^{-cx}(1 - cx), \qquad \max_{x \geq 0} u(x) = u(1/c) = 1/(ec).$$

Thus, for $n \geq N_1$,

$$\frac{R}{R-1} \exp(\beta\sqrt{n} - c_1 n) \leq \frac{R}{R-1} e^{-c_1 n/2} \leq \frac{2R}{ec_1(R-1)} \frac{1}{n},$$

$$\sigma\Gamma(\sigma) e^{-c_2 n} \leq \frac{\sigma\Gamma(\sigma)}{ec_2} \frac{1}{n}.$$

If we also require $n \geq 4/\alpha$, the right-hand side of Proposition 3.5 is at most

$$\frac{1}{n}\left[\Gamma(R)T_1 + K_G + 8\Gamma(R)D^2 + \frac{2R}{ec_1(R-1)} + \frac{\sigma\Gamma(\sigma)}{ec_2}\right].$$

Every quantity in brackets depends only on C, q, R, σ fixed in advance, and not on the matrix being estimated. To obtain one constant valid for all $n \geq 1$, the finitely many smaller dimensions must also be covered. From $|1 + E_{ij}| \leq 1 + C$,

$$|f(1)| \leq \frac{1}{n!} \sum_{\pi \in \mathfrak{S}_n} (1 + C)^n = (1 + C)^n, \qquad 1 \leq G(1) \leq \Gamma(1).$$

Let $N = \lceil \max\{N_1, 4/\alpha\} \rceil$ and enlarge the uniform constant to at least

$$\max_{1 \leq n < N} n\left[(1 + C)^n + \Gamma(1)\right]$$

covers the remaining orders as well; the maximum of an empty set is taken to be zero. □

Quantifying the existing permanent proof.

Divoux and Wang [12, Proposition 3.1] prove a uniform permanent approximation with relative error $o(1)$ for nonnegative doubly stochastic matrices. In the nonnegative setting used in our counting argument, their proof can be refined to give an $O(N^{-1})$ error. This rate is a refinement of their estimates, rather than the statement of their proposition. Theorem 3.1 here allows an arbitrary real doubly centered matrix E, without requiring $J + E$ to be nonnegative. We retain that formulation and its detailed proof; the two theorem statements do not have identical hypotheses.

The refinement concerns two estimates in [12, Sections 3.2–3.3]. Their finite coefficient sum is already bounded by

$$\sum_{1 \leq E \leq K\sqrt{N}} \frac{(cE/N)^{\lceil E/3 \rceil}}{1 - cE/N}.$$

Put $\eta_N = cK/\sqrt{N}$. For sufficiently large N, this is at most

$$\frac{c}{N(1 - \eta_N)} \sum_{E \geq 1} E\eta_N^{\lceil E/3 \rceil - 1} = O(N^{-1}).$$

The remaining coefficients satisfy the exponential tail estimate in their proof. In the subsequent Gamma average, $G \sim \Gamma(N + 1, 1)$ and $Z = N/G$ satisfy

$$\mathbb{E}(Z-1)=0, \qquad \mathbb{E}(Z-1)^2=\frac{1}{N-1}.$$

For $\Delta(s)=\det(I-s^2B^TB)^{-1/2}$, the fixed operator-norm gap and Frobenius-norm bound give a uniform bound on Δ'' over the fixed window used there. A second-order Taylor expansion, together with their estimate outside the window, gives an $O(N^{-1})$ error. Thus this permanent error in the nonnegative doubly stochastic case can be obtained by refining the earlier proof. The global path estimate still requires control of the errors from scaling, degree imbalance, and exceptional vertices.

4. Diagonal scaling and nonprincipal submatrices

This section returns to the graph notation: $s=S\mathbf{1}$ is the score vector, $a=s/(n-1)$, and we assume $n\geq 2$. Theorem 3.1 applies to matrices whose row and column sums agree after scalar normalization. Tournament matrices generally have different row and column sums: if $C=2A/(n-1)$, then

$$C\mathbf{1}=\mathbf{1}+a, \qquad C^{\mathsf{T}}\mathbf{1}=\mathbf{1}-a.$$

We must therefore control the permanent and its Gaussian factor while correcting the marginals. We first construct a local diagonal scaling and estimate its total logarithmic cost. We then compare Gram matrices to control the effects of deleting rows and columns. Finally, paired score corrections bring tournament submatrices within the hypotheses of the local theorem, thereby proving Lemma 4.4.

For a general order $p\geq 1$, retain $P_p=J_p/p=\mathbf{1}_p\mathbf{1}_p^{\mathsf{T}}/p$ and set $\Pi_p=I_p-P_p$, the orthogonal projection that removes the all-ones direction. The total mass of a matrix is $\mathfrak{m}(X)=\sum_{i,j}X_{ij}$. A doubly stochastic matrix is a square matrix with nonnegative entries and every row sum and column sum equal to one. Constants in this section are uniform once the density and centered-kernel spectral-gap parameters have been fixed.

Definition 4.1 (Marginal errors).

For any real $p\times p$ matrix Q ($p\geq 1$; its mass need not yet equal p), define

$$\alpha(Q)=Q\mathbf{1}_p-\mathbf{1}_p, \qquad \beta(Q)=Q^{\mathsf{T}}\mathbf{1}_p-\mathbf{1}_p,$$

$$g(Q)=(\alpha(Q),\beta(Q))\in\mathbb{R}^{2p}, \qquad \varepsilon(Q)=\|g(Q)\|_\infty,$$

$$\|g(Q)\|_2^2=\sum_i\alpha(Q)_i^2+\sum_j\beta(Q)_j^2.$$

Likewise, (x,y) denotes concatenation of two vectors, so $\|(x,y)\|_\infty=\max(\|x\|_\infty,\|y\|_\infty)$. When the mass is p, the errors on each side sum to zero. When the mass differs from p, these definitions still apply, but normalization is required before invoking Lemma 4.1.

Write n for the full order and $m=n-t$ for the retained order. In Lemmas 4.1 and 4.2, p denotes a general matrix order; in the application we take $p=m$. The matrices $C, C', \widehat{C}$ have order n, while $X, \widetilde{X}, B_X, Z$ have order m. Their total masses are

$$\mathfrak{m}(C')=n+\nu, \quad \mathfrak{m}(\widehat{C})=n, \quad \mathfrak{m}(X)=m+\kappa, \quad \mathfrak{m}(\widetilde{X})=\mathfrak{m}(B_X)=m.$$

Here $Z=\Pi_m[(S-I)/(n-1)][R,T]\Pi_m$, and B_X is the scaled doubly stochastic matrix. In Theorem 3.1, the perturbation is mB_X-J_m and its normalization is B_X-P_m.

4.1 Local Scaling and Its Cost

We seek a matrix of the form $B_{ij} = X_{ij}e^{x_i+y_j}$ with every row sum and column sum equal to one. Replacing (x, y) by $(x+c\mathbf{1}, y-c\mathbf{1})$ leaves B unchanged; the condition $\sum x_i = \sum y_j$ removes this one-dimensional freedom. The singular-value spectral gap controls the linearized balancing equations on the remaining subspace. The entry bound further provides dimension-independent control in the infinity norm, making a contraction argument applicable.

Lemma 4.1 (Local scaling).

Let $p \geq 1$, and let X have total mass p and be a real $p \times p$ matrix. Write

$$\alpha = X\mathbf{1}_p - \mathbf{1}_p, \qquad \beta = X^{\mathsf{T}}\mathbf{1}_p - \mathbf{1}_p, \qquad \varepsilon = \max(\|\alpha\|_\infty, \|\beta\|_\infty).$$

Assume $|X_{ij}| \leq C/p$, that the sum of absolute values in each row and each column is at most 2, and that

$$X_0 = X - \frac{\alpha\mathbf{1}_p^{\mathsf{T}} + \mathbf{1}_p\beta^{\mathsf{T}}}{p}, \qquad E = X_0 - P_p$$

satisfies $|E_{ij}| \leq C_0/p$ and $\|E\|_{\text{op}} \leq q < 1$, where $C, C_0 \geq 0$ and $0 \leq q < 1$ are fixed parameters. Define

$$L_0 = \frac{3}{2} + C_0 + \frac{C_0^2}{1-q},$$

$$\varepsilon_0 = \min\left\{\frac{1}{256L_0^2}, \frac{1-q}{2(32L_0+2)}\right\}.$$

If $\varepsilon \leq \varepsilon_0$, the condition $\sum_i x_i = \sum_j y_j$ is satisfied by finite real potential vectors x, y such that every row sum and column sum of $B = \text{diag}(e^x)X\,\text{diag}(e^y)$ equals 1. Moreover,

$$\|(x, y)\|_\infty \leq 4L_0\varepsilon, \qquad |B_{ij}| \leq 2C/p, \qquad \|B - P_p\|_{\text{op}} \leq (1+q)/2.$$

If X is entrywise nonnegative, then B is doubly stochastic and has the same zero pattern as X.

Proof.

The entries of each marginal-error vector sum to zero. Consequently, every row sum and column sum of X_0 is 1, and E is doubly centered. Set $u = (\mathbf{1}_p, -\mathbf{1}_p)$ and $N = uu^{\mathsf{T}}/(2p)$. The balanced Hessian H_0 has diagonal blocks I and off-diagonal blocks X_0, X_0^{T}, respectively. Its gauge-fixed inverse can be written explicitly. Put

$$R_L = (I - EE^{\mathsf{T}})^{-1}, \qquad R_R = (I - E^{\mathsf{T}}E)^{-1}, \qquad O = ER_R.$$

The four blocks of $(\mathsf{H}_0 + N)^{-1}$ are then

$$M_{LL} = R_L - P_p/4, \qquad M_{RR} = R_R - P_p/4,$$

$$M_{LR} = -O - P_p/4, \qquad M_{RL} = -O^{\mathsf{T}} - P_p/4.$$

Indeed, on the common constant direction, the inverse eigenvalue is $1/2$; on the gauge direction, it is 1; and on the two zero-sum subspaces, the usual block inverse applies. The dense entry bound gives

$$|(R_L - I)_{ij}|,\ |(R_R - I)_{ij}| \leq \frac{C_0^2}{p(1-q^2)},$$

$$|O_{ij}| \le \frac{C_0}{p} + \frac{C_0^2 q}{p(1-q^2)}.$$

For the last estimate, decompose $O = E + EE^{\mathsf{T}}ER_R$ and control the two-norms of the row and column at the two ends. The vector one-norm is

$$\|x\|_1 = \sum_i |x_i|,$$

The corresponding induced matrix one-norm and infinity norm are $\|M\|_{1\to 1} = \max_j \sum_i |M_{ij}|$ and $\|M\|_{\infty\to\infty} = \max_i \sum_j |M_{ij}|$, respectively. Both induced norms of the inverse above are therefore at most L_0.

Let H_X have diagonal blocks $\mathrm{diag}(X\mathbf{1}_p)$ and $\mathrm{diag}(X^{\mathsf{T}}\mathbf{1}_p)$, respectively, and off-diagonal blocks X, X^{T}. Both induced norms of the difference $\mathsf{H}_X - \mathsf{H}_0$ are at most 3ε. The Neumann series shows that both induced norms of $(\mathsf{H}_X + N)^{-1}$ are at most $2L_0$.

Write $z = (x, y)$ and $g = (\alpha, \beta)$. The balancing equation is

$$0 = g + \mathsf{H}_X z + \mathcal{N}_X(z).$$

Here $\mathcal{N}_X(z)$ is the concatenation of the row sums and column sums of $X_{ij}(e^{x_i+y_j} - 1 - x_i - y_j)$. On the ball $\|z\|_\infty \le R \le 1/4$, the absolute marginal bounds give

$$\|\mathcal{N}_X(z)\|_\infty \le 8R^2, \qquad \|D\mathcal{N}_X(z)\|_{\infty\to\infty} \le 16R.$$

All these vectors are orthogonal to u. On the gauge subspace $u^\perp$, consider

$$\mathcal{T}(z) = -(\mathsf{H}_X + N)^{-1}(g + \mathcal{N}_X(z)).$$

Take $R = 4L_0\varepsilon$. The norm of the image of the origin is at most $R/2$, and the Lipschitz constant is at most $128L_0^2\varepsilon \le 1/2$. Thus the map preserves the gauge ball and is a contraction, yielding the required finite potential vectors. If $\varepsilon = 0$, simply take $z = 0$.

Finally, $R \le 1/4$ gives the entry bound. The assumptions on absolute row and column sums imply

$$\|B - X\|_{1\to 1},\ \|B - X\|_{\infty\to\infty} \le 32L_0\varepsilon.$$

We also have $\|X - X_0\|_{\mathrm{op}} \le 2\varepsilon$, so the second condition in ε_0 establishes the spectral gap of the centered kernel. Positive diagonal factors preserve both nonnegativity and the zero pattern. This proves the lemma.

Finding a balanced point alone does not suffice to estimate the permanent: restoring the diagonal factors requires multiplying the permanent by $\exp(-\sum x_i - \sum y_j)$. The next lemma controls this total cost and also gives the Frobenius displacement bound needed to compare Gaussian factors.

Lemma 4.2 (Displacement and capacity).

Under the preceding hypotheses, assume in addition that $X \ge 0$. Let $\theta_X = \sum_i x_i + \sum_j y_j$ denote the total scaling potential. There is a constant depending only on the fixed density and spectral-gap parameters such that

$$\|(x, y)\|_2 \le K\|(\alpha, \beta)\|_2, \qquad \|B - X\|_F \le \frac{K}{\sqrt{p}}\|(\alpha, \beta)\|_2,$$

$$0 \le \theta_X \le K\|(\alpha, \beta)\|_2^2.$$

General mass. For a nonnegative matrix X of arbitrary positive total mass, suppose that $\widetilde{X} = pX/\mathfrak{m}(X)$ satisfies the hypotheses of Lemma 4.1, and apply the preceding conclusions to $\widetilde{X}$. If B is its scaling, the following exact identities hold:

$$\theta_X = \theta_{\widetilde{X}} + p \log \frac{p}{\mathfrak{m}(X)}, \qquad \operatorname{per} X = e^{-\theta_X} \operatorname{per} B.$$

In particular, if $\kappa = \mathfrak{m}(X) - p$, then $\operatorname{per} X \le e^{\kappa} \operatorname{per} B$. When the mass differs from p, θ_X itself need not be nonnegative.

Proof.

The convex potential is

$$\Phi_X(x, y) = \sum_{i,j} X_{ij} e^{x_i + y_j} - \sum_i x_i - \sum_j y_j.$$

Its gradient at the origin is $g = (\alpha, \beta)$, and a balanced point is a minimizer. If $R = \|(x, y)\|_\infty$, then the Hessian along tz, $0 \le t \le 1$, is bounded below by the Hessian at the balanced matrix B multiplied by e^{-2R}. On the gauge subspace, the smallest eigenvalue of the latter is at least $1 - \|B - P_p\|_{\mathrm{op}}$. Thus there is a fixed positive lower bound λ along the entire segment. Integrating the gradient gives $\|z\|_2 \le \lambda^{-1} \|g\|_2$. The dense entry bound together with $|e^{x_i + y_j} - 1| \le K |x_i + y_j|$ then gives the Frobenius displacement estimate.

Since $\Phi_X(0) = p$ and $\Phi_X(z) = p - \theta_X$, convexity gives $\theta_X \ge 0$. Strong convexity gives $\theta_X \le \|g\|_2^2/(2\lambda)$. The general-mass identity follows by absorbing the scalar $p/\mathfrak{m}(X)$ into the two diagonal potentials. Finally, $p \log(\mathfrak{m}(X)/p) \le \kappa$ gives the permanent upper bound.

From the mass identity $\sum_{ij} X_{ij} = p$,

$$\sum_i \alpha_i = \sum_j \beta_j = 0, \quad X_0 \mathbf{1} = X\mathbf{1} - \alpha - \mathbf{1} \frac{\sum_j \beta_j}{p} = \mathbf{1}, \quad X_0^{\mathsf{T}} \mathbf{1} = \mathbf{1}.$$

Since $P_p \mathbf{1} = \mathbf{1}$, we have $E\mathbf{1} = E^{\mathsf{T}} \mathbf{1} = 0$. The same calculation also gives

$$\Pi_p X \Pi_p = X - P_p - \frac{\alpha \mathbf{1}^{\mathsf{T}} + \mathbf{1} \beta^{\mathsf{T}}}{p} = E.$$

On the constant subspace, set $c = (\mathbf{1}, \mathbf{1})$ and $u = (\mathbf{1}, -\mathbf{1})$. Then

$$\mathsf{H}_0 c = 2c, \quad \mathsf{H}_0 u = 0, \quad Nc = 0, \quad Nu = u.$$

The inverse therefore has eigenvalues $1/2, 1$ on these two lines, respectively. On the two zero-sum subspaces, direct multiplication verifies

$$\begin{pmatrix} I & E \\ E^{\mathsf{T}} & I \end{pmatrix} \begin{pmatrix} R_L & -E R_R \\ -E^{\mathsf{T}} R_L & R_R \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix},$$

where we use

$$R_L - E R_R E^{\mathsf{T}} = I, \quad R_R - E^{\mathsf{T}} R_L E = I, \quad R_L E = E R_R.$$

The last identity can be checked term by term using the Neumann series. Writing the projection onto the constant subspace back in block form gives the four blocks, including $R_L - P_p/4$ and $-E R_R - P_p/4$.

Because every row and column of E has two-norm at most $C_0/\sqrt{p}$,

$$R_L - I = E(I - E^{\mathsf{T}} E)^{-1} E^{\mathsf{T}}, \quad |(R_L - I)_{ij}| \le \frac{C_0^2}{p(1 - q^2)},$$

$$O - E = EE^{\mathsf{T}}E(I - E^{\mathsf{T}}E)^{-1}, \quad |(O-E)_{ij}| \le \frac{C_0^2 q}{p(1-q^2)}.$$

The absolute sum of each $2p$-dimensional block row is therefore at most

$$1 + \frac{1}{4} + \frac{1}{4} + C_0 + \frac{C_0^2(1+q)}{1-q^2} = \frac{3}{2} + C_0 + \frac{C_0^2}{1-q} = L_0.$$

The matrix is symmetric, so its induced one-norm is the same. Also,

$$\|X - X_0\|_{\infty\to\infty}, \|X - X_0\|_{1\to 1} \le 2\varepsilon, \quad \|\mathsf{H}_X - \mathsf{H}_0\|_{\infty\to\infty}, \|\mathsf{H}_X - \mathsf{H}_0\|_{1\to 1} \le 3\varepsilon.$$

Set $M = (\mathsf{H}_0 + N)^{-1}$ and $D = \mathsf{H}_X - \mathsf{H}_0$. Then

$$(\mathsf{H}_X + N)^{-1} = (I + MD)^{-1}M = \sum_{h=0}^{\infty}(-MD)^h M,$$

$$\|(\mathsf{H}_X + N)^{-1}\| \le \frac{L_0}{1 - 3L_0\varepsilon} \le 2L_0.$$

Here either the one-norm or the infinity norm above may be used; $\varepsilon \le 1/(256L_0^2)$ guarantees $3L_0\varepsilon < 1/2$. For $|h| \le 2R \le 1/2$, the Taylor integral remainder is

$$e^h - 1 - h = h^2 \int_0^1 (1-t)e^{th}\,dt, \qquad |e^h - 1 - h| \le \tfrac{1}{2}e^{1/2}h^2.$$

Each absolute row sum is at most two, and $|x_i + y_j| \le 2R$, so

$$\|\mathcal{N}_X(z)\|_\infty \le 2 \cdot \tfrac{1}{2}e^{1/2}(2R)^2 \le 8R^2.$$

In the derivative, the sum of absolute coefficients for a given row potential and all column potentials is at most

$$2\sum_j |X_{ij}|\,|e^{x_i+y_j} - 1| \le 2 \cdot 2 \cdot 2Re^{1/2} \le 16R.$$

The totals on the two sides agree, so $u^{\mathsf{T}}g = u^{\mathsf{T}}\mathcal{N}_X(z) = 0$. Also, $\mathsf{H}_X u = 0$, and hence the inverse preserves $u^\perp$. Setting $R = 4L_0\varepsilon$ gives

$$\|\mathcal{T}(0)\|_\infty \le 2L_0\varepsilon = R/2, \quad \mathrm{Lip}(\mathcal{T}) \le 2L_0 \cdot 16R = 128L_0^2\varepsilon \le 1/2.$$

Consequently, $\|\mathcal{T}(z)\|_\infty \le R/2 + \|z\|_\infty/2 \le R$. Starting from $z_0 = 0$ and $z_{h+1} = \mathcal{T}(z_h)$,

$$\|z_{h+1} - z_h\|_\infty \le 2^{-h}\|z_1 - z_0\|_\infty, \quad \sum_{h=0}^{\infty}\|z_{h+1} - z_h\|_\infty < \infty.$$

The iteration therefore converges to a finite real vector, and its limit satisfies the balancing equation. Since $R \le 1/4$, we have $e^{x_i+y_j} \le e^{1/2} < 2$, giving $|B_{ij}| \le 2C/p$. At the same time,

$$\|B - X\|_{\infty\to\infty}, \|B - X\|_{1\to 1} \le 2 \cdot 2Re^{1/2} \le 8R = 32L_0\varepsilon,$$

$$\|B - P_p\|_{\mathrm{op}} \le q + (32L_0 + 2)\varepsilon \le (1+q)/2.$$

For nonnegative X, the Hessian quadratic form in any direction $h = (v, w)$ is

$$h^{\mathsf{T}} \nabla^2 \Phi_X(z) h = \sum_{ij} X_{ij} e^{x_i + y_j} (v_i + w_j)^2 \geq 0.$$

At the balanced point z_*, the Hessian is $\left(\begin{smallmatrix} I & B \\ B^{\mathsf{T}} & I \end{smallmatrix}\right)$. On the gauge subspace, decompose into the constant direction and the zero-sum directions. The former has eigenvalue two, and the latter have lower bound $1 - \|B - P_p\|_{\mathrm{op}}$. If $0 \leq t \leq 1$,

$$X_{ij} e^{t(x_i + y_j)} = B_{ij} e^{-(1-t)(x_i + y_j)} \geq e^{-2R} B_{ij}.$$

We may therefore take $\lambda = e^{-2R}(1 - \|B - P_p\|_{\mathrm{op}}) > 0$. Integrating the gradient along the segment gives

$$0 = \nabla \Phi_X(z_*) = g + \int_0^1 \nabla^2 \Phi_X(t z_*) z_* \, dt,$$

$$\lambda \|z_*\|_2^2 \leq -\langle g, z_* \rangle \leq \|g\|_2 \|z_*\|_2, \quad \|z_*\|_2 \leq \lambda^{-1} \|g\|_2.$$

For the Frobenius displacement,

$$\begin{aligned} \|B - X\|_F^2 &= \sum_{ij} X_{ij}^2 (e^{x_i + y_j} - 1)^2 \\ &\leq K \frac{C^2}{p^2} \sum_{ij} (x_i + y_j)^2 \\ &\leq K \frac{C^2}{p^2} \left(2p \sum_i x_i^2 + 2p \sum_j y_j^2 \right) \leq \frac{K'}{p} \|g\|_2^2. \end{aligned}$$

At balance, $\sum_{ij} B_{ij} = p$, so $\Phi_X(z_*) = p - \theta_X$. Strong convexity gives

$$p - \theta_X \geq p + \langle g, z_* \rangle + \frac{\lambda}{2} \|z_*\|_2^2,$$

$$0 \leq \theta_X \leq -\langle g, z_* \rangle - \frac{\lambda}{2} \|z_*\|_2^2 \leq \frac{\|g\|_2^2}{2\lambda}.$$

The last inequality follows from $0 \leq \|\sqrt{\lambda} z_* + g/\sqrt{\lambda}\|_2^2$. If the mass is $p + \kappa$, set $c = p/(p + \kappa)$; add $\frac{1}{2} \log c$ to the potentials on both sides to obtain

$$\theta_X = \theta_{\widetilde{X}} + p \log c, \qquad \operatorname{per} X = e^{-\theta_X} \operatorname{per} B,$$

$$-\theta_X \leq p \log(1 + \kappa/p) \leq \kappa.$$

Here κ may be negative, but $p + \kappa > 0$ is required.

To shorten the formulas in this calculation, temporarily write $P = P_p$, $L = R_L$, $R = R_R$, and $O = E R_R$. Centering gives

$$EP = PE = E^{\mathsf{T}} P = P E^{\mathsf{T}} = 0, \quad LP = PL = RP = PR = P, \quad OP = PO = O^{\mathsf{T}} P = P O^{\mathsf{T}} = 0.$$

After adding the gauge projection, the matrix is

$$\mathsf{H}_0 + N = \begin{pmatrix} I + P/2 & E + P/2 \\ E^{\mathsf{T}} + P/2 & I + P/2 \end{pmatrix},$$

The proposed inverse is $M = \begin{pmatrix} L - P/4 & -O - P/4 \\ -O^{\mathsf{T}} - P/4 & R - P/4 \end{pmatrix}$. Multiplying the upper-left block term by term gives

$$\begin{aligned}[(\mathsf{H}_0 + N)M]_{LL} &= (I + P/2)(L - P/4) + (E + P/2)(-O^{\mathsf{T}} - P/4) \\ &= L - P/4 + P/2 - P/8 - EO^{\mathsf{T}} - P/8 \\ &= L - EO^{\mathsf{T}}.\end{aligned}$$

Since $O^{\mathsf{T}} = E^{\mathsf{T}}L$,

$$L - EO^{\mathsf{T}} = L - EE^{\mathsf{T}}L = (I - EE^{\mathsf{T}})L = I.$$

The upper-right block is

$$\begin{aligned}[(\mathsf{H}_0 + N)M]_{LR} &= (I + P/2)(-O - P/4) + (E + P/2)(R - P/4) \\ &= -O - P/4 - P/8 + ER + P/2 - P/8 \\ &= -O + ER = 0.\end{aligned}$$

The lower-left block is

$$\begin{aligned}[(\mathsf{H}_0 + N)M]_{RL} &= (E^{\mathsf{T}} + P/2)(L - P/4) + (I + P/2)(-O^{\mathsf{T}} - P/4) \\ &= E^{\mathsf{T}}L + P/2 - P/8 - O^{\mathsf{T}} - P/4 - P/8 \\ &= E^{\mathsf{T}}L - O^{\mathsf{T}} = 0.\end{aligned}$$

The lower-right block is

$$\begin{aligned}[(\mathsf{H}_0 + N)M]_{RR} &= (E^{\mathsf{T}} + P/2)(-O - P/4) + (I + P/2)(R - P/4) \\ &= -E^{\mathsf{T}}O - P/8 + R - P/4 + P/2 - P/8 \\ &= R - E^{\mathsf{T}}O \\ &= R - E^{\mathsf{T}}ER = (I - E^{\mathsf{T}}E)R = I.\end{aligned}$$

Thus M is the required inverse.

The row-side and column-side remainders are, respectively,

$$\mathcal{N}_{L,i}(x, y) = \sum_j X_{ij}(e^{x_i + y_j} - 1 - x_i - y_j),$$

$$\mathcal{N}_{R,j}(x, y) = \sum_i X_{ij}(e^{x_i + y_j} - 1 - x_i - y_j).$$

Its row-side partial derivatives are

$$\frac{\partial \mathcal{N}_{L,i}}{\partial x_k} = \mathbf{1}\{i = k\} \sum_j X_{ij}(e^{x_i + y_j} - 1), \quad \frac{\partial \mathcal{N}_{L,i}}{\partial y_j} = X_{ij}(e^{x_i + y_j} - 1).$$

The column-side partial derivatives are

$$\frac{\partial \mathcal{N}_{R,j}}{\partial x_i} = X_{ij}(e^{x_i + y_j} - 1), \quad \frac{\partial \mathcal{N}_{R,j}}{\partial y_k} = \mathbf{1}\{j = k\} \sum_i X_{ij}(e^{x_i + y_j} - 1).$$

Thus the absolute sum of each row-side Jacobian row is at most

$$\left|\sum_j X_{ij}(e^{x_i+y_j}-1)\right| + \sum_j |X_{ij}||e^{x_i+y_j}-1| \le 2\sum_j |X_{ij}||e^{x_i+y_j}-1|.$$

When $\|(x,y)\|_\infty \le R \le 1/4$, the right-hand side is at most

$$2\sum_j |X_{ij}| \cdot 2Re^{2R} \le 8Re^{1/2} \le 16R.$$

The column side has the same bound, using the assumption that each absolute column sum is at most two. Integrating the vector-valued function along the segment joining the two points gives

$$\mathcal{N}(z) - \mathcal{N}(w) = \int_0^1 D\mathcal{N}(w + t(z-w))(z-w)\,dt,$$

Its Lipschitz constant is therefore at most $16R$. Multiplication by the gauge-fixed inverse bound $2L_0$ gives the contraction constant $32L_0R = 128L_0^2\varepsilon$.

Set $E_B = B - P_p$, where $P_p = J_p/p$. For $h = (v,w) \in u^\perp$, we have $\sum_i v_i = \sum_j w_j$, so there is a unique decomposition

$$v = c\mathbf{1} + v_0, \quad w = c\mathbf{1} + w_0, \quad \mathbf{1}^\mathsf{T} v_0 = \mathbf{1}^\mathsf{T} w_0 = 0.$$

Its squared norm is

$$\|h\|_2^2 = 2pc^2 + \|v_0\|_2^2 + \|w_0\|_2^2.$$

Since $B\mathbf{1} = B^\mathsf{T}\mathbf{1} = \mathbf{1}$, the mixed constant terms vanish. Computing term by term gives

$$\begin{aligned} h^\mathsf{T}\begin{pmatrix} I & B \\ B^\mathsf{T} & I \end{pmatrix} h &= \|v\|^2 + \|w\|^2 + 2v^\mathsf{T} B w \\ &= 4pc^2 + \|v_0\|^2 + \|w_0\|^2 + 2v_0^\mathsf{T} E_B w_0 \\ &\ge 4pc^2 + \|v_0\|^2 + \|w_0\|^2 - 2q_B\|v_0\|\|w_0\| \\ &\ge 4pc^2 + (1-q_B)(\|v_0\|^2 + \|w_0\|^2) \\ &\ge (1-q_B)\|h\|^2, \end{aligned}$$

Here $q_B = \|E_B\|_{\mathrm{op}} < 1$, and the penultimate step uses $2ab \le a^2 + b^2$. This gives the strong-convexity constant used in the displacement and capacity lemma.

4.2 Gaussian Deletion and Centering

Deleting t rows from a dense normalized matrix removes a total squared row norm of order t/n. Applying the log-determinant derivative to the corresponding Gram-matrix loss preserves this order. Switching to the Gram matrix on the other side then handles column deletion. This yields a linear deletion error even when the retained submatrix is nonprincipal.

Definition 4.2 (Gaussian determinant factor).

Whenever $\|Z\|_{\mathrm{op}} < 1$ for a real matrix Z, define

$$\mathcal{G}(Z) = \det(I - Z^\mathsf{T} Z)^{-1/2}.$$

Using ZZ^T gives the same value; adjoining zero rows or zero columns also leaves this value unchanged.

Lemma 4.3 (Gram deletion and stability).

Suppose $\|Z\|_{\mathrm{op}} \le q_* < 1$. Delete the row set I and the column set J, leaving the sets R, T. Then

$$0 \le \log \mathcal{G}(Z) - \log \mathcal{G}(Z[R,T]) \le \frac{\sum_{i\in I} \|Z_{i,\cdot}\|_2^2 + \sum_{j\in J} \|Z_{\cdot,j}\|_2^2}{2(1-q_*^2)}.$$

Double centering. At order $m \ge 1$, assume $\|W\|_{\mathrm{op}} \le q_*$ for a square retained matrix W. Set $u = \mathbf{1}_m/\sqrt{m}$. Then

$$0 \le \log \mathcal{G}(W) - \log \mathcal{G}(\Pi_m W \Pi_m) \le \frac{\|W^{\mathsf{T}}u\|_2^2 + \|Wu\|_2^2}{2(1-q_*^2)}.$$

Stability. For two real matrices of the same dimensions whose operator norms are both at most q_*, we also have

$$|\log \mathcal{G}(U) - \log \mathcal{G}(V)| \le \frac{\|U\|_F + \|V\|_F}{2(1-q_*^2)} \|U - V\|_F.$$

Proof.

Let P_R be the diagonal projection onto the retained row coordinates. Deleting rows reduces the column-side Gram matrix by the positive semidefinite matrix $Z^{\mathsf{T}}(I - P_R)Z$, whose trace is the sum of squared norms of the deleted rows. For $F(H) = -\frac{1}{2}\log\det(I-H)$ and a positive semidefinite direction D,

$$DF(H)[D] = \tfrac{1}{2}\operatorname{tr}((I-H)^{-1}D) \le \frac{\operatorname{tr} D}{2(1-q_*^2)}.$$

Integrate this bound, and then use the row-side Gram matrix to handle column deletion. Projecting away the two all-ones directions in turn gives the centering bound by the same method. For the final conclusion, integrate along the segment joining the two Gram matrices and use

$$\|U^{\mathsf{T}}U - V^{\mathsf{T}}V\|_* \le (\|U\|_F + \|V\|_F)\|U - V\|_F,$$

where $\|\cdot\|_*$ denotes the trace norm. The required operator spectral gap holds along the entire Gram-matrix segment.

Now let S be a tournament sign matrix of order $n \ge 4$, with $s = S\mathbf{1}_n$ and $\tau = \|s\|_2^2/(n-1)^2$. Set

$$Y = \frac{S - I}{n-1}, \qquad D_n(S) = \det(I - S^{\mathsf{T}}S/n^2)^{-1/2}.$$

Skew-symmetry and the paired singular values give

$$\|Y\|_{\mathrm{op}}^2 \le q_Y^2 = \frac{n(n-1)/2 + 1}{(n-1)^2} < 1.$$

Take an integer $0 \le t < n$, delete any t rows and t columns, and write $m = n - t \ge 1$ and $Z = \Pi_m Y[R,T]\Pi_m$. Every row and column of Y has squared two-norm $n/(n-1)^2$. Moreover,

$$\|Y\mathbf{1}_n\|_2^2 = \|Y^{\mathsf{T}}\mathbf{1}_n\|_2^2 = \tau + \frac{n}{(n-1)^2},$$

Here $\mathbf{1}_I, \mathbf{1}_J \in \mathbb{R}^n$ are indicator column vectors in the original vertex universe $[n]$, for the sets I, J: a coordinate is one when it belongs to the corresponding set and zero otherwise.

$$\|Y\mathbf{1}_J\|_2^2,\ \|Y^{\mathsf{T}}\mathbf{1}_I\|_2^2 \le \frac{t^2 n}{(n-1)^2}.$$

The preceding lemma therefore gives

$$0 \le \log \mathcal{G}(Y) - \log \mathcal{G}(Z) \le \frac{tn/(n-1)^2 + 2[\tau + n(1+t^2)/(n-1)^2]/m}{1 - q_Y^2}.$$

The full-matrix Gram identity is $Y^{\mathsf{T}}Y = (S^{\mathsf{T}}S + I)/(n-1)^2$. Its difference from $S^{\mathsf{T}}S/n^2$ is positive semidefinite and has trace

$$\vartheta_n = \frac{2n-1}{n(n-1)} + \frac{n}{(n-1)^2}.$$

Consequently, $0 \le \log \mathcal{G}(Y) - \log D_n(S) \le 43/(4n)$. If $m \ge n/2$, the first loss is at most $(24t + 18\tau + 8)/n$. Both are nonnegative losses relative to the same full-matrix kernel, so

$$|\log \mathcal{G}(Z) - \log D_n(S)| \le \frac{24t + 18\tau + 11}{n}.$$

To make the constants explicit, note that $1 - q_Y^2 = n(n-3)/(2(n-1)^2)$. Substitution and $t \le n/2$ directly verify the constants above. The estimate applies to independently chosen row and column deletion sets I, J; skew-symmetry is used only for the original full matrix S.

For $0 \le H \le q_*^2 I$, spectral decomposition or the determinant derivative formula gives

$$\frac{d}{dt}\left[-\frac{1}{2}\log\det(I - H - tD)\right] = \frac{1}{2}\operatorname{tr}((I - H - tD)^{-1}D).$$

If $D \ge 0$ and the entire segment remains bounded above by $q_*^2 I$, the right-hand side lies in $[0, \operatorname{tr} D/(2(1 - q_*^2))]$. For row deletion,

$$H_0 = Z^{\mathsf{T}}P_R Z, \quad D = Z^{\mathsf{T}}(I - P_R)Z \ge 0, \quad \operatorname{tr} D = \sum_{i \in I}\sum_j Z_{ij}^2.$$

For column deletion, switch to $(P_R Z)(P_R Z)^{\mathsf{T}}$. The squared norms of the deleted columns are no larger than their original squared norms, so the two losses can be added. Centering uses the projection $I - uu^{\mathsf{T}}$; the first corresponding Gram loss is $W^{\mathsf{T}}uu^{\mathsf{T}}W$, with trace $\|W^{\mathsf{T}}u\|^2$. The other step contributes at most $\|Wu\|^2$.

For arbitrary U, V,

$$U^{\mathsf{T}}U - V^{\mathsf{T}}V = U^{\mathsf{T}}(U - V) + (U - V)^{\mathsf{T}}V,$$

$$\|U^{\mathsf{T}}U - V^{\mathsf{T}}V\|_* \le \|U\|_F\|U - V\|_F + \|U - V\|_F\|V\|_F.$$

Along the Gram segment $H_t = (1-t)V^{\mathsf{T}}V + tU^{\mathsf{T}}U$, we still have $0 \le H_t \le q_*^2 I$. Using $|\operatorname{tr}(AB)| \le \|A\|_{\mathrm{op}}\|B\|_*$ and integrating then gives the Lipschitz constant in the lemma.

Since $S^{\mathsf{T}} = -S$,

$$(S - I)^{\mathsf{T}}(S - I) = (-S - I)(S - I) = -S^2 + I = S^{\mathsf{T}}S + I.$$

Consequently,

$$q_Y^2 = \frac{n(n-1)/2 + 1}{(n-1)^2}, \qquad 1 - q_Y^2 = \frac{n(n-3)}{2(n-1)^2} > 0 \quad (n \ge 4).$$

Each row has exactly n entries of modulus $1/(n-1)$, and its squared norm is $n/(n-1)^2$. Also,

$$Y\mathbf{1} = (s - \mathbf{1})/(n-1), \quad Y^{\mathsf{T}}\mathbf{1} = (-s - \mathbf{1})/(n-1),$$

$$\|s \mp \mathbf{1}\|^2 = \|s\|^2 + n \mp 2\sum_i s_i = \|s\|^2 + n.$$

For the indicator vector of the deleted columns, each row satisfies $|(Y\mathbf{1}_J)_i| \le t/(n-1)$, so the squared norm is at most $nt^2/(n-1)^2$. The all-ones direction of the retained matrix $W = Y[R,T]$ satisfies

$$\begin{aligned}\|W\mathbf{1}_m/\sqrt{m}\|^2 &\le \frac{1}{m}\|Y(\mathbf{1}_n - \mathbf{1}_J)\|^2 \\ &\le \frac{2}{m}\left[\tau + \frac{n}{(n-1)^2} + \frac{nt^2}{(n-1)^2}\right].\end{aligned}$$

The other direction is analogous. Substitution into the deletion and centering bounds gives the loss L above. Multiply it by n and substitute the spectral gap:

$$nL \le \frac{2nt}{n-3} + \frac{4(n-1)^2\tau}{m(n-3)} + \frac{4n(1+t^2)}{m(n-3)}.$$

When $m = n - t \ge n/2$,

$$\frac{n}{n-3} \le 4, \quad \frac{4(n-1)^2}{m(n-3)} \le 18, \quad \frac{4n}{m(n-3)} \le 8,$$

$$\frac{4nt^2}{m(n-3)} \le 16t.$$

The last inequality follows from $t/m \le 1$ and $n/(n-3) \le 4$. Hence $nL \le 24t + 18\tau + 8$.

The difference between the full-order Gram matrices is

$$\left(\frac{1}{(n-1)^2} - \frac{1}{n^2}\right) S^{\mathsf{T}}S + \frac{I}{(n-1)^2} \ge 0,$$

Its trace is ϑ_n. The logarithmic loss is at most

$$\frac{\vartheta_n}{2(1-q_Y^2)} = \frac{3n^2 - 3n + 1}{n^2(n-3)} \le \frac{43}{4n}.$$

The last inequality is equivalent to $31n^2 - 117n - 4 \ge 0$; at $n = 4$, its left-hand side is $24 > 0$, and it increases thereafter. If $a, b \ge 0$, then $|a - b| \le \max(a, b)$. Both losses are relative to $\mathcal{G}(Y)$, so

$$|\log \mathcal{G}(Z) - \log D_n(S)| \le \max\left(\frac{24t + 18\tau + 8}{n}, \frac{43}{4n}\right) \le \frac{24t + 18\tau + 11}{n}.$$

4.3 Preprocessing Tournament Submatrices

The row and column errors of $C = 2A/(n-1)$ have opposite signs. Multiplying row i by $(1+a_i)^{-1}$ corrects the original row sum; multiplying column i by $(1-a_i)^{-1}$ corrects the original column sum. Applying both corrections simultaneously leaves marginal residuals, which we estimate below. The main effect is to cancel the first-order change in total mass and produce the score penalty Γ in the restored product.

Fix $0 \le a_0 < 1$ and finite positive parameters A_0, B_0. In this subsection, assume

$$a_i = \frac{s_i}{n-1}, \qquad \max_i |a_i| \le a_0, \qquad \tau = \sum_i a_i^2 \le A_0 \log n, \qquad t \le B_0 \log n.$$

Define

$$C = \frac{J - I + S}{n-1} = \frac{2A}{n-1}, \qquad \ell_i = \frac{1}{1+a_i}, \qquad r_i = \frac{1}{1-a_i}, \qquad g_i = \frac{1}{1-a_i^2}.$$

Set $C' = \operatorname{diag}(\ell) C \operatorname{diag}(r)$, $F = C' - C$, and $\nu = \mathfrak{m}(C') - n$. To compute the total mass, write $v_i = a_i/(1-a_i^2)$, $w_i = a_i^2/(1-a_i^2)$, and also $V = \sum_i v_i$, $W = \sum_i w_i$. Expanding $\ell = \mathbf{1} - v + w$ and $r = \mathbf{1} + v + w$ gives the exact identity

$$\nu = \frac{W^2 - V^2 + W + 2w^{\mathsf{T}} S v}{n-1}.$$

More precisely, $\mathbf{1}^{\mathsf{T}} S v = -(n-1)W$ cancels the first-order mass change. The fixed score bound gives $W \le K\tau$, $|V| \le K\tau$, and $\|v\|_2 \le K\sqrt{\tau}$. Estimating Sv row by row yields

$$|\nu| \le K\mathcal{M}_\tau, \qquad \mathcal{M}_\tau = \frac{\tau + \tau^2}{n} + \frac{\tau^{3/2}}{\sqrt{n}} = o(1).$$

The marginal identities are

$$C'\mathbf{1}_n - \mathbf{1}_n = \operatorname{diag}(\ell) C (r - \mathbf{1}_n),$$

$$(C')^{\mathsf{T}}\mathbf{1}_n - \mathbf{1}_n = \operatorname{diag}(r) C^{\mathsf{T}} (\ell - \mathbf{1}_n).$$

Every row and column of C has two-norm $O(n^{-1/2})$, and $\|C\|_{\mathrm{op}} = O(1)$. Thus the marginal error is $O(\sqrt{\tau/n})$ in the infinity norm and $O(\sqrt{\tau})$ in the two-norm. A direct entrywise estimate also gives

$$\|F\|_F \le K\sqrt{\tau/n}.$$

Normalize $\widehat{C} = nC'/\mathfrak{m}(C')$. Denote its row and column errors by e_i, f_j, so $\sum e_i = \sum f_j = 0$. For arbitrary deletion sets of common size t, denoted by I, J, write $R = [n] \setminus I$, $T = [n] \setminus J$, $m = n - t$, and set

$$X = \frac{n}{m}\widehat{C}[R,T], \qquad \kappa = \mathfrak{m}(X) - m, \qquad \widetilde{X} = \frac{m}{\mathfrak{m}(X)} X.$$

The exact mass identity for deleting rows and columns is

$$\kappa = \frac{n}{m}\left[-\sum_{i \in I} e_i - \sum_{j \in J} f_j + \sum_{i \in I, j \in J} \widehat{C}_{ij} - \frac{t^2}{n} \right]. \tag{4.1}$$

The marginal error in each retained row before the final normalization is

$$(X\mathbf{1}_m)_i - 1 = \frac{n}{m}\left[e_i + \frac{t}{n} - \sum_{j \in J} \widehat{C}_{ij} \right],$$

The column side has the corresponding identity of the same form. Consequently,

$$|\kappa| \le K\left[t\sqrt{\tau/n} + t^2/n \right],$$

$$\varepsilon(\widetilde{X}) \le K(\sqrt{\tau/n} + t/n), \qquad \|g(\widetilde{X})\|_2 \le K(\sqrt{\tau} + t/\sqrt{n}).$$

Here $g(\widetilde{X})$ is the concatenated vector of row and column marginal errors. The entries of $\widetilde{X}$ lie in $[0, K/m]$. To apply Lemma 4.1, we now verify the centered-kernel conditions for $\widetilde{X}$. Write

$$\eta = \frac{n^2}{\mathfrak{m}(C')\mathfrak{m}(X)} = 1 + O((t+1)/n),$$

Then

$$\Pi_m \widetilde{X} \Pi_m = \eta \left[Z + \Pi_m F[R,T] \Pi_m \right].$$

Since $\mathfrak{m}(C') = n + o(1)$ and $\mathfrak{m}(X) = m + o(1)$, we have $\eta = (n/m)(1 + o(1))$. Compression does not increase the operator norm of Y, and the Frobenius estimate above bounds the second term. The centered kernel therefore has operator norm at most $1/\sqrt{2} + o(1)$ and entries $O(1/m)$. The marginal error of the original matrix $\widetilde{X}$ tends to zero in the infinity norm. Since $\widetilde{X}$ has mass m, the matrix E in Lemma 4.1 equals $\Pi_m \widetilde{X} \Pi_m$. Fix $q = 3/4$ and entry bounds C, C_0. The resulting L_0, ε_0 depend only on a_0, A_0, B_0. Lemma 4.1 thus applies uniformly for sufficiently large n, giving a doubly stochastic matrix B_X and the spectral-norm bound $7/8$.

The two-norm displacement lemma gives

$$\|B_X - \widetilde{X}\|_F \le K(\sqrt{\tau/n} + t/n).$$

Combined with the stronger centered-kernel bound, this shows that the centered operator norm of B_X is still at most $1/\sqrt{2} + o(1)$; for example, it is eventually at most $4/5$. The entries continue to satisfy the bound K/m.

Apply the Gaussian deletion estimate to Z and use the logarithmic Lipschitz bound for the perturbations due to prescaling, normalization, and diagonal scaling. Since $B_X - P_m = \Pi_m B_X \Pi_m$, this gives

$$|\log \mathcal{G}(B_X - P_m) - \log D_n(S)| \le K \left[\sqrt{\tau/n} + (t+1)/n \right].$$

The term τ/n from centering has been absorbed into $\sqrt{\tau/n}$. The deletion error is linear in t/n, as needed for summing against the subsequent subset weights.

We can now collect the errors under the fixed dimension conventions. All constants and the final applicability threshold depend only on the fixed parameters a_0, A_0, B_0. The full-order mass error contributes $O(|\nu|) \le K\mathcal{M}_\tau$ to the restoration logarithm. The mass change from deletion is at most $K[t\sqrt{\tau/n} + t^2/n]$, and the Gaussian logarithmic error is $K[\sqrt{\tau/n} + (t+1)/n]$. Applying Theorem 3.1 at order $m \ge n/2$ adds $O(1/m) = O(1/n)$. The final scalar restoration contributes $e^{-1} \exp(O((t+1)/n))$. With total mass $m + \kappa$, the matrix X uses the capacity estimate only in the form $e^{-\theta_X} \le e^{\kappa}$; one cannot assert that θ_X is nonnegative for this general mass. Section 4.4 restores these factors exactly and obtains the error exponent in Lemma 4.4.

Verify entry by entry that

$$1 - v_i + w_i = 1 - \frac{a_i}{1 - a_i^2} + \frac{a_i^2}{1 - a_i^2} = \frac{1}{1 + a_i} = \ell_i,$$

$$1 + v_i + w_i = \frac{1}{1 - a_i} = r_i, \qquad a_i v_i = w_i, \quad \ell_i r_i = g_i = 1 + w_i.$$

Set $L = \mathbf{1} - v + w$ and $R = \mathbf{1} + v + w$ (here L, R denote vectors only within this calculation). Then

$$L^{\mathsf{T}} J R = (n - V + W)(n + V + W) = (n + W)^2 - V^2,$$

$$L^{\mathsf{T}} I R = \sum_i g_i = n + W.$$

Expand the skew-symmetric part term by term:

$$\begin{aligned} L^{\mathsf{T}}SR &= \mathbf{1}^{\mathsf{T}}Sv + \mathbf{1}^{\mathsf{T}}Sw - v^{\mathsf{T}}S\mathbf{1} - v^{\mathsf{T}}Sv - v^{\mathsf{T}}Sw \\ &\quad + w^{\mathsf{T}}S\mathbf{1} + w^{\mathsf{T}}Sv + w^{\mathsf{T}}Sw \\ &= -2(n-1)W + 2w^{\mathsf{T}}Sv. \end{aligned}$$

Here $\mathbf{1}^{\mathsf{T}}Sv = -(n-1)a^{\mathsf{T}}v = -(n-1)W$; $\mathbf{1}^{\mathsf{T}}Sw + w^{\mathsf{T}}S\mathbf{1} = 0$; $v^{\mathsf{T}}Sv = w^{\mathsf{T}}Sw = 0$; and $-v^{\mathsf{T}}Sw = w^{\mathsf{T}}Sv$. Thus

$$\begin{aligned} \mathfrak{m}(C') &= \frac{(n+W)^2 - V^2 - (n+W) - 2(n-1)W + 2w^{\mathsf{T}}Sv}{n-1} \\ &= n + \frac{W^2 - V^2 + W + 2w^{\mathsf{T}}Sv}{n-1}. \end{aligned}$$

Since $\sum_i a_i = 0$,

$$V = \sum_i \left(a_i + \frac{a_i^3}{1-a_i^2} \right) = \sum_i \frac{a_i^3}{1-a_i^2}, \quad |V| \le \frac{a_0}{1-a_0^2}\tau, \quad W \le \frac{\tau}{1-a_0^2},$$

$$\|v\|_2 \le \frac{\sqrt{\tau}}{1-a_0^2}, \quad \|Sv\|_\infty \le \sqrt{n}\|v\|_2, \quad |w^{\mathsf{T}}Sv| \le \|w\|_1\|Sv\|_\infty \le K\sqrt{n}\tau^{3/2}.$$

Substitution gives $|\nu| \le K[(\tau+\tau^2)/n + \tau^{3/2}/\sqrt{n}]$.
Since $\mathrm{diag}(\ell)C\mathbf{1} = \mathbf{1}$,

$$C'\mathbf{1} - \mathbf{1} = \mathrm{diag}(\ell)C(r - \mathbf{1}).$$

Moreover, $|r_i - 1| \le K|a_i|$, $|\ell_i - 1| \le K|a_i|$, and $\|C_{i,\cdot}\|_2 \le K/\sqrt{n}$. Hence

$$\|C'\mathbf{1} - \mathbf{1}\|_\infty \le K\sqrt{\tau/n}, \quad \|C'\mathbf{1} - \mathbf{1}\|_2 \le K\|C\|_{\mathrm{op}}\sqrt{\tau} \le K'\sqrt{\tau}.$$

The column side is analogous. Entry by entry,

$$F_{ij} = C_{ij}(\ell_i r_j - 1), \quad |\ell_i r_j - 1| \le K(|a_i| + |a_j|),$$

$$\|F\|_F^2 \le \frac{K}{n^2}\sum_{ij}(a_i^2 + a_j^2) = \frac{2K\tau}{n}.$$

At total mass n, the normalized matrix $\widehat{C}$ has retained-block mass given directly by inclusion–exclusion:

$$\begin{aligned} \sum_{i\in R, j\in T} \widehat{C}_{ij} &= n - \sum_{i\in I}(1+e_i) - \sum_{j\in J}(1+f_j) + \sum_{i\in I, j\in J}\widehat{C}_{ij} \\ &= n - 2t - \sum_I e_i - \sum_J f_j + \sum_{I,J}\widehat{C}_{ij}. \end{aligned}$$

Since $m^2/n = n - 2t + t^2/n$, multiplying by n/m and then subtracting m gives equation (4.1). By $\widehat{C}_{ij} \le K/n$,

$$|\kappa| \le \frac{n}{m}\left(2t\max(\|e\|_\infty, \|f\|_\infty) + (K+1)t^2/n\right) \le K'[t\sqrt{\tau/n} + t^2/n].$$

The error in a retained row likewise satisfies, term by term,

$$\frac{n}{m}\left(1+e_i-\sum_{j\in J}\widehat{C}_{ij}\right)-1=\frac{n}{m}\left(e_i+t/n-\sum_{j\in J}\widehat{C}_{ij}\right).$$

The normalization multiplier $m/(m+\kappa)$ differs from one by $O(|\kappa|/m)$. Thus the marginal infinity norm and two-norm are $K(\sqrt{\tau/n}+t/n)$ and $K(\sqrt{\tau}+t/\sqrt{n})$, respectively.

Successive substitution of the definitions gives

$$\widetilde{X}=\frac{m}{m+\kappa}\frac{n}{m}\frac{n}{n+\nu}C'[R,T]=\frac{n^2}{(n+\nu)(m+\kappa)}C'[R,T]=\eta C'[R,T].$$

Also, $C=J/(n-1)+Y$. Multiplying on the left and right by Π_m removes J and yields the centered-kernel identity. Since $m=n-t$,

$$\eta=\frac{n}{m}(1+\nu/n)^{-1}(1+\kappa/m)^{-1}=1+O((t+1)/n).$$

Consequently,

$$\|\Pi_m\widetilde{X}\Pi_m-Z\|_F\le|\eta-1|\|Z\|_F+|\eta|\|F[R,T]\|_F\le K[(t+1)/n+\sqrt{\tau/n}].$$

Adding the displacement from diagonal scaling gives

$$\|(B_X-P_m)-Z\|_F\le K[(t+1)/n+\sqrt{\tau/n}].$$

All the matrices above have Frobenius norms $O(1)$, and their operator norms are eventually at most the fixed value $4/5<1$. Adding the Gram Lipschitz bound and the deletion bound gives

$$\begin{aligned}|\log\mathcal{G}(B_X-P_m)-\log D_n(S)|&\le K\left[\frac{t+\tau+1}{n}+\sqrt{\tau/n}\right]\\&\le K'\left[\frac{t+1}{n}+\sqrt{\tau/n}\right],\end{aligned}$$

For sufficiently large n, we have $\tau/n\le 1$, and hence $\tau/n\le\sqrt{\tau/n}$.

First write

$$e'=C'\mathbf{1}-\mathbf{1},\quad f'=(C')^{\mathsf{T}}\mathbf{1}-\mathbf{1},\quad \widehat{C}=\frac{n}{n+\nu}C'.$$

Coordinatewise,

$$\begin{aligned}e_i&=(\widehat{C}\mathbf{1})_i-1=\frac{n}{n+\nu}(1+e'_i)-1=\frac{ne'_i-\nu}{n+\nu},\\f_j&=(\widehat{C}^{\mathsf{T}}\mathbf{1})_j-1=\frac{n}{n+\nu}(1+f'_j)-1=\frac{nf'_j-\nu}{n+\nu}.\end{aligned}$$

Since $\sum_i e'_i=\sum_j f'_j=\nu$, these formulas also verify

$$\sum_i e_i=\frac{n\nu-n\nu}{n+\nu}=0,\quad \sum_j f_j=0.$$

When $|\nu|\le n/2$,

$$\|e\|_\infty\le 2\|e'\|_\infty+2|\nu|/n,\quad \|e\|_2\le 2\|e'\|_2+2|\nu|/\sqrt{n},$$

$$\|f\|_\infty \le 2\|f'\|_\infty + 2|\nu|/n, \quad \|f\|_2 \le 2\|f'\|_2 + 2|\nu|/\sqrt{n}.$$

For $0 < \tau \le A_0 \log n$, divide $|\nu|/n$ by $\sqrt{\tau/n}$. The mass bound already obtained gives at most a constant times

$$\frac{\sqrt{\tau} + \tau^{3/2}}{n^{3/2}} + \frac{\tau}{n} = o(1).$$

Dividing $|\nu|/\sqrt{n}$ by $\sqrt{\tau}$ gives the same expression. When $\tau = 0$, we have $a = 0$, so $\nu = 0$, and no division is needed. This yields the infinity-norm bound $K\sqrt{\tau/n}$ and the two-norm bound $K\sqrt{\tau}$ on both sides, respectively.

Suppose

$$h = X\mathbf{1}_m - \mathbf{1}_m, \quad k = X^\mathsf{T}\mathbf{1}_m - \mathbf{1}_m, \quad \widetilde{X} = \frac{m}{m+\kappa}X.$$

The normalized errors on the two sides are then exactly

$$\widetilde{\alpha}_i = \frac{mh_i - \kappa}{m+\kappa}, \qquad \widetilde{\beta}_j = \frac{mk_j - \kappa}{m+\kappa}.$$

Using $\sum_i h_i = \sum_j k_j = \kappa$ again verifies $\sum_i \widetilde{\alpha}_i = \sum_j \widetilde{\beta}_j = 0$. When $|\kappa| \le m/2$,

$$\|\widetilde{\alpha}\|_\infty \le 2\|h\|_\infty + 2|\kappa|/m, \quad \|\widetilde{\alpha}\|_2 \le 2\|h\|_2 + 2|\kappa|/\sqrt{m},$$

$$\|\widetilde{\beta}\|_\infty \le 2\|k\|_\infty + 2|\kappa|/m, \quad \|\widetilde{\beta}\|_2 \le 2\|k\|_2 + 2|\kappa|/\sqrt{m}.$$

Here k denotes the column-error vector. The retained row and column errors are, respectively,

$$h_i = \frac{n}{m}\left(e_i + t/n - \sum_{j\in J}\widehat{C}_{ij}\right), \qquad k_j = \frac{n}{m}\left(f_j + t/n - \sum_{i\in I}\widehat{C}_{ij}\right).$$

Since $\widehat{C}_{ij} \le K/n$,

$$\|h\|_\infty, \|k\|_\infty \le K'[\sqrt{\tau/n} + t/n],$$

$$\|h\|_2, \|k\|_2 \le K'[\sqrt{\tau} + t/\sqrt{n}].$$

The mass term $|\kappa|/\sqrt{m} \le K[t\sqrt{\tau}/n + t^2/n^{3/2}]$ is absorbed into the two-norm bound above because $t/n \to 0$; the infinity-norm mass term is likewise absorbed into its corresponding bound. This verifies all the marginal-error hypotheses of the local scaling lemma, one marginal at a time.

The normalized mass is m, so the centered matrix is exactly

$$E = \widetilde{X} - P_m - \frac{\widetilde{\alpha}\mathbf{1}^\mathsf{T} + \mathbf{1}\widetilde{\beta}^\mathsf{T}}{m} = \Pi_m \widetilde{X} \Pi_m.$$

If $0 \le \widetilde{X}_{ij} \le C_1/m$ and the infinity norms of both marginal error vectors are at most ϵ_n, then

$$|E_{ij}| \le \frac{C_1 + 1 + 2\epsilon_n}{m} \le \frac{C_1 + 3}{m} \quad (\epsilon_n \le 1).$$

Nonnegativity makes each absolute row sum or column sum equal to the corresponding row sum or column sum, which is at most $1 + \epsilon_n \le 2$. Also,

$$\|E\|_{\mathrm{op}} \le |\eta|\|Z\|_{\mathrm{op}} + |\eta|\|F[R,T]\|_F \le \frac{1}{\sqrt{2}} + o(1).$$

First fix $q = 3/4$ and the entry bound $C_0 = C_1 + 3$; these determine L_0, ε_0. Then choose a uniform order threshold such that

$$\|E\|_{\mathrm{op}} \le 3/4, \quad \epsilon_n \le \varepsilon_0, \quad m \ge n/2, \quad |\nu| \le n/2, \quad |\kappa| \le m/2.$$

All these conditions follow from bounds involving only fixed parameters and n, so the threshold is fixed before choosing the particular graph and deletion sets. The local theorem initially gives $\|B_X - P_m\|_{\mathrm{op}} \le 7/8$; more precisely,

$$\|B_X - P_m\|_{\mathrm{op}} \le \|\Pi_m \widetilde{X} \Pi_m\|_{\mathrm{op}} + \|B_X - \widetilde{X}\|_F \le 1/\sqrt{2} + o(1) \le 4/5.$$

This estimate is again uniform over tournaments and deletion sets.

Take a sufficiently large odd integer n and the corresponding regular carousel tournament, with adjacency matrix A_n and sign matrix S_n^{cir}. All scores are then zero, all the preceding prescaling factors are one, and

$$C = \frac{2A_n}{n-1} = \frac{J_n - I_n + S_n^{\mathrm{cir}}}{n-1}, \qquad C\mathbf{1}_n = C^{\mathsf{T}}\mathbf{1}_n = \mathbf{1}_n.$$

Consequently, $C' = X = \widetilde{X} = C$ (here no rows or columns are deleted, so $t = 0$). The potential vectors $x = y = 0$ already satisfy the balancing equation; the unique small solution in local scaling gives $B_X = C$. On the constant-vector direction, $C - P_n$ vanishes; on its orthogonal complement,

$$(C - P_n)|_{\mathbf{1}_n^\perp} = \left.\frac{S_n^{\mathrm{cir}} - I_n}{n-1}\right|_{\mathbf{1}_n^\perp}, \qquad (S_n^{\mathrm{cir}} - I_n)^{\mathsf{T}}(S_n^{\mathrm{cir}} - I_n) = I_n - (S_n^{\mathrm{cir}})^2.$$

The carousel transformation in Section 8.2 preserves all frequencies of the sign matrix, and the largest positive frequency is $\cot(\pi/(2n))$. Therefore,

$$\begin{aligned}\|B_X - P_n\|_{\mathrm{op}} &= \frac{\sqrt{1 + \cot^2(\pi/(2n))}}{n-1} \\ &= \frac{1}{(n-1)\sin(\pi/(2n))} \\ &= \frac{2n}{\pi(n-1)} \frac{\pi/(2n)}{\sin(\pi/(2n))} \longrightarrow \frac{2}{\pi} < \frac{1}{\sqrt{2}}.\end{aligned}$$

Here we use $\sin u/u \to 1$. Thus the centered norm need not tend to zero; the fixed bound $4/5$ suffices for Theorem 3.1.

4.4 Proof of the Nonprincipal Permanent Bound

We apply the scaling and deletion estimates to tournament submatrices. The deleted row and column sets may differ, since a cycle cover can enter and leave the exceptional set through different core vertices.

Lemma 4.4 (Nonprincipal permanent bound).

Fix $0 \le b < 1$ and positive constants A_0, B_0. For a tournament core of order $N \ge 2$, set $a = S\mathbf{1}/(N-1)$ and $\tau = \sum_i a_i^2$. Let $I, J \subseteq [N]$ be the deleted row and column sets, respectively. Here I, J are sets, distinguished from the identity and all-ones matrices by the set operations below. Assume

$$\max_i |a_i| \le b, \qquad \tau \le A_0 \log N, \qquad |I| = |J| = t \le B_0 \log N.$$

Define $\ell_i = (1 + a_i)^{-1}$, $r_i = (1 - a_i)^{-1}$, and

$$\Gamma = \prod_i (1 - a_i^2), \qquad \mathcal{M}_\tau = \frac{\tau + \tau^2}{N} + \frac{\tau^{3/2}}{\sqrt{N}}.$$

Let $R = [N] \setminus I$ and $T = [N] \setminus J$ be the retained row and column sets, respectively; here T is a column index set. Then

$$\operatorname{per} A[R,T] \le e^{-1} D_N(S) \Gamma \left(\prod_{i \in I} \ell_i \right) \left(\prod_{j \in J} r_j \right) \frac{(N-t)!}{2^{N-t}} \exp(\varepsilon_{\tau,t}),$$

$$\varepsilon_{\tau,t} \le K \left[\mathcal{M}_\tau + \sqrt{\tau/N} + \frac{t+1}{N} + t\sqrt{\tau/N} + \frac{t^2}{N} \right].$$

For every sufficiently large N, this estimate holds simultaneously for every admissible tournament and every pair I, J. The constant K and the threshold depend only on b, A_0, B_0.

The factor Γ records the cost of degree imbalance. For a principal deleted set $I = J = U$, the restored vertex weight is $\ell_i r_i = (1 - a_i^2)^{-1}$; when the deleted sets differ, the row and column factors are retained separately. The error contains a term of order $\sqrt{\tau/N}$. In the final case, it is therefore essential to retain $\Gamma \le e^{-\tau}$.

We now prove Lemma 4.4, using n for the full core order N and taking $a_0 = b$. The preceding construction has already produced the scaled matrix B_X. It remains to restore all diagonal factors and scalar factors and then compare the resulting Gaussian determinant with $D_n(S)$.

Proof.

The construction above gives the entry bound and spectral gap required by Theorem 3.1. Hence

$$\operatorname{per} B_X = \frac{m!}{m^m} \mathcal{G}(B_X - P_m)(1 + O(1/m)).$$

The row and column scaling factors are restored exactly as

$$\prod_{i \in R} (1 + a_i) \prod_{j \in T} (1 - a_j) = \Gamma \prod_{i \in I} \ell_i \prod_{j \in J} r_j.$$

Thus deleting rows restores the ℓ factors, and deleting columns restores the r factors. The complete scalar identity is

$$\operatorname{per} A[R,T] = \Gamma \prod_{i \in I} \ell_i \prod_{j \in J} r_j \left[\frac{m(n-1)\mathfrak{m}(C')}{2n^2} \right]^m e^{-\theta_X} \operatorname{per} B_X. \tag{4.2}$$

After substituting $m!/m^m$ into equation (4.2), the remaining scalar factor is

$$\frac{m!}{2^m} (1 - 1/n)^m \left[\frac{\mathfrak{m}(C')}{n} \right]^m.$$

The difference between its logarithm and the logarithm of $e^{-1} m!/2^m$ is $O((t+1)/n + |\nu|)$. The capacity bound gives $e^{-\theta_X} \le e^{\kappa}$. Substituting the mass error, Gaussian error, and baseline permanent approximation error gives the required exponent bound. The following calculation gives the restoration factors in detail, uniformly over the tournament and deletion sets.

The factors associated with the deleted rows and columns are, respectively,

$$\begin{aligned}\operatorname{per} C[R,T] &= \left(\prod_{i\in R}\ell_i^{-1}\right)\left(\prod_{j\in T}r_j^{-1}\right)\operatorname{per} C'[R,T]\\ &= \left(\prod_{i=1}^{n}(1+a_i)\prod_{j=1}^{n}(1-a_j)\right)\left(\prod_{i\in I}(1+a_i)^{-1}\right)\left(\prod_{j\in J}(1-a_j)^{-1}\right)\operatorname{per} C'[R,T]\\ &= \Gamma\prod_{i\in I}\ell_i\prod_{j\in J}r_j\operatorname{per} C'[R,T].\end{aligned}$$

From $X = n^2C'[R,T]/[m(n+\nu)]$, we obtain

$$\operatorname{per} C'[R,T] = [m(n+\nu)/n^2]^m \operatorname{per} X.$$

Using $A = (n-1)C/2$ and $\operatorname{per} X = e^{-\theta_X}\operatorname{per} B_X$ then gives equation (4.2). After substituting the permanent approximation, the scalar factor is

$$\left[\frac{m(n-1)(n+\nu)}{2n^2}\right]^m \frac{m!}{m^m} = \frac{m!}{2^m}(1-1/n)^m(1+\nu/n)^m.$$

Set $u = 1/n$. Since

$$|\log(1-u)+u| \le \frac{u^2}{2(1-u)},$$

$$|m\log(1-1/n)+1| \le \left|1-\frac{m}{n}\right| + \frac{m}{2n^2(1-1/n)} \le \frac{t}{n} + \frac{1}{2(n-1)}.$$

When $|\nu|/n \le 1/2$, we have $|m\log(1+\nu/n)| \le 2|\nu|$. The capacity contribution satisfies $-\theta_X \le \kappa$. The Gaussian factor satisfies $\mathcal{G}(B_X - P_m) \le D_n(S)e^{K[\sqrt{\tau/n}+(t+1)/n]}$. Finally, $1 + O(1/m) \le e^{K/n}$ after increasing the uniform threshold. Thus the error exponent is bounded term by term as follows:

$$\begin{aligned}\varepsilon_{\tau,t} \le K\Bigg[&\underbrace{\mathcal{M}_\tau}_{\text{full-order mass}} + \underbrace{t\sqrt{\tau/n}+t^2/n}_{\text{deletion mass}}\\ &+ \underbrace{\sqrt{\tau/n}+(t+1)/n}_{\text{Gaussian and scalar}} + \underbrace{1/n}_{\text{permanent error}}\Bigg].\end{aligned}$$

Combining terms of the same order gives Lemma 4.4. □

4.5 Two-Sided Estimates for Small Scores

When every score is $o(\sqrt{n})$, the original marginals are already sufficiently close to one for local scaling to apply directly. The capacity bound then controls the restoration factor from both sides and gives the approximation needed in Theorem 8.1.

Lemma 4.5 (Permanent approximation for small scores).

Suppose $d = \|S\mathbf{1}_n\|_\infty = o(\sqrt{n})$. Fix $0 < B_0 < \infty$, delete any $t \le B_0\log n$ rows and t columns, and write $m = n - t$. Uniformly over these choices,

$$\operatorname{per} A[R,T] = e^{-1}D_n(S)\frac{m!}{2^m}\left[1 + O_{B_0}\left(\frac{(d+t+1)^2}{n}\right)\right]. \tag{4.3}$$

More precisely, for fixed $B_0 > 0$ there are constants $C_{\text{err}}, \delta > 0$ and an integer $N_0 \ge 4$ such that the following holds uniformly for $n \ge N_0$. Let S be a tournament sign matrix and let $I, J \subseteq [n]$ satisfy

$|I| = |J| = t \le B_0 \log n$. Put $d = \|S\mathbf{1}_n\|_\infty$ and $\eta = (d+t+1)^2/n$. If $\eta \le \delta$, the absolute relative error in equation (4.3) is at most $C_{\mathrm{err}}\eta$. Here $R = [n] \setminus I$, $T = [n] \setminus J$, and $m = n - t$; the two deleted sets may be chosen independently. In the asymptotic range of the lemma, $\eta \to 0$.

Proof.

Increase N_0 so that $2t \le n$ throughout the logarithmic deletion window, giving $m \ge n/2 > 0$. Paired prescaling is not needed here. Set $C = 2A/(n-1)$ and $X = (n/m)C[R,T]$. The row and column errors of the full matrix C are $s_i/(n-1)$ and $-s_i/(n-1)$, respectively. Thus the exact mass identity equation (4.1) and the marginal formulas in Section 4.3 give

$$|\kappa| = O(t(d+t)/n), \qquad \varepsilon(X) = O((d+t)/n), \qquad \|g(X)\|_2 = O((d+t)/\sqrt{n}).$$

The mass error satisfies $|\kappa| \le K\eta$. Decreasing δ if necessary uniformly ensures $\mathfrak{m}(X) = m + \kappa > 0$. Normalize $\widetilde{X} = mX/\mathfrak{m}(X)$. Its density and centered-kernel spectral-gap parameters are fixed, so the local scaling lemma applies. The two-norm displacement and capacity estimates give

$$\|B_X - \widetilde{X}\|_F = O((d+t)/n), \qquad 0 \le \theta_{\widetilde{X}} = O((d+t)^2/n).$$

The correction for general mass then gives $|\theta_X| = O((d+t)^2/n)$. We also have $\tau \le nd^2/(n-1)^2$. The finite-deletion estimate, normalization factors, and diagonal-scaling perturbation together give

$$|\log \mathcal{G}(B_X - P_m) - \log D_n(S)| = O((d+t+1)^2/n).$$

The exact restoration now contains no score product and is

$$\operatorname{per} A[R,T] = \left[\frac{m(n-1)}{2n}\right]^m e^{-\theta_X} \operatorname{per} B_X.$$

Apply the uniform permanent theorem and use $(1-1/n)^m = e^{-1}\exp(O((t+1)/n))$. Every logarithmic error is controlled by a fixed multiple of $\eta = (d+t+1)^2/n$. Choose δ sufficiently small and exponentiate these bounds. Throughout the preceding finite range, this uniformly bounds the absolute value of the relative error by $C_{\mathrm{err}}\eta$. In the asymptotic range $d = o(\sqrt{n})$, $t \le B_0 \log n$, we have $\eta \to 0$. This proves the uniform approximation for submatrices with few deletions used in Theorem 8.1.

Since $|s_i| \le d$, the absolute marginal error of the full matrix is at most $d/(n-1)$. The deletion-mass formula gives

$$|\kappa| \le \frac{n}{m}\left[\frac{2td}{n-1} + \frac{2t^2}{n-1} + \frac{t^2}{n}\right] \le K\frac{t(d+t)}{n}.$$

The error in a retained row is at most $K(d+t)/n$. There are $2m$ row and column errors in total, so

$$\|g(X)\|_2 \le \sqrt{2m}\,K(d+t)/n \le K'(d+t)/\sqrt{n}.$$

Normalization preserves this order. Capacity and displacement give

$$0 \le \theta_{\widetilde{X}} \le K(d+t)^2/n, \quad \|B_X - \widetilde{X}\|_F \le \frac{K}{\sqrt{m}}\frac{d+t}{\sqrt{n}} \le K'(d+t)/n.$$

Since $|\kappa|/m \le 1/2$,

$$\left|m \log \frac{m}{m+\kappa}\right| = m|\log(1+\kappa/m)| \le 2|\kappa|,$$

It follows that $|\theta_X| \le K[(d+t)^2 + t(d+t)]/n \le K'(d+t)^2/n$. If $d = t = 0$, all these quantities are zero.

Gaussian deletion gives $K(t+\tau+1)/n$, while

$$\tau = \frac{\sum_i s_i^2}{(n-1)^2} \le \frac{nd^2}{(n-1)^2} \le \frac{4d^2}{n}.$$

The centered-kernel displacement is $K(d+t+1)/n$, so the total Gaussian logarithmic error is at most

$$K\left[\frac{t+1}{n} + \frac{d^2}{n^2} + \frac{d+t+1}{n}\right] \le K' \frac{(d+t+1)^2}{n} = K'\eta.$$

After complete restoration, we may write

$$\frac{\operatorname{per} A[R,T]}{e^{-1}D_n(S)m!/2^m} = e^u(1+v), \qquad |u| \le K\eta, \quad |v| \le K/n \le K\eta.$$

After choosing a fixed $\delta \le 1/(2K)$,

$$|e^u(1+v)-1| \le |e^u-1| + e^{|u|}|v| \le e^{K\delta}(|u|+|v|) \le 2Ke^{K\delta}\eta.$$

Here K, δ, N_0 depend only on B_0 and the fixed analytic parameters.

There is no paired prescaling here: $C = J/(n-1) + Y$, $X = (n/m)C[R,T]$, and $\widetilde{X} = mX/(m+\kappa)$. Therefore,

$$\widetilde{X} = \frac{n}{m+\kappa}C[R,T], \qquad \Pi_m \widetilde{X} \Pi_m = \frac{n}{m+\kappa}Z.$$

The scalar displacement is

$$\left|\frac{n}{m+\kappa} - 1\right| = \frac{|t-\kappa|}{|m+\kappa|} \le \frac{4(t+|\kappa|)}{n} \quad (m \ge n/2,\ |\kappa| \le m/2).$$

Hence

$$\begin{aligned}\|(B_X - P_m) - Z\|_F &\le \|\Pi_m(B_X - \widetilde{X})\Pi_m\|_F + \left|\frac{n}{m+\kappa} - 1\right| \|Z\|_F \\ &\le K\frac{d+t}{n} + K\frac{t+|\kappa|}{n} \\ &\le K'\left[\frac{d+t}{n} + \frac{t(d+t)}{n^2}\right].\end{aligned}$$

Since $t/n \le 1/2$, the last term is absorbed into $K(d+t)/n$. If $d = t = 0$, this displacement is zero. Including the $O(n^{-1})$ difference between Y and S/n gives the uniform error bound $O((d+t+1)^2/n)$.

For $d, t \ge 0$, each of the following terms is at most a fixed multiple of $\eta = (d+t+1)^2/n$:

$$\frac{t+1}{n} \le \eta, \quad \frac{d+t}{n} \le \eta, \quad \frac{(d+t)^2}{n} \le \eta, \quad \frac{t(d+t)}{n} \le \eta, \quad \frac{d^2}{n^2} \le \eta.$$

Thus each logarithmic error is $O(\eta)$. The inequality $|e^u - 1| \le e^{|u|}|u|$ gives the same order for the relative error. □

5. Path convolution and a determinant representation

5.1 Positive path convolution

Lemma 5.1 (Path convolution).

For every tournament,

$$H(T) = \sum_{U \subseteq V(T)} \det(I + A[U]) \operatorname{per} A[U^c]. \tag{5.1}$$

Proof.

The permanent of the adjacency matrix counts directed cycle covers. The convolution equation (5.1) converts cycle-cover counts on complementary vertex sets into the path count. It is the tournament specialization of Proposition 2 of Irving and Omar [1], where the complementary adjacency matrix is $\overline{A} = J - A = I + A^{\mathsf{T}}$ and includes the diagonal entries.

The identity can also be checked directly using generating functions. Set $X = \operatorname{diag}(z_1, \dots, z_n)$, the diagonal matrix with diagonal entries $z_1, \dots, z_n$ and zero off-diagonal entries. More generally, $\operatorname{diag}(v)$ has the entries of the vector v on its diagonal. Multiplying by the corresponding variable whenever a vertex is visited gives the generating function for all walks:

$$1 + \mathbf{1}^{\mathsf{T}}(I - XA)^{-1}X\mathbf{1} = \frac{\det(I + X\overline{A})}{\det(I - XA)}.$$

The equality follows from the rank-one determinant lemma. Extracting the coefficient of $z_1 \cdots z_n$ selects exactly the Hamiltonian paths. The trace of a matrix is the sum of its diagonal entries,

$$\operatorname{tr} M = \sum_i M_{ii}.$$

For the denominator, the formal identity $\det(I - XA)^{-1} = \exp(\sum_{r \geq 1} \operatorname{tr}((XA)^r)/r)$ shows that the coefficients of squarefree monomials, in which each variable has exponent zero or one, count disjoint directed cycle covers and thus equal permanents. Combining this with the principal-minor expansion of the numerator gives

$$H(T) = \sum_U \det \overline{A}[U] \operatorname{per} A[U^c].$$

Substituting $\overline{A}[U] = I + A[U]^{\mathsf{T}}$ proves the identity.

Write $T[U]$ for the subtournament induced by the vertex set U, and let $k = |U|$. The symmetric part of $I + A[U]$ is the positive definite matrix $(I + J)/2$. Thus its real eigenvalues are positive, its nonreal eigenvalues occur in conjugate pairs, and its determinant is positive. The squared row norms are $1 + d_i^+(T[U])$, whose mean is $(k + 1)/2$. Hadamard's inequality and the arithmetic–geometric mean inequality give

$$0 < \det(I + A[U]) \leq h_k, \qquad h_k = \left(\frac{k+1}{2}\right)^{k/2}.$$

Set $h_0 = 1$ and define

$$w_k = \frac{2^k h_k}{k!}.$$

For every fixed $c > 0$ and nonnegative integer r,

$$\sum_{k \geq 0} k^r c^k w_k < \infty.$$

Indeed, Stirling's formula gives $\log(c^k w_k) = -(k/2)\log k + O_c(k)$, where the subscript c allows the implicit constant to depend on the value of c fixed in advance. Write $\lceil x \rceil$ for the least integer at least x. For sufficiently large n, choose

$$k_n = \left\lceil \frac{4 \log n}{\log \log n} \right\rceil,$$

Then $\sum_{k>k_n} c^k w_k = n^{-2+o(1)}$.

The Brégman bound stated in [2, Lemma 2.1], the degree-balancing argument of [2, Corollary 2.3], and Stirling's formula give a uniform constant C such that every m-vertex tournament satisfies

$$\operatorname{per} A_T \le C\sqrt{m+1}\,\frac{m!}{2^m}.$$

Increasing C also covers $m = 0$. Lemma 7.2 will prove a version with a variance penalty. Consequently, the contribution from $|U| > k_n$ in the path convolution, after division by μ_n, is at most

$$\frac{C}{2}\sqrt{n+1}\sum_{k>k_n} w_k = n^{-3/2+o(1)} = o(n^{-1}).$$

This bound is uniform over all tournaments. We call the terms with $|U| \le k_n$ short terms. The permanents in the path convolution need to be estimated accurately only for these terms.

Below, $[m]F$ denotes a coefficient in the formal series F, namely the coefficient of the monomial m, and we abbreviate $z_U = \prod_{i\in U} z_i$. The formal inverse matrix expands as

$$(I - XA)^{-1} = \sum_{r\ge 0} (XA)^r,$$

$$\mathbf{1}^{\mathsf{T}}(XA)^r X\mathbf{1} = \sum_{v_0,\dots,v_r} z_{v_0} A_{v_0 v_1} z_{v_1} \cdots A_{v_{r-1}v_r} z_{v_r}.$$

In the monomial $z_1 \cdots z_n$, each variable appears exactly once. It therefore requires $r = n-1$ and distinct vertices $v_0, \dots, v_{n-1}$. Its coefficient is precisely $H(T)$. Set $M = I - XA$. Then

$$M + X\mathbf{1}\mathbf{1}^{\mathsf{T}} = I + X(J - A) = I + X\overline{A},$$
$$\det(M + X\mathbf{1}\mathbf{1}^{\mathsf{T}}) = \det M\,[1 + \mathbf{1}^{\mathsf{T}} M^{-1} X \mathbf{1}].$$

The principal-minor expansion gives

$$\det(I + X\overline{A}) = \sum_{U\subseteq[n]} \left(\prod_{i\in U} z_i\right) \det \overline{A}[U].$$

In $\operatorname{tr}((XA)^r)/r$, a directed cycle on r distinct vertices has r starting points and receives weight one after division by r. The factor $1/h!$ in the exponential expansion removes the orderings of h disjoint cycles. Thus, for any W,

$$\left[\prod_{i\in W} z_i\right] \det(I - XA)^{-1} = \sum_{\sigma: W\to W \text{ bijection}} \prod_{i\in W} A_{i,\sigma(i)} = \operatorname{per} A[W].$$

Therefore

$$\begin{aligned} H(T) &= [z_1 \cdots z_n] \left(\sum_U z_U \det \overline{A}[U] \right) \det(I - XA)^{-1} \\ &= \sum_U \det \overline{A}[U] \operatorname{per} A[U^c] \\ &= \sum_U \det(I + A[U]^{\mathsf{T}}) \operatorname{per} A[U^c] \\ &= \sum_U \det(I + A[U]) \operatorname{per} A[U^c]. \end{aligned}$$

For $k = 0$, the empty-determinant convention immediately gives $\det(I + A[\varnothing]) = h_0 = w_0 = 1$. All subsequent calculations involving $1/k$ assume $k \geq 1$. Set $M = I + A[U]$. Here $\operatorname{Re} z$ denotes the real part of a complex number, and the superscript $*$ denotes conjugate transpose; real matrices satisfy $M^* = M^{\mathsf{T}}$. For any nonzero complex vector z,

$$\operatorname{Re}(z^* M z) = z^* \frac{M + M^{\mathsf{T}}}{2} z = \frac{1}{2}(\|z\|_2^2 + |\mathbf{1}^{\mathsf{T}} z|^2) > 0.$$

If $Mz = \lambda z$, then $\operatorname{Re} \lambda > 0$. The real eigenvalues are positive, and each nonreal conjugate pair has product $|\lambda|^2 > 0$. Hence $\det M > 0$. On the other hand,

$$\begin{aligned} |\det M| &\leq \prod_{i \in U} \|M_{i,\cdot}\|_2 = \prod_{i \in U} (1 + d_i^+(T[U]))^{1/2} \\ &\leq \left(\frac{1}{k} \sum_{i \in U} (1 + d_i^+(T[U])) \right)^{k/2} \\ &= \left(\frac{k + k(k-1)/2}{k} \right)^{k/2} = \left(\frac{k+1}{2} \right)^{k/2} = h_k. \end{aligned}$$

Using $k! \geq (k/e)^k$, for $k \geq 1$ we have

$$c^k w_k \leq \left(\frac{2ec}{k} \sqrt{\frac{k+1}{2}} \right)^k \leq \left(\frac{2ec}{\sqrt{k}} \right)^k.$$

For fixed r and sufficiently large k, the right-hand side multiplied by k^r is at most 2^{-k}. Every fixed moment therefore converges. More precisely,

$$\log(c^k w_k) = -\tfrac{1}{2} k \log k + O_c(k), \qquad k_n \log k_n = (4 + o(1)) \log n.$$

The ratio of successive terms follows directly from the definition:

$$\frac{c^{k+1} w_{k+1}}{c^k w_k} = \frac{2c}{k+1} \sqrt{\frac{k+2}{2}} \left(1 + \frac{1}{k+1} \right)^{k/2} \leq \frac{K_c}{\sqrt{k+1}}.$$

The tail sum is therefore at most twice its first term and hence is $n^{-2+o(1)}$. The number of vertex subsets of size k is $\binom{n}{k}$, where for integers $0 \leq k \leq n$ we define

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

After normalization, the convolution terms of size k satisfy

$$\frac{\binom{n}{k} h_k \, C\sqrt{n-k+1}(n-k)!2^{-(n-k)}}{n!2^{-(n-1)}} = \frac{C}{2}\sqrt{n-k+1}\frac{2^k h_k}{k!} \le \frac{C}{2}\sqrt{n+1}\, w_k.$$

Here $\binom{n}{k}(n-k)!/n! = 1/k!$, and the power of two is 2^{k-1}. Summation gives the asserted error from long terms.

We now derive the formal identity used above. Temporarily introduce another variable t and write $M = XA$. A determinant with constant term one has a formal logarithm. Differentiation of the determinant gives

$$\frac{d}{dt}\log\det(I - tM) = -\operatorname{tr}[(I-tM)^{-1}M].$$

Substitute the formal geometric series into the right-hand side:

$$\begin{aligned} -\operatorname{tr}[(I-tM)^{-1}M] &= -\operatorname{tr}\left[\left(\sum_{h=0}^{\infty} t^h M^h\right) M\right] \\ &= -\sum_{h=0}^{\infty} t^h \operatorname{tr}(M^{h+1}) \\ &= -\sum_{r=1}^{\infty} t^{r-1} \operatorname{tr}(M^r). \end{aligned}$$

Formal integration term by term, with $\log\det(I - 0M) = 0$, gives

$$\log\det(I - tM) = -\sum_{r=1}^{\infty} \frac{t^r}{r} \operatorname{tr}(M^r).$$

Rearranging, exponentiating and setting $t = 1$, as a formal series in the variables z_i, yields

$$\det(I - XA)^{-1} = \exp\left(\sum_{r=1}^{\infty} \frac{\operatorname{tr}((XA)^r)}{r}\right).$$

The index expansion of each trace term is

$$\operatorname{tr}((XA)^r) = \sum_{i_1,\ldots,i_r} z_{i_1} A_{i_1 i_2} z_{i_2} A_{i_2 i_3} \cdots z_{i_r} A_{i_r i_1}.$$

In a squarefree monomial, $i_1, \ldots, i_r$ must be distinct. The same cycle has r cyclic starting points, each giving the same product, so the factor $1/r$ counts them as one cycle. Selecting exactly h disjoint cycles in the exponential produces

$$\frac{1}{h!}\left(\sum_{\mathcal{C}} \text{weight}(\mathcal{C})\right)^h$$

The same collection of h cycles is selected in $h!$ orders. The factor $1/h!$ cancels this repetition exactly. If two cycles intersect, the variable at a common vertex occurs at least twice and cannot contribute to a squarefree coefficient. This gives the cycle-cover count term by term.

5.2 The emergence of the spectral factor

First suppose $d = o(\sqrt{n})$. By Lemma 4.5, for $|U| = k \le k_n$ we have

$$\operatorname{per} A[U^c] = e^{-1} D_n(S) \frac{(n-k)!}{2^{n-k}} \left(1 + O\left(\frac{(d+k+1)^2}{n}\right)\right).$$

For integers $0 \le k \le n$, define the falling factorial by

$$(n)_k = \prod_{h=0}^{k-1} (n-h),$$

Thus $(n)_0 = 1$, and $(n)_k > 0$ in this range. The normalization satisfies $((n-k)!/2^{n-k})/\mu_n = 2^{k-1}/(n)_k$. Substituting the leading term into Lemma 5.1 therefore gives the normalized sum

$$\frac{e^{-1} D_n(S)}{2} \sum_{|U| \le k_n} \frac{2^{|U|}}{(n)_{|U|}} \det(I + A[U]),$$

The relative error is uniform over tournaments and sets $|U| \le k_n$. Replacing $(n)_k$ by n^k in the short sum and adding the tail yields the principal-minor identity below; Section 8.1 estimates the resulting errors.

$$\sum_U (2/n)^{|U|} \det(I + A[U]) = \det\left(I + \frac{2}{n}(I + A)\right).$$

Since $2A = J - I + S$, the rank-one term J asymptotically contributes a factor of two, while the scalar term contributes e. More precisely, the small-score assumption gives

$$\det\left(I + \frac{2}{n}(I + A)\right) = 2e \det(I + S/n) \left(1 + O(n^{-1} + d^2/n^2)\right).$$

The factor e^{-1} in the permanent estimate comes from restoring $(1 - 1/n)^{n-k}$ and reflects the zero diagonal. The factor $1/2$ comes from the path normalization above. They cancel the factor $2e$ in the generating determinant, leaving $D_n(S) \det(I + S/n) = \rho_n(S)$. The two determinants have separate origins: the first approximates cycle-cover counts, and the second sums the weights in the path convolution. Theorem 8.1 will complete the error summation and prove

$$\frac{H(T)}{\mu_n} = \rho_n(S) + O((d+1)^2/n) \qquad \text{when } d = o(\sqrt{n}).$$

To obtain an upper bound for all tournaments, Section 7 separately handles large score variance and extreme degrees, retaining the score penalty in the remaining class. The next lemma controls this same spectral factor for every tournament.

6. The spectral bound and equality

The spectral-ratio bound in this section is due to Divoux and Wang [12, Proposition 7.1], using the comparison of Fang and Huang [11]. The determinant polynomial and the carousel switching are also in [12, Section 7 and Lemma 9.1]. We give the details and follow the equality conditions needed for the fixed-base corollary.

6.1 The spectral radius and convex optimization

Lemma 6.1 (Spectral cap).

Every tournament sign matrix satisfies

$$\|S\|_{\mathrm{op}} \le \cot\left(\frac{\pi}{2n}\right) < \frac{2n}{\pi}, \qquad \rho_n(S) \le C_*.$$

Proof.

We first give the phase-ordering proof of the operator bound [4, Theorem 3.1] and then carry out the required convex optimization. A Hermitian matrix is a complex square matrix satisfying $H^* = H$. The Hermitian matrix iS has spectrum symmetric about zero, so its largest eigenvalue is $\|S\|_{\mathrm{op}}$. For an arbitrary complex vector z, apply the same signed permutation to z, S so that the phases of the nonzero coordinates lie in $[0, \pi)$ and occur in increasing order. Zero coordinates may be placed arbitrarily. For $i < j$, we then have $c_{ij} = \mathrm{Im}(\overline{z}_i z_j) \ge 0$, where Im takes the imaginary part and $\overline{z}_i$ is the complex conjugate of z_i. Let T_n^0 be the transitive sign matrix with every entry above the diagonal equal to one. Then

$$z^* iSz = -2\sum_{i<j} S_{ij} c_{ij} \le 2\sum_{i<j} c_{ij} = z^*(-iT_n^0)z.$$

Maximizing the Rayleigh quotient gives $\|S\|_{\mathrm{op}} \le \|T_n^0\|_{\mathrm{op}}$. To compute the latter norm, subtract consecutive row eigenvalue equations to obtain $(\lambda - 1)v_i = (\lambda + 1)v_{i+1}$. The first row equation shows that the geometric ratio $r = (\lambda - 1)/(\lambda + 1)$ satisfies $r^n = -1$. Hence all eigenvalues are $i \cot((2j-1)\pi/(2n))$, $1 \le j \le n$, giving the stated norm. For $n \ge 2$, the final strict inequality uses $\tan u > u$; the case $n = 1$ follows directly.

Define $a_* = 4/\pi^2$. The operator bound above gives $0 \le x_j \le a_*$ and $\sum_j x_j < 1/2$. The function

$$\phi(x) = \log \frac{1+x}{1-x}$$

is increasing and convex on $[0, 1)$. Moving mass from two interior coordinates toward one capped coordinate and one remainder coordinate cannot decrease $\sum_j \phi(x_j)$. Add zero coordinates when necessary and relax the total mass to $1/2$. Since $1/4 < a_* < 1/2$, the relaxed maximum is attained at $a_*, 1/2 - a_*, 0, \ldots$. Exponentiating gives C_*. Every spectrum realizable by a tournament matrix lies in this relaxed region, so its maximum is an upper bound for $\rho_n(S)$.

The same eigenvalue calculation also gives the following identity; see also [4, Theorem 2.1]:

$$\det(I + zT_n^0) = \frac{(1+z)^n + (1-z)^n}{2}.$$

Section 8 will use this determinant polynomial to compute the spectral factor of the carousel construction.

Write $T = T_n^0$ and suppose $Tv = \lambda v$. The equations for row i and row $i+1$, respectively, are

$$\lambda v_i = \sum_{j>i} v_j - \sum_{j<i} v_j, \qquad \lambda v_{i+1} = \sum_{j>i+1} v_j - \sum_{j<i+1} v_j.$$

Subtracting gives

$$\lambda(v_i - v_{i+1}) = v_i + v_{i+1}, \qquad (\lambda - 1)v_i = (\lambda + 1)v_{i+1}.$$

A skew-symmetric matrix has purely imaginary eigenvalues, so $\lambda + 1 \ne 0$. Set $r = (\lambda - 1)/(\lambda + 1)$. Then $v_i = v_1 r^{i-1}$ and $v_1 \ne 0$. Since $r = 1$ would imply $-1 = 1$,

$$\lambda = \sum_{h=1}^{n-1} r^h = \frac{r - r^n}{1 - r}, \qquad \lambda = \frac{1 + r}{1 - r}.$$

Comparison yields $r - r^n = 1 + r$, or $r^n = -1$. Set $\theta_j = (2j - 1)\pi/(2n)$. Then

$$r_j = e^{2i\theta_j}, \qquad \lambda_j = \frac{1 + e^{2i\theta_j}}{1 - e^{2i\theta_j}} = \frac{2e^{i\theta_j} \cos\theta_j}{-2ie^{i\theta_j} \sin\theta_j} = i \cot\theta_j.$$

These n roots are distinct and therefore give all eigenvalues. The largest modulus is $\cot(\pi/(2n))$. For $0 < u < \pi/2$,

$$\tan u - u = \int_0^u (\sec^2 v - 1)\, dv = \int_0^u \tan^2 v\, dv > 0,$$

Thus $\cot u < 1/u$.

After the signed permutation, write $z_i = r_i e^{i\theta_i}$ with $r_i \geq 0$ and $0 \leq \theta_1 \leq \cdots \leq \theta_n < \pi$. A signed permutation is an orthogonal matrix Q. Simultaneously replacing S by $Q^{\mathsf{T}} S Q$ and z by $Q^{\mathsf{T}} z$ preserves the quadratic form and norms. Expanding pair by pair gives

$$\begin{aligned} z^* i S z &= \sum_{i<j} i S_{ij} (\overline{z}_i z_j - \overline{z}_j z_i) \\ &= -2 \sum_{i<j} S_{ij} r_i r_j \sin(\theta_j - \theta_i) \\ &\leq 2 \sum_{i<j} r_i r_j \sin(\theta_j - \theta_i) = z^* (-i T_n^0) z. \end{aligned}$$

Differentiate ϕ twice:

$$\phi'(x) = \frac{2}{1 - x^2} > 0, \qquad \phi''(x) = \frac{4x}{(1 - x^2)^2} \geq 0.$$

If $0 < x \leq y < a_*$, choose $0 \leq t \leq \min(x, a_* - y)$. Then

$$\frac{d}{dt}[\phi(x - t) + \phi(y + t)] = -\phi'(x - t) + \phi'(y + t) \geq 0.$$

Taking $t = \min(x, a_* - y)$ sends at least one coordinate to 0 or a_*. After finitely many such steps, at most one coordinate remains in the interior. Since $2a_* > 1/2$ and $a_* < 1/2$, at total mass $1/2$ the final vector is

$$(a_*, 1/2 - a_*, 0, \ldots), \qquad \rho_n(S) \leq e^{\phi(a_*) + \phi(1/2 - a_*)} = C_*.$$

Finally, consider the determinant polynomial. If $p_n(z) = \det(I + z T_n^0)$, the eigenvalues above show that it has the same complex roots and degree as $q_n(z) = ((1 + z)^n + (1 - z)^n)/2$. Alternatively, applying the same eigenvalue calculation to each even-order principal minor gives $\det T_{2h}^0 = 1$ and $\det T_{2h+1}^0 = 0$. With $\lfloor x \rfloor$ denoting the greatest integer at most x, the principal-minor expansion directly gives

$$p_n(z) = \sum_{h=0}^{\lfloor n/2 \rfloor} \binom{n}{2h} z^{2h} = \frac{1}{2} \sum_{k=0}^{n} \binom{n}{k} (z^k + (-z)^k) = q_n(z).$$

The even-order determinant can also be obtained as one from the roots of q_{2h}: $\det T_{2h}^0 = \prod_{j=1}^{2h} i \cot\theta_j = 1$. Pair the positive-frequency cot factors with tan factors at complementary angles to obtain this product.

For order $2m$, the positive frequencies are

$$\lambda_j = \cot\theta_j, \quad \theta_j = \frac{(2j-1)\pi}{4m}, \quad 1 \le j \le m.$$

The index of the complementary angle is $m+1-j$, because

$$\theta_{m+1-j} = \frac{(2m+1-2j)\pi}{4m} = \frac{\pi}{2} - \theta_j.$$

Hence $\cot\theta_{m+1-j} = \tan\theta_j = (\cot\theta_j)^{-1}$, and therefore

$$\left(\prod_{j=1}^{m} \cot\theta_j\right)^2 = \prod_{j=1}^{m} \cot\theta_j \cot\theta_{m+1-j} = 1.$$

Since each $\lambda_j > 0$, their product is one. Pairing the eigenvalues as $i\lambda_j, -i\lambda_j$ gives

$$\det T_{2m}^0 = \prod_{j=1}^{m} (i\lambda_j)(-i\lambda_j) = \prod_{j=1}^{m} \lambda_j^2 = 1.$$

For odd order $2m+1$, the middle eigenvalue is $i\cot(\pi/2) = 0$, so the determinant vanishes. Every principal submatrix retains the transitive form with ones above the diagonal. Thus all even-order principal minors equal one, and all odd-order principal minors equal zero. □

6.2 Product comparison and equality

Theorem 6.2 (Fang–Huang).

We use the following form of Fang–Huang, Corollary 3.4. Define the comparison matrix K_d by

$$(K_d)_{ij} = \begin{cases} 1, & i+j \le d, \\ 0, & i+j = d+1, \\ (-1)^{i+j-d-2}, & i+j \ge d+2. \end{cases}$$

Let $p \ge 1$, let $x \in \mathbb{R}^d$, and let $S_1, \dots, S_p$ be tournament sign matrices of order d. Write x^* for the vector obtained by arranging the absolute values of the coordinates of x in decreasing order. Then

$$(S_1 \cdots S_p x)^* \prec_w K_d^p x^*.$$

The right-hand side is a nonnegative decreasing vector. Weak majorization $u \prec_w v$ means

$$\sum_{i=1}^{r} u_i \le \sum_{i=1}^{r} v_i \quad (1 \le r \le d).$$

Moreover, vector equality in the outermost comparison implies equality for every suffix product. To apply this to Frobenius norms, we first record the squared-norm inequality under weak majorization, including its equality case.

Lemma 6.3 (Truncated sums under weak majorization).

Write $(a-t)_+ = \max(a-t, 0)$. For nonnegative decreasing vectors satisfying $u \prec_w v$ and $t \ge 0$,

$$\sum_{i=1}^{d} (u_i - t)_+ = \max_{0 \le r \le d} \left(\sum_{i=1}^{r} u_i - rt \right) \le \max_{0 \le r \le d} \left(\sum_{i=1}^{r} v_i - rt \right) = \sum_{i=1}^{d} (v_i - t)_+.$$

Proof.

The first equality follows by taking $r = \#\{i : u_i > t\}$: adding coordinates at most t, or removing coordinates greater than t, cannot increase the sum. For $a \geq 0$, we also have

$$2\int_0^\infty (a-t)_+\,dt = 2\int_0^a (a-t)\,dt = 2[at - t^2/2]_0^a = a^2.$$

Integrating coordinate by coordinate gives

$$\|u\|_2^2 = 2\int_0^\infty \sum_i (u_i - t)_+\,dt \leq 2\int_0^\infty \sum_i (v_i - t)_+\,dt = \|v\|_2^2.$$

Suppose the squared norms are equal. The function $g(t) = \sum_i (v_i - t)_+ - \sum_i (u_i - t)_+$ is continuous and nonnegative, with integral zero. If $g(t_0) > 0$, then $g(t) > g(t_0)/2$ on an interval of positive length, a contradiction. Thus $g \equiv 0$. Comparing right derivatives gives

Here $\#U$ denotes the cardinality of a finite set U, and $\mathbf{1}_{\mathcal{E}}$ is the indicator of the condition $\mathcal{E}$.

$$\#\{i : u_i > t\} = \#\{i : v_i > t\} =: N(t).$$

For coordinate j,

$$u_j = \int_0^\infty \mathbf{1}_{\{\#\{i:u_i>t\}\geq j\}}\,dt = \int_0^\infty \mathbf{1}_{\{N(t)\geq j\}}\,dt = v_j.$$

Equality of squared norms therefore forces $u = v$. Taking $x = e_j$ and using $e_j^* = e_1$ yields

$$\|S^p e_j\|_2^2 \leq \|K_d^p e_1\|_2^2.$$

Equality also forces $(S^p e_j)^* = K_d^p e_1$.

6.3 Moments of the comparison matrix

Write $\operatorname{tr} M = \sum_j M_{jj}$.

Let T_d be the transitive tournament sign matrix, with all entries above the diagonal equal to 1. Let R_0 be the reversal permutation matrix and $D_0 = \operatorname{diag}(1, -1, \ldots)$. By the definition of K_d,

$$(D_0 K_d R_0)_{ij} = (-1)^{i-1}(K_d)_{i,d+1-j} = \begin{cases} (-1)^{i-1}, & i < j, \\ 0, & i = j, \\ (-1)^j, & i > j. \end{cases}$$

Here $R_0 e_j = e_{d+1-j}$ and $(D_0)_{jj} = (-1)^{j-1}$. Since $(K_d)_{ij}$ depends only on $i + j$, we have $K_d^T = K_d$.

Write the sequence of increasing odd indices followed by decreasing even indices as

$$(\ell_1, \ldots, \ell_d) = (1, 3, \ldots, d, d-1, d-3, \ldots, 2), \qquad \Lambda e_j = e_{\ell_j}.$$

Each index occurs once. Hence $\Lambda^T \Lambda = \Lambda \Lambda^T = I$, and $\ell_1 = 1$.

Permute the rows and columns in this order. Odd indices are increasing, so the corresponding upper-triangular entries are $(-1)^{i-1} = 1$; even indices are decreasing, so those entries are $(-1)^j = 1$. An odd index precedes an even index, and the corresponding entry is $(-1)^{i-1}$ if $i < j$, or $(-1)^j$ if $i > j$, again equal to 1. Hence

$$T_d = \Lambda^T D_0 K_d R_0 \Lambda.$$

Set $U = D_0\Lambda$. Since $R_0R_0^T = I$, $\Lambda\Lambda^T = I$ and $K_d^T = K_d$, multiplication gives

$$\begin{aligned} T_dT_d^T &= \Lambda^T D_0 K_d R_0 \Lambda\Lambda^T R_0^T K_d D_0 \Lambda \\ &= \Lambda^T D_0 K_d^2 D_0 \Lambda = U^T K_d^2 U. \end{aligned}$$

The first index in this ordering is 1, so $Ue_1 = e_1$. Also $T_d^T = -T_d$, hence

$$T_d^2 = -U^T K_d^2 U, \qquad T_d^{2p} = (-1)^p U^T K_d^{2p} U$$

because each adjacent pair UU^T in the product equals I. Consequently,

$$(-1)^p (T_d^{2p})_{11} = e_1^T K_d^{2p} e_1 = (K_d^p e_1)^T (K_d^p e_1) = \|K_d^p e_1\|_2^2.$$

The signed cyclic shift Z in Appendix A.2 satisfies $Z^T T_d Z = T_d$, and therefore $Z^T T_d^{2p} Z = T_d^{2p}$. Comparing diagonal entries gives $(T_d^{2p})_{j+1,j+1} = (T_d^{2p})_{jj}$ for $j < d$. At $j = d$, the two signs cancel and give $(T_d^{2p})_{11} = (T_d^{2p})_{dd}$. Thus all diagonal entries agree, and

$$\begin{aligned} \|T_d^p\|_F^2 &= \operatorname{tr}((T_d^p)^T T_d^p) = (-1)^p \operatorname{tr} T_d^{2p} \\ &= d(-1)^p (T_d^{2p})_{11} = d\|K_d^p e_1\|_2^2. \end{aligned}$$

Sum the d column inequalities from the preceding section:

$$\|S^p\|_F^2 = \sum_{j=1}^{d} \|S^p e_j\|_2^2 \le d\|K_d^p e_1\|_2^2 = \|T_d^p\|_F^2 \qquad (p \ge 1).$$

In particular, the inequality holds for every odd p, as required in the next section.

6.4 The extremum of the spectral ratio

Let S be any tournament sign matrix of order d, with nonzero eigenvalues $\pm i\lambda_j$, where $\lambda_j > 0$, and put $x_j = \lambda_j^2/d^2$. Since $2\sum_j \lambda_j^2 = \|S\|_F^2 = d(d-1)$, we have $x_j \le (d-1)/(2d) < 1/2$. In the skew-symmetric canonical form, a zero block contributes 1 to both determinants, while a nonzero two-dimensional block contributes $1 + x_j$ and $1 - x_j$, respectively. Hence

$$\det(I + S/d) = \prod_j (1 + x_j), \qquad \det(I + iS/d) = \prod_j (1 - x_j) > 0.$$

For $0 \le x < 1$, integrate the geometric series:

$$\begin{aligned} \log\frac{1+x}{1-x} &= \int_0^x \left(\frac{1}{1+t} + \frac{1}{1-t}\right) dt \\ &= 2\int_0^x \frac{dt}{1-t^2} = 2\sum_{r=0}^{\infty} \int_0^x t^{2r}\, dt = 2\sum_{r=0}^{\infty} \frac{x^{2r+1}}{2r+1}. \end{aligned}$$

The interchange follows from uniform convergence of the geometric series on $[0, x]$. The skew-symmetric canonical form also gives $\|S^p\|_F^2 = 2\sum_j \lambda_j^{2p}$. Substituting term by term yields

$$\begin{aligned} \log \rho_d(S) &= 2\sum_j \sum_{r=0}^{\infty} \frac{\lambda_j^{4r+2}}{(2r+1)d^{4r+2}} \\ &= \sum_{r=0}^{\infty} \frac{\|S^{2r+1}\|_F^2}{(2r+1)d^{4r+2}} \le \sum_{r=0}^{\infty} \frac{\|T_d^{2r+1}\|_F^2}{(2r+1)d^{4r+2}} = \log \rho_d(T_d). \end{aligned}$$

Put $r_d = \rho_d(T_d)$. If equality holds above, every nonnegative difference in the series vanishes. The term $r = 1$ gives $\|S^3\|_F^2 = \|T_d^3\|_F^2$. Since this difference is itself the sum of the nonnegative column differences, equality holds in each column. The equality criterion of Section 6.2 and the suffix equality in Fang–Huang's theorem then give

$$(S^3 e_j)^* = K_d^3 e_1, \qquad (S^2 e_j)^* = K_d^2 e_1.$$

Take squared norms in the second identity and sum over j:

$$\operatorname{tr} S^4 = \|S^2\|_F^2 = d\|K_d^2 e_1\|_2^2 = \|T_d^2\|_F^2 = \operatorname{tr} T_d^4.$$

By Appendix A.3, equality of the fourth moments is equivalent to $S = \Omega^T T_d \Omega$ for a signed permutation matrix Ω. Conversely, $\Omega^T \Omega = I$ implies

$$I + tS = \Omega^T (I + tT_d)\Omega, \qquad \det(I + tS) = (\det \Omega)^2 \det(I + tT_d) = \det(I + tT_d).$$

Lemma 6.4 (The spectral ratio and equality).

Take $t = 1/d$ and $t = i/d$ separately to obtain $\rho_d(S) = r_d$. Thus

$$\rho_d(S) \le r_d, \quad \text{with equality if and only if } S = \Omega^T T_d \Omega.$$

Proof (Ordinary isomorphism in the regular case).

Write $\Omega = D_\varepsilon P$, where P is a permutation matrix and $D_\varepsilon = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_d)$ with $\varepsilon_i \in \{-1, 1\}$. Put $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)^T$. Then

$$S = P^T D_\varepsilon T_d D_\varepsilon P, \qquad PSP^T = D_\varepsilon T_d D_\varepsilon.$$

Relabel by P and continue to write S for the resulting matrix. If the tournament is regular, then $S\mathbf{1} = 0$, and $D_\varepsilon^2 = I$ gives

$$D_\varepsilon T_d \varepsilon = 0, \qquad T_d \varepsilon = 0.$$

The difference between rows i and $i + 1$ has entry 1 in each of columns $i, i + 1$ and zero elsewhere. Therefore

$$0 = (T_d\varepsilon)_i - (T_d\varepsilon)_{i+1} = \varepsilon_i + \varepsilon_{i+1}.$$

It follows that $\varepsilon_i = (-1)^{i-1}\varepsilon_1$. The common sign ε_1 cancels in $D_\varepsilon T_d D_\varepsilon$, leaving the alternating switching. Appendix A.2 identifies this tournament with Car_d. Thus, among regular tournaments, equality holds only for the carousel, up to isomorphism.

6.5 A gap from the sixth moment

$\binom{n}{k} = n!/(k!(n-k)!)$, with value 0 for $k > n$.

Lemma 6.5 (Strict sixth-moment deficit).

If regular S is not ordinarily isomorphic to the carousel, the preceding equality argument gives

$$\Delta_3 := \|T_d^3\|_F^2 - \|S^3\|_F^2 > 0.$$

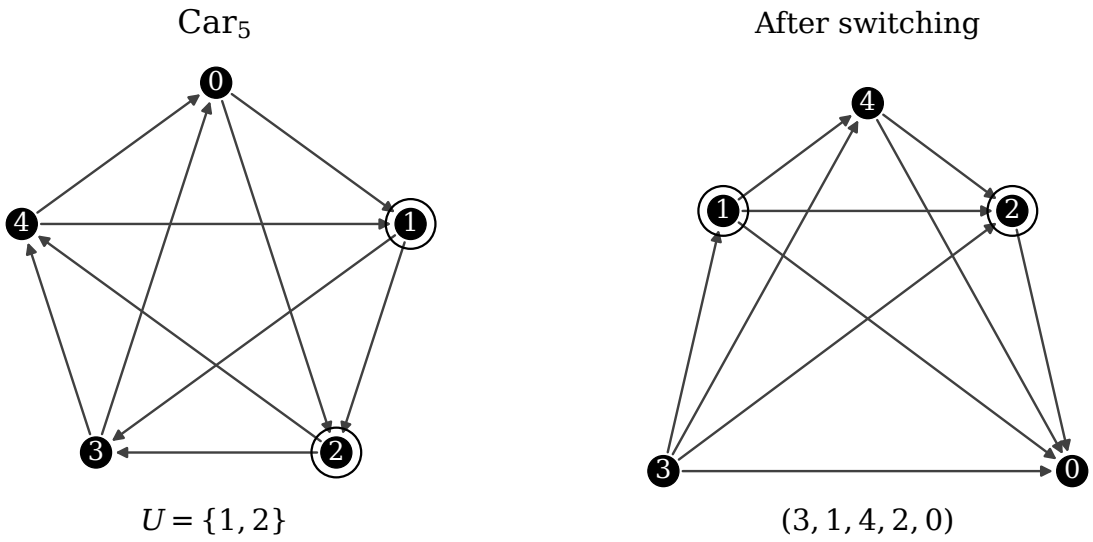


Figure 3: Reversing all arcs between $U = \{1, 2\}$ and its complement in Car_5 gives the transitive order $(3, 1, 4, 2, 0)$. Circled vertices belong to U. This switching conjugates the sign matrix as $S \mapsto DSD$ and preserves the spectral ratio; it need not preserve the Hamiltonian path count.

Proof.

Modulo 2, $S \equiv J + I$. Since $J^2 = dJ$ and $J^3 = d^2J$,

$$S^3 \equiv (J + I)^3 = I + 3J + 3dJ + d^2J \equiv I + (1 + d + d^2)J \equiv I + J \pmod 2.$$

The last step uses that $d(d + 1)$ is even. Also $(S^3)^T = -S^3$, so its diagonal entries vanish and its off-diagonal entries are odd. If $z = 2k + 1$,

$$z^2 = 4k(k + 1) + 1 \equiv 1 \pmod 8.$$

Pair the skew-symmetric positions in the squared norm:

$$\|S^3\|_F^2 = 2\sum_{i<j}((S^3)_{ij})^2 \equiv 2\binom{d}{2} = d(d - 1) \pmod{16}.$$

This also applies to T_d. Thus Δ_3 is a positive multiple of 16, and $\Delta_3 \geq 16$. Retaining just term $r = 1$ in the logarithmic series gives

$$\log \frac{r_d}{\rho_d(S)} \geq \frac{\Delta_3}{3d^6} \geq \frac{16}{3d^6}.$$

Exponentiate and rearrange:

$$\rho_d(S) \leq r_d e^{-16/(3d^6)}, \qquad r_d - \rho_d(S) \geq r_d(1 - e^{-16/(3d^6)}) > 0.$$

The gap depends on the base order d. Section 11.7 obtains a corresponding block threshold by comparing the limits.

7. The upper bound for arbitrary tournaments

In this section, $V(T)$ denotes the real-valued variance below. In the earlier expression $U \subseteq V(T)$, the symbol $V(T)$ denotes the vertex set; its use within a set-inclusion relation distinguishes the two meanings. Write

$$V(T)=\sum_i\left(d_i^+-\frac{n-1}{2}\right)^2=\frac{1}{4}\|S\mathbf{1}\|_2^2,\qquad \tau=\frac{4V(T)}{(n-1)^2}.$$

The proof has three cases. If $V(T)\ge 16n^2\log n$, the variance penalty in Brégman's inequality makes the path count negligible. Otherwise, vertices with degrees outside $[(n-1)/20,19(n-1)/20]$ number at most $O(\log n)$. If any such vertices exist, separating them from the rest gives a path count below the random expectation. In the remaining tournaments, normalized scores stay a fixed distance from ±1, and $\tau=O(\log n)$. For them, the permanent estimate below retains the factor $e^{-\tau}$ and controls the final error by $O(n^{-1})$.

The constant K in this section may increase from one occurrence to the next; after the degree thresholds above are fixed, it is absolute. Section 5.1 has already bounded the long terms by $o(n^{-1})\mu_n$, so here we estimate only short terms $|U|\le k_n$.

The next bound records the counting estimates with the elementary spectral-radius bound. Its upper constant exceeds the leading constant already established in [12]; it is used only as an intermediate estimate.

Proposition 7.1 (Auxiliary counting bound). There are absolute constants K,n_0 such that every $n\ge n_0$ satisfies

$$(L-K/n)\mu_n\le P(n)\le(C_*+K/n)\mu_n,\qquad C_*=\frac{3\pi^4+4\pi^2-32}{\pi^4+4\pi^2-32}.$$

7.1 Applying the nonprincipal permanent estimate

The estimate of Section 4.4 allows different deleted row and column sets. We use it below to control edges between the exceptional set and the core. When the deleted sets agree, the score product is retained outside the path convolution.

7.2 Large score variance

Lemma 7.2 (Permanent bound with a variance penalty).

There is an absolute constant K such that, for every order $m\ge1$, a tournament Q with outdegrees d_i and degree variance $V(Q)=\sum_i(d_i-(m-1)/2)^2$ satisfies

$$\operatorname{per}A_Q\le K\sqrt{m+1}\frac{m!}{2^m}\exp\left(-\frac{V(Q)}{8m^2}\right).$$

Proof.

Concavity of the row-degree factor strengthens the Brégman bound [2]. If any $d_i=0$, the bound is immediate. Otherwise, define $f(k)=\log(k!)/k$ for positive integers. Direct calculation gives, for $k\ge2$,

$$2f(k)-f(k-1)-f(k+1)=\frac{2[\log k-f(k-1)]-k\log(1+1/k)}{k(k+1)}.$$

Applying the arithmetic–geometric mean inequality to $1,\dots,k-1$ gives $\log k-f(k-1)\ge\log 2$. Also, $(1+1/k)^k<e<3$, so the numerator is at least $\log(4/3)\ge1/4$. Consequently, on the degree interval $[1,m-1]$, the piecewise linear interpolation of the integer-point function $f(k)+k^2/(8m^2)$ is concave. Jensen's inequality at the mean $\eta=(m-1)/2$ gives

$$\sum_i f(d_i)\le m\widetilde{f}(\eta)-\frac{V(Q)}{8m^2}+\frac{1}{32m},$$

Here $\widetilde{f}$ is the linear interpolation. The last term handles a half-integer mean and can be omitted when the mean is an integer. Stirling's formula gives

$$\exp(m\widetilde{f}(\eta)) \le K\sqrt{m+1}\frac{m!}{2^m}.$$

Combining this with Brégman's row-degree bound proves the variance estimate.

We apply this bound to the tournament associated with each short deleted set. We must therefore check that its variance remains sufficiently large after deletion.

After deleting a size-k set U, each remaining centered degree changes by at most $k/2$, and the sum of the original squared deviations at deleted vertices is at most $kn^2/4$. Applying Cauchy–Schwarz to the cross term gives

$$V(T-U) \ge V(T) - \frac{kn^2}{4} - k\sqrt{nV(T)}.$$

If $V(T) \ge 16n^2 \log n$, then uniformly for every $k \le k_n$,

$$V(T-U) \ge (1-o(1))V(T).$$

Thus the variance penalty in every short permanent is at most $n^{-2+o(1)}$, absorbing the preceding $\sqrt{n}$ factor. Summation with the summable determinant weights bounds the total normalized short contribution by $n^{-3/2+o(1)}$. Adding the long-deletion tail gives, uniformly over this graph class,

$$V(T) \ge 16n^2 \log n \quad \Longrightarrow \quad H(T)/\mu_n = o(1/n).$$

Write $L_k = \log(k!)$. Then

$$\begin{aligned} 2f(k) - f(k-1) - f(k+1) &= \frac{2(L_{k-1} + \log k)}{k} - \frac{L_{k-1}}{k-1} - \frac{L_{k-1} + \log k + \log(k+1)}{k+1} \\ &= \frac{2\log k - 2L_{k-1}/(k-1) - k\log(1+1/k)}{k(k+1)}. \end{aligned}$$

AM–GM gives

$$((k-1)!)^{1/(k-1)} \le \frac{1 + \cdots + (k-1)}{k-1} = k/2,$$

Hence $\log k - f(k-1) \ge \log 2$. Also, $k\log(1+1/k) < 1 < \log 3$, and

$$\log(4/3) = \int_1^{4/3} \frac{du}{u} \ge \frac{1/3}{4/3} = \frac{1}{4}.$$

The second difference is therefore at least $1/[4k(k+1)] \ge 1/(4m^2)$. Set $h(k) = f(k) + k^2/(8m^2)$. Then

$$2h(k) - h(k-1) - h(k+1) \ge \frac{1}{4m^2} - \frac{2}{8m^2} = 0.$$

If η is an integer, $\widetilde{h}(\eta) = \widetilde{f}(\eta) + \eta^2/(8m^2)$. If $\eta = k + 1/2$, then

$$\frac{k^2 + (k+1)^2}{2} = \eta^2 + \frac{1}{4}, \quad \widetilde{h}(\eta) = \widetilde{f}(\eta) + \frac{\eta^2 + 1/4}{8m^2}.$$

Jensen's inequality therefore gives

$$\sum_i f(d_i) + \frac{\sum_i d_i^2}{8m^2} = \sum_i h(d_i) \le m\widetilde{h}(\eta)$$
$$\le m\widetilde{f}(\eta) + \frac{m\eta^2}{8m^2} + \frac{1}{32m},$$
$$\sum_i f(d_i) \le m\widetilde{f}(\eta) - \frac{\sum_i (d_i-\eta)^2}{8m^2} + \frac{1}{32m}.$$

For $m = 2r+1$, the logarithmic form of Stirling's formula gives

$$mf(r) = \frac{2r+1}{r}\left[r\log r - r + \tfrac{1}{2}\log(2\pi r) + O(1/r)\right],$$

$$\log(m!/2^m) = m\log(m/2) - m + \tfrac{1}{2}\log(2\pi m) + O(1/m).$$

Subtracting and using $r = (m-1)/2$ gives $\frac{1}{2}\log m + O(1)$. For $m = 2r$, $\widetilde{f}(\eta) = [f(r-1) + f(r)]/2$; applying the same formula to each term gives the same difference. Exponentiating therefore yields $K\sqrt{m+1}$. If $d_i = 0$, each term in the permanent contains a zero factor. The cases $m = 1, 2$ are incorporated directly into the constant.

Write $b_i = d_i^+ - (n-1)/2$. For a retained vertex, let e_i count its outgoing edges to U. Then

$$b_i' = d_i^+ - e_i - (n-k-1)/2 = b_i + \delta_i, \qquad \delta_i = k/2 - e_i, \quad |\delta_i| \le k/2.$$

Thus

$$V(T-U) = \sum_{i\notin U}(b_i+\delta_i)^2$$
$$\ge V(T) - \sum_{i\in U} b_i^2 - 2\left|\sum_{i\notin U} b_i\delta_i\right|$$
$$\ge V(T) - kn^2/4 - k\sum_{i\notin U}|b_i|$$
$$\ge V(T) - kn^2/4 - k\sqrt{nV(T)}.$$

When $V \ge 16n^2\log n$, the relative losses are respectively at most

$$\frac{k}{64\log n} = o(1), \qquad k\sqrt{n/V} \le \frac{k}{4\sqrt{n\log n}} = o(1).$$

It follows that $V(T-U)/(8(n-k)^2) \ge (2-o(1))\log n$, so the penalty is $n^{-2+o(1)}$. Multiplying by $\sqrt{n}$ and $\sum_k w_k < \infty$ gives $n^{-3/2+o(1)}$.

To estimate the factorial factor in the variance bound, apply the Stirling formula from Appendix B.3. With c denoting its constant,

$$f(r) = \log r - 1 + \frac{\log r}{2r} + \frac{c}{r} + O(r^{-2}).$$

First take $m = 2r+1$, so that the average outdegree is the integer r. Multiplication by m gives

$$mf(r) = m\log r - m + \left(1+\frac{1}{2r}\right)\log r + 2c + \frac{c}{r} + O(r^{-1}).$$

The logarithm of the natural scale is

$$\log(m!/2^m) = m\log(m/2) - m + \frac{1}{2}\log m + c + O(m^{-1}).$$

Denote their difference by Δ_m:

$$\begin{aligned}\Delta_m &= m\log\frac{2r}{m} + \log r - \frac{1}{2}\log m + c + O(\log m/m)\\ &= m\log(1-1/m) + \frac{1}{2}\log m - \log 2 + \log(1-1/m) + c + O(\log m/m).\end{aligned}$$

Since $m\log(1-1/m) = -1 + O(1/m)$ and all other terms except $\frac{1}{2}\log m$ are uniformly bounded, $\Delta_m \le \frac{1}{2}\log m + K$.

Next take $m = 2r$ with $r \ge 2$. The mean is $r - 1/2$, and linear interpolation gives

$$m\widetilde{f}((m-1)/2) = r[f(r-1) + f(r)].$$

Substitution term by term yields

$$\begin{aligned}r[f(r-1)+f(r)] = r\log(r-1) + r\log r - 2r\\ + \frac{r}{2(r-1)}\log(r-1) + \frac{1}{2}\log r\\ + c\frac{r}{r-1} + c + O(r^{-1}).\end{aligned}$$

Subtracting $\log((2r)!/2^{2r}) = 2r\log r - 2r + \frac{1}{2}\log(2r) + c + O(r^{-1})$ gives

$$\begin{aligned}\Delta_{2r} = r\log(1-1/r) + \frac{r}{2(r-1)}\log(r-1)\\ - \frac{1}{2}\log 2 + c\frac{r}{r-1} + O(r^{-1}).\end{aligned}$$

The first term is nonpositive. The second satisfies

$$\frac{r}{2(r-1)}\log(r-1) \le \frac{1}{2}\log r + \frac{\log r}{2(r-1)} \le \frac{1}{2}\log r + \frac{1}{2},$$

Here $\log r \le r - 1$. The constant term $cr/(r-1)$ has absolute value at most $2|c|$. Thus again $\Delta_m \le \frac{1}{2}\log m + K$. Exponentiating in both parity cases gives

$$\exp(m\widetilde{f}((m-1)/2)) \le e^K\sqrt{m}\,m!/2^m.$$

The finitely many values of m outside the expansion range are covered by increasing the same K. □

7.3 Exceptional vertices

Assume from now on that $V(T) < 16n^2\log n$. Define the exceptional set

$$F = \left\{i : \frac{d_i^+}{n-1} \notin [1/20, 19/20]\right\}, \qquad f = |F|, \qquad M = T - F, \qquad N = n - f.$$

Each exceptional vertex contributes to $V(T)$ at least a fixed positive multiple of n^2, so $f = O(\log n)$. A nonexceptional vertex originally has absolute score at most $0.9(n-1)$, and deleting F changes its score by at most f. Consequently, uniformly for all sufficiently large n,

$$\max_i |(S_M\mathbf{1})_i/(N-1)| < 0.95, \qquad \tau_M = O(\log n).$$

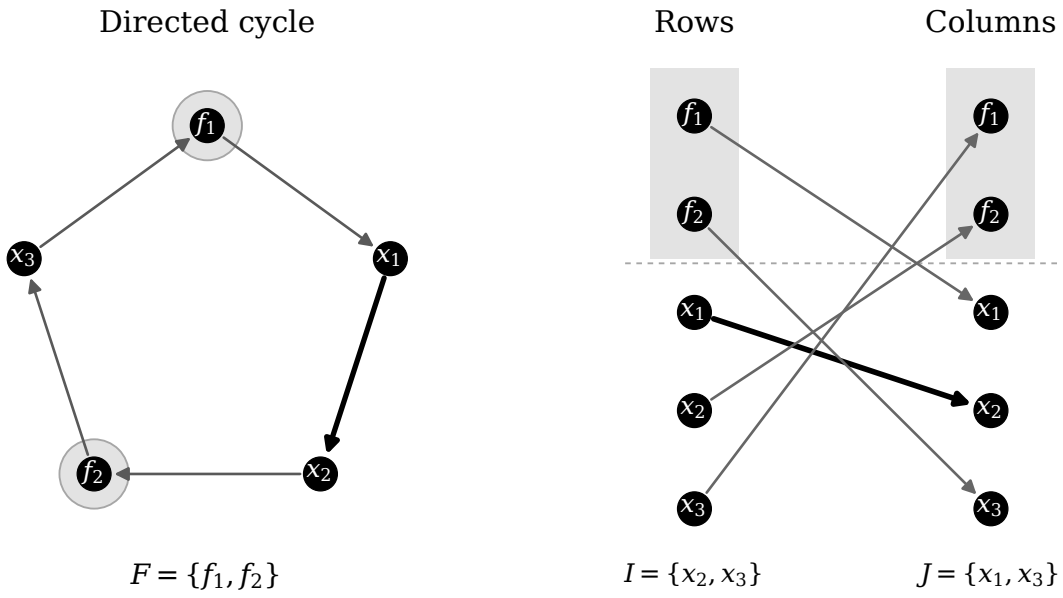


Figure 4: A directed five-cycle in a permanent term and its row-column matching. Shaded vertices form the exceptional set $F = \{f_1, f_2\}$. Cross-block arcs occupy core rows $I = \{x_2, x_3\}$ and core columns $J = \{x_1, x_3\}$, leaving the nonprincipal entry $A[x_1, x_2]$ shown in bold. Only the arcs of this permanent term are drawn.

For the squared score sum, use $\|S_M\mathbf{1}\|_2 \le \|S\mathbf{1}\|_2 + f\sqrt{N}$ and divide by $N-1$. The same argument remains valid after further deleting a short core subset. Throughout these deletions the original exceptional set F is retained, allowing the constants to be chosen uniformly.

First classify cycle covers according to their connections between F and the core M, thereby estimating per A_T. An exceptional vertex has few neighbors in one direction. A cycle cover must use edges in both directions unless that vertex is matched within F, and the latter choice incurs a cost of order $1/N$. Use the paired core weights ℓ, r from Lemma 4.4, whose total deviation satisfies

$$\sum_{i\in M} |\ell_i - 1| + \sum_{i\in M} |r_i - 1| \le K\sqrt{N\log n}.$$

For $x \in F$, write $q_x = N^{-1}|\{j \in M : x \to j\}|$. Its distance from $[0, 1/20] \cup [19/20, 1]$ is $O(f/n)$. The weighted sums of cross-edge neighbors on the two sides are bounded by $N(q_x + \beta)$ and $N(1 - q_x + \beta)$, where $\beta = O(\sqrt{\log n/n})$. Set

$$u_x = 2(q_x + \beta), \qquad v_x = 2(1 - q_x + \beta).$$

Uniformly for every x, we may choose

$$u_x v_x \le c_n = 19/100 + o(1), \qquad u_x, v_x \le L_n = 2 + o(1).$$

Decompose the permanent according to the permutation edges lying inside F. If there are s internal edges, their row set R and column set C both have size s, and there are exactly $t = f - s$ cross edges in each direction. The deleted core row and column sets I, J both have size t, but need not coincide. The exact four-block expansion is

$$\operatorname{per} A_T = \sum_{s=0}^{f} \sum_{\substack{R,C\subseteq F\\ |R|=|C|=s}} \sum_{\substack{I,J\subseteq M\\ |I|=|J|=f-s}} \operatorname{per} A[R,C]\ \operatorname{per} A[F\setminus R,J]\ \operatorname{per} A[I,F\setminus C]\ \operatorname{per} A[M\setminus I,M\setminus J].$$

Every contributing permutation belongs to exactly one class. First apply the uniform bound of Lemma 4.4 to the core permanent and then separate the two cross-edge sums. Deleting a core row contributes ℓ_i, and deleting a core column contributes r_j. For fixed R, C, the outgoing cross-edge sum is exactly

$$\begin{aligned}&\sum_{\substack{J\subseteq M\\ |J|=t}} \operatorname{per} A[F\setminus R,J]\prod_{j\in J} r_j\\ &= \sum_{\phi:F\setminus R\hookrightarrow M}\ \prod_{x\in F\setminus R} A_{x,\phi(x)}r_{\phi(x)} \le \prod_{x\in F\setminus R}\sum_{j\in M} A_{xj}r_j.\end{aligned}$$

Independently, the incoming cross-edge sum is

$$\begin{aligned}&\sum_{\substack{I\subseteq M\\ |I|=t}} \operatorname{per} A[I,F\setminus C]\prod_{i\in I} \ell_i\\ &= \sum_{\psi:F\setminus C\hookrightarrow M}\ \prod_{y\in F\setminus C} A_{\psi(y),y}\ell_{\psi(y)} \le \prod_{y\in F\setminus C}\sum_{i\in M} A_{iy}\ell_i.\end{aligned}$$

Here $\hookrightarrow$ means an injection. Each injection is counted exactly once by its image set and its corresponding permanent term, with no extra factor $t!$. The image sets I, J need neither coincide nor be disjoint: they belong to the row and column copies of the core, respectively. Since every factor is nonnegative, the injectivity restrictions may be dropped. The product of the two upper bounds is

$$N^{2t}2^{-2t}\left(\prod_{x\notin R} u_x\right)\left(\prod_{y\notin C} v_y\right).$$

The permanent inside the exceptional set is at most $s!$. After division by $n!/2^n$, the full factor restoring factorials and powers of two is

$$2^s N^{2t}\frac{(N-t)!}{(N+f)!} = \left(\frac{2}{N}\right)^s \exp(O(f^2/N)).$$

Here $N+f-(N-t) = f+t = 2f-s$, so the factorial ratio contributes $N^{-(2f-s)}\exp(O(f^2/N))$. Multiplication by N^{2t} leaves exactly N^{-s}.

Let $h = |R\cap C|$. The paired product satisfies

$$\left(\prod_{x\notin R} u_x\right)\left(\prod_{y\notin C} v_y\right) \le c_n^{f-2s}L_n^{2s}(c_n/L_n^2)^h \le c_n^{f-2s}L_n^{2s}.$$

The first inequality follows by counting vertices carrying two factors, one factor or no factor; it remains valid even when $f-2s$ is negative. For $t \le f = O(\log n)$, the error in Lemma 4.4 is uniformly $o(1)$, and its factor $\Gamma \le 1$ may be discarded. Using $\binom{f}{s}^2 s! \le f^{2s}/s!$, the full sum gives

$$\frac{\operatorname{per} A_T}{n!/2^n} \le (1+o(1))e^{-1}D_N(S_M)c_n^f\exp(O(f^2/N))\exp\left(\frac{2L_n^2f^2}{c_n^2N}\right).$$

All additional exponents here are $o(1)$. More explicitly, after deleting a short core subset U, the core order is $N' = n - f - |U| \sim n$, meaning $N'/n \to 1$, while the exceptional set remains the original F. The normalized core scores are still eventually at most 0.95, their squared sum is $O(\log n)$, and the cross-edge deletion size is $t \le f = O(\log n)$. Hence the error exponent in Lemma 4.4 is

$$O\left(\frac{(\log n)^{3/2}}{\sqrt{n}} + \frac{(\log n)^2}{n}\right) = o(1),$$

The constants are independent of U, I, J. The cross-edge neighbor proportions change from those in the original graph by only $O((f + |U|)/n)$, and the total one-norm deviation of the weights remains $O(\sqrt{n \log n})$. Thus the cross-edge sum estimates above also hold uniformly. The core Gaussian factor becomes $\exp(O(|U|/n))$ times the original one: applying the Gaussian row-and-column deletion lemma to S_M/N gives logarithmic loss $O(|U|/n)$, and changing the normalization scale from N to $N - |U|$ has the same order of cost.

In the path convolution, the proportion of size-k subsets intersecting F is at most fk/n. The general permanent bound therefore bounds their full normalized short contribution by

$$K \frac{f}{\sqrt{n}} \sum_k k w_k = o(1).$$

For subsets contained in the core, restore the falling factorial and sum the positive principal minors. For every core order $N \le n$, the rank-one determinant identity gives

$$\det\left(I_N + \frac{2}{n}(I_N + A_M)\right) \le 2e \det(I_N + S_M/N).$$

To prove this, first factor out $(1 + 1/n)^N \le e$. The remaining all-ones rank-one factor is at most $1 + N/(n+1) < 2$, because the real quadratic form of the inverse of $I + S_M/(n+1)$ is at most the squared vector norm. Finally, the skew-frequency determinant product increases monotonically as the scale increases from $1/(n+1)$ to $1/N$.

Restoring the falling factorial for short subsets contributes $\exp(O(k_n^2/n)) = 1 + o(1)$. Since eventually $c_n < 1/4$, the uniform estimates above give

$$\frac{H(T)}{\mu_n} \le (1/4)^f \rho_N(S_M) + o(1) \le C_*/4 + o(1) < 1, \qquad f \ge 1.$$

Since $P(n) \ge \mu_n$, a tournament with exceptional vertices cannot maximize the path count once n is sufficiently large. A weaker bound for short convolution terms avoiding F is $(10/3)4^{-f}(1 + \varepsilon_n)$. Denote the combined contribution of the other short terms and the long terms by δ_n. Then

$$\frac{H(T)}{\mu_n} \le \frac{10}{3} 4^{-f}(1 + \varepsilon_n) + \delta_n, \qquad \varepsilon_n, \delta_n \longrightarrow 0. \tag{7.1}$$

Both errors tend to zero uniformly over the low-variance class. Take n large enough that $1 + \varepsilon_n \le 21/20$ and that the short terms meeting F and the long terms each contribute at most $1/25$. For $f \ge 1$, equation (7.1) gives

$$\frac{10}{3} \cdot \frac{1}{4} \cdot \frac{21}{20} + \frac{1}{25} + \frac{1}{25} = \frac{191}{200} < 1.$$

Either choice leaves the final upper-bound constant unchanged.

An exceptional vertex satisfies

$$|d_i^+ - (n-1)/2| > \frac{9}{20}(n-1), \quad f\frac{81}{400}(n-1)^2 \le V(T) < 16n^2 \log n,$$

Hence $f < (6400/81)(n/(n-1))^2 \log n$. For the core,

$$\max_i |s_i^M| \le 0.9(n-1) + f, \quad \frac{0.9(n-1)+f}{n-f-1} \longrightarrow 0.9,$$

$$\tau_M \le \frac{2\|s\|_2^2 + 2f^2 N}{(N-1)^2} = O(\log n) + O((\log n)^2/n) = O(\log n).$$

Since $|\ell_i - 1| \le |a_i|/(1-b)$ and $|r_i - 1| \le |a_i|/(1-b)$,

$$\sum_i (|\ell_i - 1| + |r_i - 1|) \le \frac{2}{1-b} \sum_i |a_i| \le \frac{2}{1-b} \sqrt{N \tau_M} = O(\sqrt{N \log n}).$$

This bounds the difference between the weighted and unweighted count of any neighbor subset by the total deviation above. Specifically,

$$\sum_j A_{xj} r_j \le N q_x + K\sqrt{N \log n}, \quad \sum_i A_{ix} \ell_i \le N(1-q_x) + K\sqrt{N \log n}.$$

For $q \in [0, 1/20] \cup [19/20, 1]$, we have $4q(1-q) \le 4(1/20)(19/20) = 19/100$. The $O(f/n)$ perturbation of the neighbor proportion and the perturbation $\beta = o(1)$ together give $u_x v_x \le 19/100 + o(1)$.

In the four-block expansion, $t = f - s$. There are s edges within the exceptional set, t from the exceptional set to the core, t from the core to the exceptional set, and $N - t$ within the core. Their total is

$$s + t + t + (N - t) = N + f = n.$$

There are no signs in a permanent expansion, and each permutation occurs once after classification. The full powers-of-two and factorial factor is

$$\frac{[(N-t)!/2^{N-t}]\,[N^{2t}/2^{2t}]}{(N+f)!/2^{N+f}} = 2^{f-t} N^{2t} \frac{(N-t)!}{(N+f)!} = 2^s N^{2t} \frac{(N-t)!}{(N+f)!}.$$

Since

$$\frac{(N-t)!}{(N+f)!} = \prod_{h=-t+1}^{f} \frac{1}{N+h} = N^{-(f+t)} \exp\left(- \sum_{h=-t+1}^{f} \log(1 + h/N) \right),$$

and $\sum |h|/N = O(f^2/N)$, this becomes

$$2^s N^{2t-(f+t)} e^{O(f^2/N)} = (2/N)^s e^{O(f^2/N)}.$$

If $h = |R \cap C|$, the vertices carrying both u, v number $f - 2s + h$, and those carrying just one factor number $2s - 2h$. Consequently,

$$c_n^{f-2s+h} L_n^{2s-2h} = c_n^{f-2s} L_n^{2s} (c_n/L_n^2)^h \le c_n^{f-2s} L_n^{2s}.$$

Summing over all choices of internal edges gives

$$\begin{aligned} \sum_{s=0}^{f} \binom{f}{s}^2 s! (2/N)^s c_n^{f-2s} L_n^{2s} &\le c_n^f \sum_{s=0}^{f} \frac{1}{s!} \left(\frac{2L_n^2 f^2}{c_n^2 N} \right)^s \\ &\le c_n^f \exp\left(\frac{2L_n^2 f^2}{c_n^2 N} \right). \end{aligned}$$

Below, $\#E$ and $|E|$ both denote the cardinality of a finite set E. When $k = 0$, the deleted set avoids the exceptional set, and the proportion is zero. For $1 \le k \le n$, the proportion satisfies

$$\frac{\#\{U : |U| = k, U \cap F \neq \varnothing\}}{\binom{n}{k}} \le \sum_{x \in F} \frac{\binom{n-1}{k-1}}{\binom{n}{k}} = fk/n.$$

After multiplication by the general convolution weight $K\sqrt{n}\, w_k$, the total is $Kf \sum_k k w_k/\sqrt{n} = o(1)$.

Set $Q = S_M/(n+1)$ and $v = \mathbf{1}_N$. Then

$$\det\left(I_N + \frac{2}{n}(I_N + A_M)\right) = (1+1/n)^N \det(I_N + Q)\left[1 + \frac{1}{n+1} v^{\mathsf{T}}(I_N + Q)^{-1} v\right].$$

Using $(I+Q)^{-1} = (I-Q)(I-Q^2)^{-1}$ and the vanishing of a skew-symmetric quadratic form,

$$v^{\mathsf{T}}(I+Q)^{-1}v = v^{\mathsf{T}}(I + Q^{\mathsf{T}}Q)^{-1}v \le \|v\|_2^2 = N.$$

Also, $(1+1/n)^N \le e^{N/n} \le e$, and

$$\det(I + S_M/(n+1)) = \prod_j \left(1 + \frac{\lambda_j^2}{(n+1)^2}\right) \le \prod_j \left(1 + \frac{\lambda_j^2}{N^2}\right).$$

The three factors are at most e, $\det(I + S_M/N)$, and 2, respectively. Multiplication by the permanent factor $e^{-1}/2$ leaves $\rho_N(S_M)$. The constants in the exceptional-set estimate satisfy

$$\frac{10}{3}\frac{1}{4}\frac{21}{20} = \frac{7}{8} = \frac{175}{200}, \qquad \frac{1}{25} + \frac{1}{25} = \frac{16}{200}, \qquad \frac{175+16}{200} = \frac{191}{200}.$$

Suppose that, after deleting the exceptional vertices and a short set, the distance of every exceptional vertex's core-neighbor proportion from $[0, 1/20] \cup [19/20, 1]$ is at most ϵ_n, where $\epsilon_n = O((f + k_n)/n)$. On $q \in [0,1]$, the derivative of $q(1-q)$ has absolute value at most one. Thus

$$q_x(1-q_x) \le \frac{19}{400} + \epsilon_n.$$

Using the same nonnegative weight deviation $\beta_n = O(\sqrt{\log n/n})$,

$$\begin{aligned} u_x v_x &= 4(q_x + \beta_n)(1 - q_x + \beta_n) \\ &= 4q_x(1-q_x) + 4\beta_n + 4\beta_n^2 \\ &\le \frac{19}{100} + 4\epsilon_n + 4\beta_n + 4\beta_n^2 =: c_n. \end{aligned}$$

Moreover, $u_x, v_x \le 2 + 2\beta_n =: L_n$. Since $\epsilon_n, \beta_n \to 0$, choose a uniform threshold such that

$$4\epsilon_n + 4\beta_n + 4\beta_n^2 < \frac{6}{100}, \qquad c_n < \frac{1}{4}.$$

Each term in the additional exponent above also tends to zero:

$$\frac{f^2}{N} = O((\log n)^2/n) \to 0, \quad \frac{2L_n^2 f^2}{c_n^2 N} = O((\log n)^2/n) \to 0.$$

For $\tau \le A \log n, t \le B \log n$, the error terms in Lemma 4.4 are bounded respectively by constant multiples of

$$\frac{\log n + (\log n)^2}{n}, \quad \frac{(\log n)^{3/2}}{\sqrt{n}}, \quad \sqrt{\frac{\log n}{n}}, \quad \frac{\log n + 1}{n}, \quad \frac{(\log n)^{3/2}}{\sqrt{n}}, \quad \frac{(\log n)^2}{n}.$$

The constants are independent of the tournament and deletion sets, so the resulting $o(1)$ error is uniform.

7.4 The score penalty in the remaining class

The remaining tournaments satisfy

$$\max_i |a_i| \le 0.9, \qquad a = s/(n-1), \qquad \tau = \sum_i a_i^2 \le K \log n.$$

Fix an absolute constant K_0 such that $\tau \le K_0 \log n$. The paired product satisfies

$$\Gamma = \prod_i (1 - a_i^2) \le e^{-\tau}.$$

Apply Lemma 4.4 to the whole tournament. The score product is common to every short term. Retaining it outside the sum absorbs errors that, individually, may exceed $1/n$.

For a principal deleted set of size $k \le k_n$, denoted by U, Lemma 4.4 gives

$$\operatorname{per} A[U^c] \le e^{-1} D_n(S) \Gamma \left(\prod_{i \in U} g_i \right) \frac{(n-k)!}{2^{n-k}} \exp(R_\tau + R_{\tau,k}),$$

$$g_i = (1 - a_i^2)^{-1} \le G := 100/19,$$

$$R_\tau \le K[\mathcal{M}_\tau + \sqrt{\tau/n} + 1/n], \qquad R_{\tau,k} \le K[k\sqrt{\tau/n} + (k + k^2)/n].$$

When accumulating errors, we must retain the common factor Γ. Define the nonnegative coefficients

$$c_k = (2/n)^k \sum_{|U|=k} \det(I + A[U]) \prod_{i \in U} g_i.$$

They satisfy $c_0 = 1$ and $c_k \le G^k w_k$. Restoring the falling factorial produces

$$\log \frac{n^k}{(n)_k} = O(k^2/n), \qquad k \le k_n.$$

Set $\delta_n = \sqrt{\tau/n} + 1/n$ and $r_k = R_{\tau,k} + \log(n^k/(n)_k)$. Taking a nonnegative error bound, uniformly for $\tau \le K_0 \log n$ we have $0 \le r_k \le K\delta_n(k + k^2)$ and $\max_{k \le k_n} r_k = o(1)$. Therefore

$$\sum_{k \le k_n} c_k e^{r_k} \le \sum_{k=0}^{n} c_k + K\delta_n \sum_{k \ge 0} G^k w_k (k + k^2) \le \left(\sum_{k=0}^{n} c_k \right) e^{K\delta_n}.$$

The last step uses $\sum_k c_k \ge 1$ and convergence of the corresponding fixed moments. The normalized short path sum is consequently at most

$$\frac{e^{-1} D_n(S) \Gamma}{2} e^{R_\tau + K\delta_n} \det\left(I + \frac{2}{n} \operatorname{diag}(g)(I + A) \right).$$

The total error is controlled by the finite moments of $G^k w_k$, rather than by the maximum deleted-set size k_n. This avoids an extra logarithmic factor in the final error.

We now remove the weights while preserving the score penalty. Set $W = I + (2/n)(I + A)$ and $\Delta = (2/n)\operatorname{diag}(g - 1)(I + A)$. The symmetric part of W is at least I, so $\|W^{-1}\|_{\mathrm{op}} \le 1$. Each row of $I + A$ has Euclidean norm at most $\sqrt{n}$, and $\sum_i (g_i - 1) \le K\tau$. Decomposing Δ as a sum of rank-one row matrices therefore controls its trace norm, the sum of its singular values. Using $\|uv^{\mathsf{T}}\|_* = \|u\|_2 \|v\|_2$, we obtain

$$\|\Delta\|_* \le K\tau/\sqrt{n}.$$

The determinant inequality $|\det(I+E)| \le \exp(\|E\|_*)$ gives

$$\det(W+\Delta) \le \det(W) e^{K\tau/\sqrt{n}}.$$

Both determinants are positive: the symmetric part of W is positive definite, and the weighted determinant has a principal-minor expansion with positive coefficients. The unweighted rank-one bound of Section 7.3, with $N = n$, is $\det(W) \le 2e \det(I + S/n)$.

Combining the factors above and adding only the long-deletion tail at the end yields the uniform score-sensitive upper bound

$$\frac{H(T)}{\mu_n} \le \rho_n(S) \exp\left\{-\tau + K\left[\frac{1+\tau+\tau^2}{n} + \frac{\tau^{3/2}+\tau}{\sqrt{n}} + \sqrt{\tau/n}\right]\right\} + o(1/n). \tag{7.2}$$

For $\tau \le K_0 \log n$ and sufficiently large order, the polynomial error terms other than the constant $1/n$ consume at most $\tau/4$ in total. Young's inequality controls the remaining square-root term:

$$K\sqrt{\tau/n} \le \tau/4 + K^2/n.$$

The total exponent in equation (7.2) is therefore at most $-\tau/2 + K/n$. Applying the spectral cap gives

$$H(T)/\mu_n \le C_* e^{K/n} + o(1/n) = C_* + O(1/n).$$

Since $g_i - 1 = a_i^2/(1-a_i^2)$,

$$\sum_i (g_i - 1) \le \frac{\tau}{1-0.9^2} = \frac{100}{19}\tau, \qquad c_k \le (2/n)^k \binom{n}{k} h_k G^k \le G^k w_k.$$

For $k \le k_n = o(\sqrt{n})$,

$$0 \le \log \frac{n^k}{(n)_k} = \sum_{h=0}^{k-1} -\log(1-h/n) \le \frac{k(k-1)}{2n(1-k_n/n)} \le Kk^2/n.$$

When $0 \le r_k \le 1$, $e^{r_k} - 1 = \int_0^{r_k} e^u\, du \le e r_k$, so

$$\sum_{k \le k_n} c_k (e^{r_k} - 1) \le K\delta_n \sum_{k \ge 0} G^k w_k (k + k^2) = K'\delta_n.$$

Write $C = \sum_{k=0}^n c_k \ge 1$. Then $C + K'\delta_n \le C(1 + K'\delta_n) \le C e^{K'\delta_n}$. The weighted principal-minor expansion gives, term by term,

$$\sum_{k=0}^n c_k = \sum_U (2/n)^{|U|} \det(I + A[U]) \prod_{i \in U} g_i = \det(I + (2/n)\operatorname{diag}(g)(I + A)).$$

Write r_i^{T} for row i of $I + A$, and let e_i be the ith standard basis vector. Then

$$\Delta = \frac{2}{n}\sum_i (g_i - 1) e_i r_i^{\mathsf{T}}, \quad \|\Delta\|_* \le \frac{2}{n}\sum_i (g_i - 1)\|e_i\|_2 \|r_i\|_2 \le \frac{2}{\sqrt{n}} \sum_i (g_i - 1) \le K\tau/\sqrt{n}.$$

If the singular values of E are $\sigma_j(E)$, then

$$|\det(I+E)| \le \prod_j (1 + \sigma_j(E)) \le \exp\left(\sum_j \sigma_j(E)\right) = e^{\|E\|_*}.$$

Use $W + \Delta = W(I + W^{-1}\Delta)$ and $\|W^{-1}\|_{\mathrm{op}} \le 1$ to obtain the stated comparison.

Finally, set

$$E_n(\tau) = K\left[\frac{\tau + \tau^2}{n} + \frac{\tau^{3/2} + \tau}{\sqrt{n}}\right].$$

For $0 \le \tau \le K_0 \log n$,

$$E_n(\tau) \le \tau K\left[\frac{1 + K_0 \log n}{n} + \frac{\sqrt{K_0 \log n} + 1}{\sqrt{n}}\right] \le \tau/4$$

uniformly for all sufficiently large n. Completing the square gives

$$0 \le \left(\frac{\sqrt{\tau}}{2} - \frac{K}{\sqrt{n}}\right)^2 = \frac{\tau}{4} - K\sqrt{\tau/n} + \frac{K^2}{n},$$

The total exponent is therefore at most $-\tau + \tau/4 + \tau/4 + (K + K^2)/n$. Also,

$$0 \le e^{K'/n} - 1 = \int_0^{K'/n} e^u \, du \le K' e^{K'}/n,$$

This yields $C_* + O(n^{-1})$, with a constant entirely independent of τ and of the tournament.

7.5 Completion of the upper bound

The three classes cover all tournaments. High-variance tournaments have normalized path count $o(1/n)$; low-variance tournaments containing exceptional vertices have count strictly below one; and every remaining tournament satisfies the bound $C_* + O(1/n)$ just proved. Hence

$$P(n) \le \big(C_* + O(1/n)\big)\mu_n.$$

The threshold in each case is independent of the particular tournament. Taking a common threshold proves the upper bound in Proposition 7.1.

8. The small-score formula and the lower bound

Theorem 8.1 (Small-score path approximation).

There is an absolute constant C with the following property. For every nonnegative function satisfying $d(n)/\sqrt{n} \to 0$, denoted by $d = d(n)$, there is a threshold $n_0(d)$ such that every tournament of order $n \ge n_0(d)$ with $\|S\mathbf{1}\|_\infty \le d(n)$ satisfies

$$\left|\frac{H(T)}{\mu_n} - \rho_n(S)\right| \le C\frac{(d(n)+1)^2}{n}.$$

8.1 Proof of the small-score approximation

Proof.

Write $d = d(n)$. Apply Lemma 4.5 to every principal deleted set $|U| = k \le k_n$. Since $d + k_n + 1 = o(\sqrt{n})$, it applies uniformly over all these subsets. After restoring the falling factorial $(n)_k$, the positive path convolution gives

$$\frac{H(T)}{\mu_n} = \frac{e^{-1}D_n(S)}{2}\sum_{k \le k_n}\left(\frac{2}{n}\right)^k \sum_{|U|=k} \det(I + A[U]) + O\left(\frac{(d+1)^2}{n}\right).$$

This error rate has no extra logarithmic factor, because the error is at most a fixed constant times

$$\frac{1}{n}\sum_{k\ge 0} w_k(d+k+1)^2 = O\left(\frac{(d+1)^2}{n}\right).$$

Restoring the falling factorial contributes $O(n^{-1}\sum k^2 w_k)$, and Section 5.1 bounds the terms with large deleted sets by $o(n^{-1})$. Extending the short generating sum above to all k costs $n^{-2+o(1)}$, because its coefficients are bounded by w_k. Thus

$$\frac{H(T)}{\mu_n} = \frac{e^{-1}D_n(S)}{2}\det\left(I+\frac{2}{n}(I+A)\right) + O\left(\frac{(d+1)^2}{n}\right).$$

Set $u = \mathbf{1}/\sqrt{n}$ and $Q = S/(n+1)$. The rank-one determinant lemma gives

$$\det\left(I+\frac{2}{n}(I+A)\right) = (1+1/n)^n \det(I+Q)\left(1+\frac{n}{n+1}u^{\mathsf{T}}(I+Q)^{-1}u\right).$$

Skew-symmetry gives

$$u^{\mathsf{T}}(I+Q)^{-1}u = u^{\mathsf{T}}(I-Q^2)^{-1}u,$$

The absolute difference of this quantity from one is at most

$$\|Qu\|_2^2 \le \frac{d^2}{(n+1)^2}.$$

Also, $(1+1/n)^n = e(1+O(n^{-1}))$. Replacing $S/(n+1)$ by S/n incurs a logarithmic determinant error $O(n^{-1})$, by the bounded sum of the squared normalized frequencies. Consequently,

$$\det\left(I+\frac{2}{n}(I+A)\right) = 2e\det(I+S/n)\left(1+O(n^{-1}+d^2/n^2)\right).$$

Substitution proves the theorem. All estimates use the same absolute constant; only the threshold beyond which the small-score condition holds depends on d.

For every k, using $d+k+1 \le (d+1)(k+1)$ gives

$$\frac{1}{n}\sum_k w_k(d+k+1)^2 \le \frac{(d+1)^2}{n}\sum_k w_k(k+1)^2.$$

Thus using k_n as a replacement for k is unnecessary, and no $\log n$ loss occurs. For $Q^{\mathsf{T}} = -Q$,

$$(I+Q)^{-1} = (I-Q)(I-Q^2)^{-1}, \qquad Q(I-Q^2)^{-1} \text{ is skew-symmetric,}$$

$$\begin{aligned} 1 - u^{\mathsf{T}}(I+Q)^{-1}u &= u^{\mathsf{T}}[I-(I+Q^{\mathsf{T}}Q)^{-1}]u \\ &= (Qu)^{\mathsf{T}}(I+QQ^{\mathsf{T}})^{-1}(Qu) \le \|Qu\|_2^2. \end{aligned}$$

Since $Qu = s/[(n+1)\sqrt{n}]$,

$$\|Qu\|_2^2 = \frac{\sum_i s_i^2}{n(n+1)^2} \le \frac{d^2}{(n+1)^2}.$$

The determinant error from changing the normalization satisfies

$$\begin{aligned}0 &\le \log\det(I+S/n) - \log\det(I+S/(n+1)) \\ &= \sum_j \int_{\lambda_j^2/(n+1)^2}^{\lambda_j^2/n^2} \frac{du}{1+u} \\ &\le \sum_j \lambda_j^2 \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \\ &= \frac{n(n-1)}{2}\frac{2n+1}{n^2(n+1)^2} = O(n^{-1}).\end{aligned}$$

and

$$0 \le 1 - n\log(1+1/n) = n\int_0^{1/n} \frac{u}{1+u}\,du \le \frac{1}{2n}.$$

The difference between the rank-one factor and two is

$$\left|1 + \frac{n}{n+1}u^{\mathsf{T}}(I+Q)^{-1}u - 2\right| \le \frac{1}{n+1} + \frac{d^2}{(n+1)^2}.$$

Multiplying in order and using $D_n(S)\det(I+S/n) \le C_*$ gives the absolute error in Theorem 8.1. □

8.2 Carousel tournaments

Corollary 8.2 (Carousel lower bound).

For the odd-order carousel tournaments and even-order one-vertex-deletion constructions defined below,

$$H(\mathrm{Car}_n)/\mu_n = L + O(n^{-1}).$$

Proof.

For odd n, take the residue classes modulo n and define the carousel tournament Car_n as follows: for $1 \le j \le (n-1)/2$, direct $i \to i+j$. It is regular, so $d = 0$. For even n, let Car_n be obtained by deleting one vertex from Car_{n+1}. Then $d = 1$.

The sign matrix of an odd-order carousel is similar to T_n^0 by a signed permutation. Explicitly, use the diagonal signs $(-1)^i$, $0 \le i < n$, to conjugate T_n^0, and reorder the indices as $0, 2, \ldots, n-1, 1, 3, \ldots, n-2$. When two indices have the same parity, the switched edge follows increasing index order; when they have different parities, its direction is reversed. Reading indices in the stated cyclic order gives the carousel orientation. Restricting the inverse signed conjugation and reordering after a one-vertex deletion gives T_n^0 on the remaining ordered indices. Thus the same signed spectral calculation remains valid after any one vertex is deleted, and both parity cases satisfy

$$\rho_n(S_{\mathrm{Car}_n}) = r_n := \frac{(n+1)^n + (n-1)^n}{(n+i)^n + (n-i)^n}.$$

Divide numerator and denominator by n^n. At each fixed $z = 1, -1, i, -i$, use the Taylor expansion of $(1+z/n)^n$. This gives

$$r_n = \frac{\cosh 1}{\cos 1} + O(n^{-1}) = L + O(n^{-1}).$$

Both parities therefore have $H(\mathrm{Car}_n)/\mu_n = L + O(n^{-1})$, and hence

$$P(n) \ge (L - O(n^{-1}))\mu_n.$$

The transition from signed similarity to the path count uses Theorem 8.1, with $d = 0$ and $d = 1$, respectively. Let the constants and thresholds for the lower and upper bounds be K_l, N_l and K_u, N_u. Taking $K = K_l + K_u$ and $n_0 = \max(N_l, N_u, 2)$ proves Proposition 7.1.

Write $n = 2m + 1$, with original indices $i = 0, \ldots, 2m$. Let $D_{ii} = (-1)^i$. For $i < j$,

$$(DT_n^0 D)_{ij} = (-1)^{i+j}.$$

Use the reordering $p_k = 2k \pmod n$, represented in $0, \ldots, n-1$. It gives the sequence $0, 2, \ldots, 2m, 1, 3, \ldots, 2m-1$. If i, j have the same parity and their positive difference is $2h$, then

$$2^{-1}(j-i) \pmod n = h \in \{1, \ldots, m\}.$$

If their parities differ and $i < j$, write $j - i = 2h + 1$. Then

$$2^{-1}(j-i) \pmod n = h + m + 1 \in \{m+1, \ldots, 2m\}.$$

In the new indices, edges with differences $1, \ldots, m$ point forward, and those with differences $m+1, \ldots, 2m$ point backward, exactly as in the carousel tournament. A one-vertex deletion selects the corresponding principal submatrix. The restricted D is still a diagonal sign matrix, and the transitive principal submatrix on the remaining ordered indices is still T_n^0. This proves that both parity constructions have the stated spectrum; it does not assert that signed transformations preserve the path count.

By $D_n(S) = \det(I + iS/n)^{-1}$ and the principal-minor polynomial,

$$\begin{aligned}\rho_n(S) &= \frac{\det(I + S/n)}{\det(I + iS/n)} \\ &= \frac{(1+1/n)^n + (1-1/n)^n}{(1+i/n)^n + (1-i/n)^n} = r_n.\end{aligned}$$

For $z \in \{1, -1, i, -i\}$ and $n \geq 2$, the power series for the principal logarithm on $|w| < 1$ gives

$$\begin{aligned}n \log(1 + z/n) &= z - \frac{z^2}{2n} + R_{n,z}, \\ |R_{n,z}| &\leq \sum_{h=3}^{\infty} \frac{1}{hn^{h-1}} \leq \frac{1}{3n^2(1-1/n)} \leq \frac{2}{3n^2}.\end{aligned}$$

Exponentiating yields $(1 + z/n)^n = e^z(1 - z^2/(2n) + O(n^{-2}))$. Adding the terms in pairs gives

$$\begin{aligned}(1+1/n)^n + (1-1/n)^n &= 2\cosh 1\,[1 - 1/(2n) + O(n^{-2})], \\ (1+i/n)^n + (1-i/n)^n &= 2\cos 1\,[1 + 1/(2n) + O(n^{-2})].\end{aligned}$$

Because $\cos 1 > 0$, the denominator is bounded below by a fixed positive constant for all sufficiently large n. Using

$$\frac{1}{1+u} = 1 - u + \frac{u^2}{1+u},$$

we obtain the more precise spectral ratio

$$r_n = L[1 - 1/n + O(n^{-2})].$$

Theorem 8.1 gives only an $O(n^{-1})$ error for the path count itself. Thus $-L/n$ here is a correction to the spectral ratio and cannot directly be taken as the correction to the path count. Finally,

$$H(\mathrm{Car}_n)/\mu_n = r_n + O(n^{-1}) = L + O(n^{-1}),$$

there is therefore a fixed K_l such that $P(n) \geq (L - K_l/n)\mu_n$. □

9. The Hessian of the original angular function

9.1 The angular function and the Hessian formula

We continue to write $I_r, J_r, \mathbf{1}_r$ for the identity matrix, the all-one matrix and the all-one column vector, omitting subscripts when their dimensions are clear.

Definition 9.1 (Sign matrix and original angular function).

Let $d \geq 3$ be odd and let A be the adjacency matrix of a regular tournament of order d.

$$S = 2A + I - J, \quad S^T = -S, \quad S\mathbf{1} = 0, \qquad C = S/d, \quad P_0 = J/d.$$

We write $\mathrm{diag}(a_1, \ldots, a_r)$ for the diagonal matrix with diagonal entries $a_1, \ldots, a_r$.

Set

$$B(y) = I - (A + \tfrac{1}{2}I)\,\mathrm{diag}(y_1, \ldots, y_d), \quad K(y) = \det B(y), \quad z = 2/d.$$

Complex logarithms use the local analytic branches specified below and agree with the real logarithm on the positive axis.

Near $y = z\mathbf{1}$, solve $K(y) = 0$ for $y_d = h(y_1, \ldots, y_{d-1})$. Wormald's angular function is

$$\widetilde{f}(\theta) = \log h(ze^{i\theta_1}, \ldots, ze^{i\theta_{d-1}})$$

Write e_j for the jth standard basis column vector.

Write $\mathbf{H} = \nabla^2 \widetilde{f}(0)$ and $\mathcal{H} = \det \mathbf{H}$. Throughout this section, $C = S/d$ denotes a matrix.

Theorem 9.2 (The Hessian formula).

The function h and the chosen logarithm branch are analytic in a neighborhood of the specified point, and

$$\mathbf{H} = E^T(I + C^2)(I - C^2)^{-1}E, \qquad E = (e_1 - e_d, \ldots, e_{d-1} - e_d).$$

Consequently,

$$\mathcal{H} = d\,\frac{\det(I + C^2)}{\det(I - C^2)} > 0, \qquad \sqrt{\frac{d}{\mathcal{H}}} = \frac{\det(I + C)}{\det(I + iC)} =: \rho_d(S).$$

The right-hand side is positive and rational, so these identities prove Conjecture 6.

9.2 Spectral gap, the constant direction and restricted determinants

Proof.

By the tournament identities and regularity,

$$A + A^T = J - I, \qquad A\mathbf{1} = \frac{d-1}{2}\mathbf{1},$$

$$S^T = 2A^T + I - J = 2(J - I - A) + I - J = -S,$$

$$S\mathbf{1} = 2\frac{d-1}{2}\mathbf{1} + \mathbf{1} - d\mathbf{1} = 0.$$

We first bound the spectral radius of C.

For a real vector $v = (v_1, \dots, v_r)^T$, its Euclidean norm is $\|v\|_2 = (\sum_{j=1}^r v_j^2)^{1/2}$.

For a real matrix M, the two norms are

$$\|M\|_F^2 = \sum_{j,k} M_{jk}^2, \qquad \|M\|_{\text{op}} = \sup_{v \neq 0} \frac{\|Mv\|_2}{\|v\|_2}.$$

Since the $d(d-1)$ off-diagonal entries s_{ij} of S all belong to $\{\pm 1\}$,

$$\|S\|_F^2 = \sum_{i \neq j} s_{ij}^2 = d(d-1), \qquad \|C\|_F^2 = \frac{d-1}{d}.$$

Apply a real orthogonal basis change to the normalized skew-symmetric matrix C. Its canonical form consists of zero blocks and two-dimensional blocks

$$J_\lambda = \begin{pmatrix} 0 & \lambda \\ -\lambda & 0 \end{pmatrix}$$

Each two-dimensional block contributes $2\lambda^2$ to the squared Frobenius norm. Therefore every frequency satisfies

$$2\lambda^2 \leq \sum_{\text{all two-dimensional blocks}} 2\lambda_j^2 = \|C\|_F^2 = \frac{d-1}{d},$$

$$\|C\|_{\text{op}}^2 = \max_j \lambda_j^2 \leq \frac{d-1}{2d} < \frac{1}{2}.$$

Put $u = \mathbf{1}/\sqrt{d}$, $P_0 = uu^T = J/d$ and $\Pi = I - P_0$. Then

$$u^T u = 1, \quad P_0^2 = P_0, \quad \Pi^2 = I - 2P_0 + P_0^2 = \Pi,$$

$$Cu = 0, \qquad u^T C = -(Cu)^T = 0, \qquad CP_0 = (Cu)u^T = 0, \quad P_0 C = u(u^T C) = 0.$$

We also have

$$P := \frac{2A + I}{d} = P_0 + C, \qquad Pu = P_0 u + Cu = u.$$

If $u^T v = 0$, then $P_0 v = u(u^T v) = 0$, and hence $Pv = Cv$. At the critical point,

$$B(z\mathbf{1}) = I - P = \Pi - C, \qquad B(z\mathbf{1})u = 0.$$

The space $u^\perp$ is orthogonal to the real vector u. The notation $\det_{u^\perp}$ means the determinant of the restriction to this invariant subspace, computed in an orthonormal basis.

On $u^\perp$ this matrix is $I - C$. If $x \in u^\perp$ satisfies $(I - C)x = 0$, then

$$0 = x^T(I - C)x = \|x\|_2^2 - x^T C x = \|x\|_2^2,$$

where $x^T C x = -x^T C x$ implies $x^T C x = 0$. Thus $x = 0$, and the restricted matrix is invertible.

On a two-dimensional block,

$$\det(I - J_\lambda) = \det \begin{pmatrix} 1 & -\lambda \\ \lambda & 1 \end{pmatrix} = 1 + \lambda^2 > 0.$$

The eigenvalue of $I - C$ in the constant direction is 1. Therefore

$$q := \det_{u^\perp}(I - C) = \det(I - C) = \prod_j (1 + \lambda_j^2) > 0.$$

Taking transposes gives

$$\det(I + C) = \det((I + C)^T) = \det(I - C) = q.$$

9.3 Logarithmic variables, block matrices and Schur elimination

To differentiate the angular function, first introduce logarithmic variables $t \in \mathbb{C}^d$ near the origin and set $y_j = ze^{t_j}$. Then

$$\mathcal{B}(t) = B(ze^{t_1}, \ldots, ze^{t_d}) = I - P\,\mathrm{diag}(e^{t_1}, \ldots, e^{t_d}).$$

Choose an orthonormal basis of $u^\perp$ as the columns of V. The matrix $O = (u, V)$ satisfies

$$V^T V = I_{d-1}, \quad V^T u = 0, \quad VV^T = \Pi, \quad O^T O = OO^T = I_d.$$

In this basis write

$$O^T \mathcal{B}(t) O = \begin{pmatrix} a(t) & b(t) \\ c(t) & D(t) \end{pmatrix},$$

$$a = u^T \mathcal{B} u, \quad b = u^T \mathcal{B} V, \quad c = V^T \mathcal{B} u, \quad D = V^T \mathcal{B} V.$$

Here a is a scalar, b and c have dimensions $1 \times (d-1)$ and $(d-1) \times 1$, and D has order $d - 1$. Skew symmetry also gives

$$u^T P = u^T (P_0 + C) = u^T u u^T + u^T C = u^T + 0 = u^T.$$

Since $\det(O)^2 = 1$, this basis change preserves the determinant. At $t = 0$,

$$a(0) = 0, \quad b(0) = 0, \quad c(0) = 0, \qquad D(0) = V^T (I - C) V.$$

By Section 9.2, $\det D(0) = q > 0$. Hence $D(t)$ is invertible in a neighborhood of the origin.

Let $\mathrm{adj}\, D$ be the adjugate matrix, with $(\mathrm{adj}\, D)_{ij} = (-1)^{i+j} \det D^{(j|i)}$, where $D^{(j|i)}$ deletes row j and column i. Expansion of the determinant gives $D\,\mathrm{adj}\, D = (\det D) I$, and hence

$$D(t)^{-1} = \frac{\mathrm{adj}\, D(t)}{\det D(t)}$$

This also shows that the entries of $D(t)^{-1}$ are analytic in this neighborhood.

Block elimination gives

$$\begin{pmatrix} 1 & -bD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} a & b \\ c & D \end{pmatrix} = \begin{pmatrix} a - bD^{-1}c & 0 \\ c & D \end{pmatrix}.$$

The left multiplier has determinant 1, and the resulting matrix is block triangular. Defining

$$\psi(t) = a(t) - b(t) D(t)^{-1} c(t)$$

therefore gives the exact identity

$$\det \mathcal{B}(t) = \det D(t)\, \psi(t).$$

On this neighborhood, $\det \mathcal{B}(t) = 0$ and $\psi(t) = 0$ are equivalent because the factor $\det D(t)$ is nonzero.

9.4 First and second derivatives

For a scalar function $F(t_1, \ldots, t_d)$, subscripts F_j, F_{jk} denote $\partial F/\partial t_j$ and $\partial^2 F/(\partial t_j \partial t_k)$; matrix, row and column derivatives are taken entry by entry. When differentiating in the original variables y, we write K_{y_j}. The matrix $\nabla^2 F$ consists of the entries F_{jk}.

The symbol δ_{jk} is 1 when $j = k$ and 0 otherwise.

Write $F_j = e_j e_j^T$ and $R(t) = D(t)^{-1}$. Only column j of the exponential matrix depends on t_j, so at the origin

$$\mathcal{B}_j = -PF_j, \qquad \mathcal{B}_{jk} = -\delta_{jk} PF_j.$$

Differentiating $DR = I$ twice gives

$$D_j R + DR_j = 0, \qquad R_j = -RD_j R,$$

$$R_{jk} = RD_k RD_j R - RD_{jk} R + RD_j RD_k R.$$

These identities hold throughout the neighborhood where D is invertible.

Using $u^T P = u^T$ and $e_j^T u = 1/\sqrt{d}$,

$$a_j(0) = -u^T PF_j u = -u_j^2 = -1/d, \qquad a_{jk}(0) = -\delta_{jk}/d.$$

The product rule gives

$$\partial_j (bRc) = b_j Rc + bR_j c + bRc_j.$$

Differentiating once more with respect to t_k, we obtain

$$\begin{aligned} \partial_k \partial_j (bRc) = {} & b_{jk} Rc + b_j R_k c + b_j Rc_k \\ & + b_k R_j c + bR_{jk} c + bR_j c_k \\ & + b_k Rc_j + bR_k c_j + bRc_{jk}. \end{aligned}$$

At the origin, $b = c = 0$, so the seven terms containing an undifferentiated factor b or c vanish. Thus

$$\partial_k \partial_j (bRc)(0) = b_j(0) R(0) c_k(0) + b_k(0) R(0) c_j(0).$$

We now calculate these factors:

$$b_j(0) = -u^T Pe_j e_j^T V = -\frac{1}{\sqrt{d}} e_j^T V,$$

$$c_k(0) = -V^T Pe_k e_k^T u = -\frac{1}{\sqrt{d}} V^T Ce_k,$$

Here $V^T P = V^T C$ follows from $V^T u = 0$. Put $U = (I - C)^{-1}$. Since C preserves $u^\perp$, we have $(I - C)V = VD(0)$. Multiplying on the left by U and on the right by $R(0)$, and using $\Pi C = C$, gives

$$(I - C)V = VD(0), \qquad UV = VR(0),$$

$$UC = U\Pi C = UVV^TC = VR(0)V^TC.$$

Consequently,

$$VR(0)V^TC = UC.$$

The two second-order products are therefore

$$b_jR(0)c_k = \frac{1}{d}e_j^TUCe_k = \frac{(UC)_{jk}}{d},$$

$$b_kR(0)c_j = \frac{1}{d}e_k^TUCe_j = \frac{(UC)_{kj}}{d}.$$

The first derivative of the product vanishes at the origin. Thus

$$\psi_j(0) = -1/d, \qquad \psi_{jk}(0) = -\frac{\delta_{jk} + (UC)_{jk} + (UC)_{kj}}{d}.$$

From $C(I-C) = (I-C)C$, multiplying on the left and right by the inverse gives $CU = UC$. Also,

$$(UC)^T = C^TU^T = -C(I+C)^{-1}.$$

The difference of the two inverses expands as

$$\begin{aligned}(I-C)^{-1} - (I+C)^{-1} &= (I-C)^{-1}[(I+C) - (I-C)](I+C)^{-1} \\ &= 2C(I-C)^{-1}(I+C)^{-1} \\ &= 2C(I-C^2)^{-1}.\end{aligned}$$

It follows that

$$\begin{aligned}I + UC + (UC)^T &= I + C[(I-C)^{-1} - (I+C)^{-1}] \\ &= I + 2C^2(I-C^2)^{-1} \\ &= (I+C^2)(I-C^2)^{-1} =: Q.\end{aligned}$$

Thus the first and second derivatives of ψ at the origin are

$$\psi_j(0) = -1/d, \qquad \nabla^2\psi(0) = -Q/d.$$

Differentiate the determinant product. Since $\psi(0) = 0$ and $\det D(0) = q$,

$$\begin{aligned}\partial_{t_j} \det \mathcal{B}(0) &= \partial_{t_j}(\det D)(0)\,\psi(0) + \det D(0)\psi_j(0) \\ &= 0 + q(-1/d) = -q/d.\end{aligned}$$

At the origin $\partial y_j/\partial t_j = z$, so

$$zK_{y_j}(z\mathbf{1}) = -q/d, \qquad K_{y_j}(z\mathbf{1}) = -\frac{q}{dz} = -q/2 \neq 0.$$

In particular, $K_{y_d}(z\mathbf{1}) \neq 0$, so y_d can be eliminated locally.

9.5 Eliminating the last variable

Write $y' = (y_1, \dots, y_{d-1})$ and $t' = (t_1, \dots, t_{d-1})$.

The determinant is linear in column d, and this column is affine in y_d. Hence

$$K(y) = K_0(y_1, \dots, y_{d-1}) + y_d K_1(y_1, \dots, y_{d-1}).$$

where

$$K_0(y') = K(y', 0), \qquad K_1(y') = K(y', 1) - K(y', 0) = K_{y_d}(y', y_d).$$

The last partial derivative does not depend on y_d, because the displayed expression is affine in y_d.

The derivative calculation in Section 9.4 gives

$$K_1(z, \dots, z) = K_{y_d}(z\mathbf{1}) = -q/2 \neq 0.$$

On a sufficiently small neighborhood, $K_1(y') \neq 0$, and therefore

$$h(y') = -\frac{K_0(y')}{K_1(y')}.$$

Since $K(z\mathbf{1}) = 0$,

$$K_0(z, \dots, z) + zK_1(z, \dots, z) = 0, \qquad h(z, \dots, z) = z.$$

Define

$$g(t') = \log\left[\frac{h(ze^{t_1}, \dots, ze^{t_{d-1}})}{z}\right].$$

Since $h(z, \dots, z) = z > 0$, we may shrink the neighborhood so that $|h(ze^{t'})/z - 1| < 1/2$, with exponentiation taken coordinatewise. On this disk, choose the analytic logarithm with $\log 1 = 0$. Then

$$g(0) = \log 1 = 0, \qquad t_d = g(t'), \qquad \psi(t', g(t')) = 0.$$

The resulting function g is analytic and satisfies the required implicit equation near the origin.

9.6 Implicit differentiation and angular variables

Let $1 \leq j, k \leq d-1$. Differentiate the identity $\psi(t', g(t')) = 0$, then evaluate at $t' = 0$ using Section 9.4. The first derivative gives

$$0 = \psi_j + \psi_d g_j, \qquad g_j(0) = -\frac{\psi_j(0)}{\psi_d(0)} = -\frac{-1/d}{-1/d} = -1.$$

Differentiating again with respect to t_k gives five terms:

$$0 = \psi_{jk} + \psi_{jd} g_k + \psi_{dk} g_j + \psi_{dd} g_j g_k + \psi_d g_{jk}.$$

At the origin, substitute the derivatives from Section 9.4 and $g_j(0) = g_k(0) = -1$ to obtain

$$0 = -\frac{Q_{jk}}{d} + \frac{Q_{jd}}{d} + \frac{Q_{dk}}{d} - \frac{Q_{dd}}{d} - \frac{g_{jk}}{d}.$$

Multiply by d and isolate the last derivative:

$$\begin{aligned} g_{jk}(0) &= -Q_{jk} + Q_{jd} + Q_{dk} - Q_{dd} \\ &= -(e_j - e_d)^T Q (e_k - e_d). \end{aligned}$$

Choose the local branch of $\log h$ whose value at $(z, \ldots, z)$ is $\log z$. By the definition of g, Wormald's angular function satisfies

$$\widetilde{f}(\theta) = \log z + g(i\theta_1, \ldots, i\theta_{d-1}).$$

Its first angular derivative is $ig_j(i\theta)$ and its second angular derivative is $i^2 g_{jk}(i\theta)$. Therefore

$$\mathbf{H}_{jk} = i^2 g_{jk}(0) = -g_{jk}(0) = (e_j - e_d)^T Q (e_k - e_d),$$

$$\mathbf{H} = E^T Q E.$$

9.7 Positive definiteness and the determinant

The notation $M \succ 0$ means that M is real symmetric and, for every nonzero real vector v, we have $v^T M v > 0$.

On each two-dimensional block from Section 9.2, $J_\lambda^2 = -\lambda^2 I_2$. Therefore

$$(I + J_\lambda^2)(I - J_\lambda^2)^{-1} = \frac{1 - \lambda^2}{1 + \lambda^2} I_2.$$

The coefficient is positive because $\lambda^2 < 1/2$. On zero blocks, Q is the identity, so $Q \succ 0$. For $v \in \mathbb{R}^{d-1}$,

$$Ev = (v_1, \ldots, v_{d-1}, -\textstyle\sum_{j=1}^{d-1} v_j)^T.$$

If $v \neq 0$, its first $d-1$ components are not all zero, so $Ev \neq 0$. Consequently,

$$v^T \mathbf{H} v = (Ev)^T Q (Ev) > 0.$$

We next compute the Gram determinant of E. The inner product of its columns j, k is

$$(e_j - e_d)^T (e_k - e_d) = \delta_{jk} + 1.$$

Thus $E^T E = I_{d-1} + J_{d-1}$. Its eigenvalue in the all-one direction is $1 + (d-1) = d$; each direction orthogonal to the all-one vector has eigenvalue 1. Therefore

$$\det(E^T E) = d \cdot 1^{d-2} = d.$$

Write $E = VL$, where V is an orthonormal basis of $u^\perp$ and L is invertible:

$$E^T Q E = L^T (V^T Q V) L, \qquad E^T E = L^T L.$$

Taking determinants gives, respectively,

$$\det(E^T Q E) = \det(L)^2 \det(V^T Q V), \qquad \det(L)^2 = d.$$

Using $Cu = 0$,

$$Qu = (I + C^2)(I - C^2)^{-1} u = u.$$

In the real orthogonal basis of Section 9.2, every block of Q is a real scalar multiple of the identity. Thus Q is real symmetric, and $u^TQV = (Qu)^TV = u^TV = 0$. Hence

$$(u, V)^TQ(u, V) = \begin{pmatrix} 1 & 0 \\ 0 & V^TQV \end{pmatrix}, \qquad \det Q = \det(V^TQV).$$

Restore the two determinant factors:

$$\mathcal{H} = d \det Q = d \frac{\det(I + C^2)}{\det(I - C^2)} > 0.$$

Matrix multiplication and transposition give

$$(I + iC)(I - iC) = I + C^2, \qquad \det(I - iC) = \det((I - iC)^T) = \det(I + iC),$$

$$(I + C)(I - C) = I - C^2, \qquad \det(I - C) = \det(I + C).$$

Therefore

$$\det(I + C^2) = \det(I + iC)^2, \qquad \det(I - C^2) = \det(I + C)^2.$$

On each skew-symmetric two-dimensional block,

$$\det(I + J_\lambda) = 1 + \lambda^2 > 0, \qquad \det(I + iJ_\lambda) = 1 - \lambda^2 > 0.$$

Thus

$$\sqrt{\frac{d}{\mathcal{H}}} = \frac{\det(I + C)}{\det(I + iC)} > 0.$$

Write $[d] = \{1, \dots, d\}$. The principal submatrix $M[U]$ uses the rows and columns in U, with convention $\det M[\varnothing] = 1$.

Write $\lfloor a \rfloor$ for the largest integer at most a. To prove rationality, expand the determinant by multilinearity, choosing in each column either the identity column or the corresponding column of C:

$$\det(I + tC) = \sum_{U \subseteq [d]} t^{|U|} \det C[U].$$

For odd $r = |U|$,

$$\det C[U] = \det(C[U]^T) = \det(-C[U]) = (-1)^r \det C[U] = -\det C[U],$$

so this principal determinant is zero. Since C has rational entries,

$$\det(I + iC) = \sum_{k=0}^{\lfloor d/2 \rfloor} \sum_{\substack{U \subseteq [d] \\ |U| = 2k}} (-1)^k \det C[U] \in \mathbb{Q}.$$

The number $\det(I + C)$ is rational as well. We have already shown that numerator and denominator are positive, so their ratio is a positive rational number. □

10. Model limits and the carousel constants

10.1 Limits in the random models

Set $0! = 1$ and $n! = 1 \cdot 2 \cdots n$.

A vertex of type j belongs to the block replacing base vertex j, which contains m vertices.

Definition 10.1 (Normalization of the random models).

We now substitute the Hessian formula into Wormald's Theorems 1, 4 and 5. Fix a regular base of order d and let the internal block order m tend to infinity, with $N = dm$. The blocks are chosen independently and uniformly from the admissible labeled tournaments. The subscripts $\mathcal{T}, \mathcal{R}$ denote independent random internal blocks and regular internal blocks, respectively; $\mathbb{E}C$ and $\mathbb{E}H$ denote the expected cycle and path counts. Define

$$\nu_N = \frac{(N-1)!}{2^N}, \qquad \mu_N = \frac{N!}{2^{N-1}}.$$

Corollary 10.2 (The four expectation limits).

When $\mathcal{H} \neq 0$, Wormald's Theorem 4 gives

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} \longrightarrow \sqrt{d/\mathcal{H}}.$$

Section 9.7 gives $\mathcal{H} > 0$ and identifies the positive square root. Hence

$$\lim_{m\to\infty} \frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} = \sqrt{\frac{d}{d\det(I+C^2)/\det(I-C^2)}} = \rho_d(S).$$

Theorem 5 replaces cycle counts in Theorem 4 by path counts. Thus

$$\lim_{m\to\infty} \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} = \sqrt{d/\mathcal{H}} = \rho_d(S).$$

Theorem 1 gives the expectation ratio between regular internal blocks and independent random internal blocks:

$$\frac{\mathbb{E}C_{\mathcal{R}}}{\mathbb{E}C_{\mathcal{T}}} \longrightarrow e^{1/d}.$$

For sufficiently large m the denominator is positive, because its normalized limit is $\rho_d(S) > 0$. Multiplication gives

$$\lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N} = \lim_{\substack{m\to\infty \\ m \text{ odd}}} \left(\frac{\mathbb{E}C_{\mathcal{R}}}{\mathbb{E}C_{\mathcal{T}}} \frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} \right) = e^{1/d}\rho_d(S).$$

Theorem 5 also replaces the counts in Theorem 1 by path counts, giving

$$\frac{\mathbb{E}H_{\mathcal{R}}}{\mathbb{E}H_{\mathcal{T}}} \longrightarrow e^{1/d}.$$

The fourth limit is therefore

$$\lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} = \lim_{\substack{m\to\infty \\ m \text{ odd}}} \left(\frac{\mathbb{E}H_{\mathcal{R}}}{\mathbb{E}H_{\mathcal{T}}} \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} \right) = e^{1/d}\rho_d(S).$$

In all four limits, d is fixed. For regular internal blocks, m tends to infinity through odd integers.

A direct application of the Divoux–Wang counting formula. The same model constants follow from [12, Proposition 5.1], without introducing Wormald's angle function. Fix a regular base tournament of order

d, put $N = dm$, and let J_r denote the all-one matrix of order r. The expected adjacency matrix of the model with independent uniform internal tournaments is

$$W = \frac{J_N - I_N + S \otimes J_m}{2}, \qquad D = \frac{2W}{N-1}, \qquad B = D - \frac{J_N}{N}.$$

A Hamilton cycle or path uses each unordered pair at most once. Independence of the random internal edges therefore gives

$$\mathbb{E}C_{\mathcal{T}} = \mathrm{cyc}(W), \qquad \mathbb{E}H_{\mathcal{T}} = \mathrm{path}(W).$$

Here cyc and path denote the corresponding weighted sums. Since $S\mathbf{1} = 0$, the matrix D is doubly stochastic, and its entries are at most $2/(N-1)$.

Decompose $\mathbb{R}^N$ orthogonally into the constant direction, the space of block-constant vectors with total sum zero, and the space of vectors whose sum in each block is zero. On these spaces, B acts respectively as

$$0, \qquad B_0 = \left.\frac{mS - I_d}{N-1}\right|_{\mathbf{1}_d^{\perp}}, \qquad -\frac{I_{N-d}}{N-1}.$$

The bound $\|S\|_{\mathrm{op}}^2 \le d(d-1)/2$ implies

$$\limsup_{m\to\infty} \|B\|_{\mathrm{op}} \le \sqrt{\frac{d-1}{2d}} < 1.$$

Thus the hypotheses of Divoux–Wang's proposition hold, and

$$\frac{N^N \,\mathrm{cyc}(D)}{(N-1)!} = (1+o(1))\frac{\det(I-B)}{\sqrt{\det(I-B^TB)}}.$$

Write R_N for the determinant ratio. The orthogonal decomposition gives

$$R_N = \frac{\det(I-B_0)}{\sqrt{\det(I-B_0^TB_0)}} \frac{\left(1+(N-1)^{-1}\right)^{N-d}}{\left(1-(N-1)^{-2}\right)^{(N-d)/2}}.$$

The second factor tends to e. On the fixed-dimensional first space, $B_0 \to C|_{\mathbf{1}_d^{\perp}}$, where $C = S/d$. Restoring the constant direction contributes a factor of one to each determinant. Consequently,

$$R_N \longrightarrow e\,\frac{\det(I-C)}{\sqrt{\det(I+C^2)}} = e\,\frac{\det(I+C)}{\det(I+iC)} = e\rho_d(S).$$

The square root is positive: an eigenvalue pair $\pm i\lambda$ of C contributes $1-\lambda^2 > 0$. Homogeneity now yields

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} = \left(\frac{N-1}{N}\right)^N \frac{N^N \,\mathrm{cyc}(D)}{(N-1)!} \longrightarrow e^{-1}\, e\rho_d(S) = \rho_d(S).$$

For paths, apply the auxiliary-vertex argument of [12, Section 10] to

$$\widehat{D} = \begin{pmatrix} 0 & \mathbf{1}_N^T/N \\ \mathbf{1}_N/N & (1-1/N)D \end{pmatrix}.$$

This matrix is doubly stochastic. Closing a path through the new vertex gives

$$\mathrm{cyc}(\widehat{D}) = N^{-2}(1-1/N)^{N-1}\,\mathrm{path}(D).$$

For $\widehat{B} = \widehat{D} - J_{N+1}/(N+1)$, we have $\|\widehat{B} - \mathrm{diag}(0, B)\|_F = O(N^{-1})$, and both matrices have trace -1. Since $\|B\|_F$ is bounded and its operator norm stays below one, the determinant comparison in [12, Section 10] shows that their factors in Proposition 5.1 have ratio $1 + o(1)$. Hence

$$\frac{\mathrm{cyc}(\widehat{D})}{\mathrm{cyc}(D)} = (1 + o(1)) \left(\frac{N}{N+1}\right)^{N+1}, \qquad \mathrm{path}(D) = (1 + o(1))N^2\,\mathrm{cyc}(D).$$

Rescaling to $W = (N-1)D/2$, we obtain

$$\frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} = (1 + o(1))\frac{N}{N-1}\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} \longrightarrow \rho_d(S).$$

Wormald's Theorems 1 and 5 multiply these limits by $e^{1/d}$ for regular internal blocks. All four model constants therefore follow from existing counting results. This argument does not determine the Hessian of the original angle function; its positive definiteness and its full matrix formula require the differentiation carried out earlier.

10.2 A closed form for the carousel base

The following closed form is the determinant evaluation of Divoux and Wang [12, Section 7 and Lemma 9.1], combined with Wormald's regular-block factor [6, Theorem 1].

Proposition 10.3 (Closed forms for the carousel model).

Let T_d be the transitive tournament sign matrix. The carousel sign matrix is similar to T_d by a signed permutation, as shown in Appendix A.2. By Appendix A.1, every even-order principal determinant of T_d is 1, and every odd-order principal determinant is 0. Therefore

$$\det(I + tT_d) = \frac{(1+t)^d + (1-t)^d}{2}.$$

When a tournament is used as the argument of ρ_d, we mean its sign matrix. The value is invariant under relabeling. Thus

$$\rho_d(\mathrm{Car}_d) = \frac{(1+1/d)^d + (1-1/d)^d}{(1+i/d)^d + (1-i/d)^d}.$$

At the first three odd orders, this gives the values $2, 44/19, 3032/1231, \ldots$ in Wormald's table. The constant for regular internal blocks is

$$\mathfrak{r}_d = e^{1/d} r_d, \qquad r_d := \rho_d(\mathrm{Car}_d).$$

10.3 Real-function form and strict monotonicity

A prime on a function of one variable denotes differentiation in its argument.

Definition 10.4 (Real functions and auxiliary variables).

We use the following elementary functions:

$$\cosh y = \frac{e^y + e^{-y}}{2}, \quad \sinh y = \frac{e^y - e^{-y}}{2}, \quad \tanh y = \frac{\sinh y}{\cosh y}, \quad \mathrm{sech}\, y = \frac{1}{\cosh y}, \quad \sec y = \frac{1}{\cos y}.$$

Set $x = 1/d$ and define on $0 < x \le 1/3$

$$\alpha(x) = \frac{\arctan x}{x}, \qquad \beta(x) = \frac{\mathrm{artanh}\, x}{x}, \qquad \mathrm{artanh}\, x = \frac{1}{2}\log\frac{1+x}{1-x}.$$

Corollary 10.5 (Monotonicity and the limit). Along odd integers $d \geq 3$, the sequence $\mathfrak{r}_d$ increases strictly to $L = \cosh 1/\cos 1$.

Proof from the explicit spectrum. By Fang–Huang [11, Lemma 4.2], the positive frequencies of the normalized transitive sign matrix, for odd d, are

$$a_{j,d} = \frac{1}{d}\cot\frac{(2j-1)\pi}{2d}, \qquad 1 \leq j \leq \frac{d-1}{2}.$$

The logarithmic expansion in Divoux–Wang [12, Section 7], together with $\|T_d\|_F^2 = d(d-1)$, gives

$$\log r_d = 2\sum_{j=1}^{(d-1)/2}\sum_{r=0}^{\infty}\frac{a_{j,d}^{4r+2}}{2r+1}, \qquad 2\sum_{j=1}^{(d-1)/2} a_{j,d}^2 = 1 - \frac{1}{d}.$$

The regular-block factor cancels the term $-1/d$, leaving

$$\log \mathfrak{r}_d = 1 + 2\sum_{j=1}^{(d-1)/2}\sum_{r=1}^{\infty}\frac{a_{j,d}^{4r+2}}{2r+1}.$$

For fixed j, put $c_j = (2j-1)\pi/2$ and $x = c_j/d$. Then $a_{j,d} = x\cot x/c_j$, while

$$\frac{d}{dx}(x\cot x) = \frac{\sin x\cos x - x}{\sin^2 x} < 0 \qquad (0 < x < \pi/2).$$

The strict inequality follows from $\sin(2x) < 2x$. Each existing frequency therefore increases strictly as d increases. Passing from d to $d+2$ also adds a positive frequency. All coefficients in the convergent series above are positive, so $\mathfrak{r}_{d+2} > \mathfrak{r}_d$. Finally, the known closed form gives

$$\mathfrak{r}_d = e^{1/d}\frac{(1+1/d)^d + (1-1/d)^d}{(1+i/d)^d + (1-i/d)^d} \longrightarrow \frac{\cosh 1}{\cos 1}.$$

□

A second proof by real-variable calculation. First express the numerator and denominator as real functions. The sum and difference of the real logarithms give

$$\log(1 \pm x) = \frac{1}{2}\log(1-x^2) \ \pm \operatorname{artanh} x.$$

Exponentiating and adding gives

$$(1+x)^{1/x} + (1-x)^{1/x} = 2(1-x^2)^{1/(2x)}\cosh\beta(x).$$

For $0 \leq x \leq 1/3$, the points $1 \pm ix$ lie in the right half-plane. Taking the analytic logarithm there that agrees with the real logarithm on the positive axis gives

$$\log(1 \pm ix) = \frac{1}{2}\log(1+x^2) \pm i\arctan x,$$

$$(1+ix)^{1/x} + (1-ix)^{1/x} = 2(1+x^2)^{1/(2x)}\cos\alpha(x).$$

Therefore

$$\mathfrak{r}(x) = e^{p(x)}\frac{\cosh\beta(x)}{\cos\alpha(x)}, \qquad p(x) = x + \frac{\log(1-x^2) - \log(1+x^2)}{2x}.$$

When $x = 1/d$, the exponent $1/x = d$ is an integer, so these complex powers equal the original polynomial powers. Also $0 < \alpha(x) \le 1 < \pi/2$, so the cosine in the denominator is strictly positive.

Subtract the two logarithmic series:

$$\log(1-x^2) - \log(1+x^2) = -2\sum_{j=0}^{\infty} \frac{x^{4j+2}}{2j+1},$$

$$p(x) = x - \sum_{j=0}^{\infty} \frac{x^{4j+1}}{2j+1} = -\sum_{j=1}^{\infty} \frac{x^{4j+1}}{2j+1},$$

$$p'(x) = -\sum_{j=1}^{\infty} \frac{4j+1}{2j+1} x^{4j} < 0.$$

Changing variables in the integrals for the arctangent and inverse hyperbolic tangent gives

$$\alpha(x) = \int_0^1 \frac{dt}{1+x^2t^2}, \qquad \beta(x) = \int_0^1 \frac{dt}{1-x^2t^2}.$$

Set $v = 1 - \alpha$ and differentiate under the integral:

$$\beta'(x) = \int_0^1 \frac{2xt^2}{(1-x^2t^2)^2}\, dt, \qquad v'(x) = \int_0^1 \frac{2xt^2}{(1+x^2t^2)^2}\, dt > 0.$$

For each $t \in [0,1]$,

$$\frac{2xt^2}{(1-x^2t^2)^2} \le \left(\frac{1+x^2}{1-x^2}\right)^2 \frac{2xt^2}{(1+x^2t^2)^2},$$

$$\left(\frac{1+x^2}{1-x^2}\right)^2 \le \left(\frac{1+1/9}{1-1/9}\right)^2 = \left(\frac{5}{4}\right)^2 = \frac{25}{16}.$$

Hence $\beta'(x) \le (25/16)v'(x)$. Since $\beta'(x) > 0$,

$$\beta(x) \le \beta(1/3) = 3 \cdot \frac{1}{2} \log \frac{1+1/3}{1-1/3} = \frac{3}{2}\log 2,$$

$$\tanh \beta(x) \le \frac{e^{3\log 2} - 1}{e^{3\log 2} + 1} = \frac{8-1}{8+1} = \frac{7}{9}.$$

Use $(1+z)^{-1} \ge 1 - z$ and integrate:

$$\alpha(x) \ge \int_0^1 (1 - x^2t^2)\, dt = 1 - \frac{x^2}{3} \ge \frac{26}{27}.$$

Appendix A.7 gives $\tan y \ge y + y^3/3$. Since $y + y^3/3$ is strictly increasing for $y \ge 0$, the bound $\alpha(x) \ge 26/27$ implies

$$\tan \alpha(x) \ge \frac{26}{27} + \frac{1}{3}\left(\frac{26}{27}\right)^3.$$

The difference between these constants is

$$\frac{26}{27} + \frac{26^3}{3 \cdot 27^3} - \frac{7}{9}\frac{25}{16} = \frac{1191008 - 1148175}{944784} = \frac{42833}{944784} > 0.$$

Consequently, differentiation of $\log \mathfrak{r}$ yields

$$\begin{aligned}\frac{d}{dx}\log \mathfrak{r}(x) &= p'(x) + \tanh\beta(x)\,\beta'(x) - \tan\alpha(x)\,v'(x)\\ &\le p'(x) + \left(\frac{7}{9}\frac{25}{16} - \frac{26}{27} - \frac{26^3}{3\cdot 27^3}\right)v'(x)\\ &= p'(x) - \frac{42833}{944784}v'(x) < 0.\end{aligned}$$

The inequality $d_1 < d_2$ implies $1/d_1 > 1/d_2$, hence $\mathfrak{r}_{d_1} < \mathfrak{r}_{d_2}$. As $x \downarrow 0$,

$$\alpha(x) \to 1, \qquad \beta(x) \to 1, \qquad p(x) \to 0,$$

$$\lim_{d\to\infty,\ d \text{ odd}} \mathfrak{r}_d = e^0\frac{\cosh 1}{\cos 1} = L.$$

□

10.4 The asymptotic expansion of the carousel constants

All expansions below are taken near $x = 0$, with absolute constants in the remainders. After substituting $x = 1/d$, the corresponding estimates are understood as $d \to \infty$.

For $0 < |x| < 1$, integrate the geometric series:

$$\frac{\arctan x}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} + O(x^6), \qquad \frac{\operatorname{artanh} x}{x} = 1 + \frac{x^2}{3} + \frac{x^4}{5} + O(x^6).$$

For example, the remainder in the second identity satisfies

$$\left|\sum_{k=3}^{\infty}\frac{x^{2k}}{2k+1}\right| \le \frac{|x|^6}{7}\sum_{j=0}^{\infty}|x|^{2j} = \frac{|x|^6}{7(1-x^2)}.$$

The absolute remainder of the first, alternating series is bounded by the same positive series. The series for p in Section 10.3 gives

$$p(x) = -x^5/3 + R_p(x), \qquad |R_p(x)| \le \frac{|x|^9}{5(1-|x|^4)}.$$

Put $b = \tanh 1$, $h = \tan 1$, $\delta_\beta = \beta - 1$ and $\delta_\alpha = \alpha - 1$. The first and second derivatives used below are

$$(\log\cosh y)' = \tanh y, \quad (\log\cosh y)'' = 1 - \tanh^2 y,$$

$$(-\log\cos y)' = \tan y, \quad (-\log\cos y)'' = 1 + \tan^2 y.$$

Differentiating once more,

$$(\log\cosh y)''' = -2\tanh y\ \operatorname{sech}^2 y, \qquad (-\log\cos y)''' = 2\tan y\ \sec^2 y.$$

The absolute value of the first expression is at most 2. The second needs to be bounded only between α and 1. Since $0 \le \alpha \le 1$, its absolute value is at most $2\tan 1\ \sec^2 1$. Write R_β, R_α for the two Taylor integral remainders. Then

$$|R_\beta| \le \frac{2}{6}|\delta_\beta|^3, \qquad |R_\alpha| \le \frac{2\tan 1\ \sec^2 1}{6}|\delta_\alpha|^3.$$

Here the factor $1/6$ comes from $\frac{1}{2}\int_0^{|\delta|}(|\delta| - t)^2 dt = |\delta|^3/6$. The expansions are

$$\log\cosh(1+\delta_\beta)-\log\cosh 1=b\delta_\beta+\frac{1-b^2}{2}\delta_\beta^2+O(\delta_\beta^3),$$

$$-\log\cos(1+\delta_\alpha)+\log\cos 1=h\delta_\alpha+\frac{1+h^2}{2}\delta_\alpha^2+O(\delta_\alpha^3).$$

Since

$$\delta_\beta=x^2/3+x^4/5+O(x^6),\quad \delta_\alpha=-x^2/3+x^4/5+O(x^6),$$

$$\delta_\beta^2=x^4/9+O(x^6),\quad \delta_\alpha^2=x^4/9+O(x^6),\quad \delta_\beta^3=O(x^6),\quad \delta_\alpha^3=O(x^6),$$

substitution gives

$$\log\cosh\beta-\log\cosh 1=\frac{b}{3}x^2+\left(\frac{b}{5}+\frac{1-b^2}{18}\right)x^4+O(x^6),$$

$$-\log\cos\alpha+\log\cos 1=-\frac{h}{3}x^2+\left(\frac{h}{5}+\frac{1+h^2}{18}\right)x^4+O(x^6).$$

Add $p(x)$ and put $L=\cosh 1/\cos 1$:

$$\log\frac{\mathfrak{r}(x)}{L}=a_2x^2+\ell_4x^4-\frac{x^5}{3}+O(x^6),$$

$$a_2=\frac{b-h}{3},\qquad \ell_4=\frac{b+h}{5}+\frac{2+h^2-b^2}{18}.$$

Denote the right-hand side by $z(x)$. Since $z=O(x^2)$,

$$e^z=1+z+\frac{z^2}{2}+O(z^3),\qquad \frac{z^2}{2}=\frac{a_2^2}{2}x^4+O(x^6),\quad z^3=O(x^6).$$

Therefore

$$\mathfrak{r}(x)=L\left[1+a_2x^2+\left(\ell_4+\frac{a_2^2}{2}\right)x^4-\frac{x^5}{3}+O(x^6)\right].$$

Define $a_4=\ell_4+a_2^2/2$ and substitute $x=1/d$:

$$\mathfrak{r}_d=L\left[1+\frac{a_2}{d^2}+\frac{a_4}{d^4}-\frac{1}{3d^5}+O(d^{-6})\right].$$

The first- and third-order terms vanish, while the fifth-order term is $-L/(3d^5)$. Numerically,

$$La_2\approx -0.757603347512237,\qquad La_4\approx 2.035222101370122.$$

Here the block limit $m\to\infty$ is taken at fixed d, followed by $d\to\infty$. The expansion describes the resulting base constants as the base order grows.

11. The unique expected extremum for a fixed base order

11.1 The random blow-up models

Definition 11.1 (The two fixed-base blow-up models).

Fix an odd integer $d \geq 3$. For any order-d tournament T^*, replace each vertex by a random tournament of order m. All edges between blocks follow T^*. The total order is $N = dm$.

Write $\mathcal{T}(T^*, m)$ for independent uniform random internal blocks, and $\mathcal{R}(T^*, m)$ for independent uniform regular internal blocks; the latter requires odd m. The expected cycle and path counts are denoted by $\mathbb{E}C, \mathbb{E}H$ and normalized by

$$\nu_N = (N-1)!/2^N, \qquad \mu_N = N!/2^{N-1}.$$

This consequence combines [12, Proposition 7.1 and Lemma 9.1], the equality classification of Fang and Huang [11, Theorem 1.4 and Corollary 3.4], and the model comparisons of Wormald [6, Theorems 1, 4 and 5]. The proof below supplies the regular-base isomorphism argument and excludes nonregular bases.

Corollary 11.2 (The unique expected extremum).

For every fixed odd $d \geq 3$, there exists M_d such that, for $m \geq M_d$, the carousel base uniquely maximizes the expected cycle and path counts in both models, up to isomorphism. In model $\mathcal{R}$, m is restricted to odd integers.

The cycle conclusion for independent internal blocks proves and strengthens Wormald's Conjecture 1. The regular-block and path conclusions are corresponding extensions.

11.2 Limiting constants for regular bases

For a regular base, Sections 9 and 10 give

$$\lim_{m\to\infty} \frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} = \lim_{m\to\infty} \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} = \rho_d(S),$$

$$\lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N} = \lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} = e^{1/d}\rho_d(S),$$

$$\rho_d(S) = \frac{\det(I + S/d)}{\det(I + iS/d)}.$$

Thus regular bases are compared by their determinant ratios $\rho_d(S)$. For nonregular bases, the model limits are not given by this formula. We instead estimate the expected cycle and path counts directly and show that both normalized expectations tend to zero.

11.3 Nonregular scores and exponential loss

Definition 11.3 (Degree parameters for a nonregular base).

Let d_i be the base outdegrees, put $b_i = d_i + 1/2$, and define

$$a_i = \frac{2b_i}{d}, \qquad \kappa = \left(\prod_{i=1}^{d} a_i\right)^{1/d}, \qquad s_i = 2d_i - (d-1).$$

Put $s = (s_1, \ldots, s_d)^T$, so that $\|s\|_2^2 = \sum_{i=1}^{d} s_i^2$. Each edge contributes exactly one outdegree. Therefore

$$\sum_i d_i = \binom{d}{2}, \qquad \sum_i a_i = \frac{2\binom{d}{2} + d}{d} = d, \qquad a_i = 1 + \frac{s_i}{d} \in (0, 2), \qquad \sum_i s_i = 0.$$

The function $f(a) = a - 1 - \log a$ satisfies $f'(a) = (a-1)/a$, so its unique minimum occurs at $a = 1$ with value $f(1) = 0$. Hence

$$\log \kappa = \frac{1}{d} \sum_i \log a_i \le \frac{1}{d} \sum_i (a_i - 1) = 0.$$

Equality requires $a_i = 1$ for every index, equivalently $d_i = (d-1)/2$ for every index. For a nonregular base, $\kappa < 1$. To quantify the loss, use a second-order expansion with integral remainder:

$$\log a = (a-1) - (a-1)^2 \int_0^1 \frac{1-t}{[1+t(a-1)]^2}\, dt.$$

Since $0 < 1 + t(a-1) < 2$,

$$\int_0^1 \frac{1-t}{[1+t(a-1)]^2}\, dt \ge \frac{1}{4} \int_0^1 (1-t)\, dt = \frac{1}{8}.$$

Substitute a_i and sum:

$$\log \kappa \le \frac{1}{d} \sum_i \left(\frac{s_i}{d} - \frac{s_i^2}{8d^2} \right) = -\frac{\|s\|_2^2}{8d^3}.$$

For odd d each s_i is even. Nonregularity implies that some $s_i \neq 0$; the zero-sum condition then requires nonzero terms of opposite signs, each with absolute value at least 2. Thus $\|s\|_2^2 \ge 2^2 + 2^2 = 8$ and

$$\kappa^N \le \exp[-N\|s\|_2^2/(8d^3)] \le e^{-N/d^3}.$$

In model $\mathcal{T}$, let W be the entrywise expectation of the adjacency matrix. Write $\mathrm{Sym}(N)$ for the permutations of $[N]$ and $\mathcal{C}_N$ for its N-cycles. For a nonnegative matrix of order N, define

$$\operatorname{per} W = \sum_{\sigma \in \mathrm{Sym}(N)} \prod_{v=1}^{N} W_{v,\sigma(v)}, \qquad C(W) = \sum_{\sigma \in \mathcal{C}_N} \prod_{v=1}^{N} W_{v,\sigma(v)}.$$

Each directed Hamiltonian cycle corresponds to one N-cycle and uses distinct unordered vertex pairs. Its random internal edges are independent, so its occurrence probability is the product of the corresponding entries of W. Summing over cycles gives

$$\mathbb{E} C_{\mathcal{T}} = C(W) \le \operatorname{per} W.$$

The inequality follows from $\mathcal{C}_N \subseteq \mathrm{Sym}(N)$ and nonnegativity of the weights. Independence is used only for a single cycle; it need not hold for a permutation containing a two-cycle.

11.4 Rounding row sums and Brégman's bound

Write $\lceil a \rceil$ for the smallest integer at least a.

Definition 11.4 (Integral targets for row sums).

Assume $m \ge 2$, with m odd for regular internal blocks. Each row of type i in W has sum

$$r_i = m d_i + \frac{m-1}{2} = m b_i - \frac{1}{2}, \qquad R_i = \lceil r_i \rceil.$$

For a row of length L, denoted w, assume $0 \le w_j \le 1$. Its current sum is $r = \sum_{j=1}^{L} w_j$, and its target integer sum R satisfies $r \le R \le L$. The amount to add is $s_0 = R - r$, and the available capacity is $\sum_j (1 - w_j) = L - r \ge s_0$. Starting at $j = 1$, choose successively

$$\delta_j = \min(1 - w_j, s_{j-1}), \qquad s_j = s_{j-1} - \delta_j, \qquad w'_j = w_j + \delta_j.$$

At each step the remaining mass satisfies

$$s_j = \max(0, s_{j-1} - (1 - w_j)) = \max\left(0, s_0 - \sum_{h=1}^{j}(1 - w_h)\right).$$

The first equality subtracts the smaller of two quantities; the second follows by substitution starting at $j = 1$.

The recursion gives $s_L = \max(0, s_0 - \sum_j (1 - w_j)) = 0$, so $0 \le w'_j \le 1$ and $\sum_j w'_j = R$. Every coefficient of the permanent is nonnegative, so these entrywise increases cannot decrease it.

Fix the other rows and write the permanent as $\sum_j w_j c_j$, where c_j is the permanent obtained by deleting the current row and column j; thus $c_j \ge 0$. If entries w_a, w_b both lie strictly between 0 and 1, label them so that $c_a \ge c_b$ and take

$$\eta = \min(1 - w_a, w_b) > 0, \qquad w'_a = w_a + \eta, \qquad w'_b = w_b - \eta.$$

The row sum is preserved, and

$$\sum_j w'_j c_j - \sum_j w_j c_j = \eta(c_a - c_b) \ge 0.$$

At least one coordinate becomes 0 or 1, reducing the number of nonintegral coordinates. After at most $L - 1$ steps there is at most one nonintegral coordinate. Since all other coordinates and the row sum R are integers, the remaining coordinate must also be integral. Perform this operation on every row to obtain a zero–one matrix $W^\sharp$. Each row of type i has R_i ones, and $\operatorname{per} W \le \operatorname{per} W^\sharp$.

The resulting matrix $W^\sharp$ has entries in $\{0, 1\}$ and the specified row sums. Its diagonal entries may be 1, which is allowed in Brégman's inequality.

Now invoke Brégman's inequality, as stated in Lemma 2.1 of Alon's paper: for a zero–one matrix with row sums $q_1, \dots, q_L > 0$,

$$\operatorname{per} W^\sharp \le \prod_{\ell=1}^{L} (q_\ell!)^{1/q_\ell}.$$

Since $r_i \ge (m - 1)/2 > 0$, we have $R_i \ge 1$. There are m rows of each type, so

$$\operatorname{per} W \le \prod_{i=1}^{d} (R_i!)^{m/R_i}.$$

Together with $\operatorname{per} W \le \operatorname{per} W^\sharp$, this gives the required bound for the weighted matrix.

11.5 Normalization with explicit constants

The desired bound is immediate if the permanent is zero. We may therefore assume it is positive when taking logarithms. Let $m \ge 2$, $N = dm$, and write $R_i = mb_i + \xi_i$. The rounding definition gives $|\xi_i| \le 1/2$ and

$$R_i \ge m/2 - 1/2 \ge m/4, \qquad R_i \le N.$$

Indeed,

$$0 \le R_i - r_i < 1, \qquad \xi_i = R_i - mb_i = (R_i - r_i) - \frac{1}{2} \in [-1/2, 1/2).$$

Use the remainder form of Stirling's bound for integers $R \geq 1$:

$$\log(R!) = R\log R - R + \frac{1}{2}\log(2\pi R) + \frac{\theta_R}{12R}, \qquad 0 < \theta_R < 1.$$

Therefore

$$\begin{aligned} m\sum_{i=1}^{d} \frac{\log(R_i!)}{R_i} &\leq m\sum_i \log R_i - N + m\sum_i \frac{\log(2\pi R_i)}{2R_i} + m\sum_i \frac{1}{12R_i^2} \\ &\leq m\sum_i \log R_i - N + 2d\log(2\pi N) + \frac{4d}{3m}. \end{aligned}$$

For the last two terms we used $m/(2R_i) \leq 2$ and $m/(12R_i^2) \leq 4/(3m)$, respectively. The lower Stirling bound also gives

$$\log(N!/2^N) \geq N\log(N/2) - N.$$

Subtracting yields

$$\log \frac{\operatorname{per} W}{N!/2^N} \leq m\sum_i \log \frac{2R_i}{N} + 2d\log(2\pi N) + \frac{4d}{3m}.$$

Set $y_i = \xi_i/(mb_i)$. Because $b_i \geq 1/2$, we have $|y_i| \leq 1/m \leq 1/2$ and

$$\frac{2R_i}{N} = a_i(1+y_i), \qquad |\log(1+y_i)| = \left|\int_0^{y_i} \frac{dt}{1+t}\right| \leq \frac{|y_i|}{1-|y_i|} \leq \frac{2}{m}.$$

Thus

$$m\sum_i \log \frac{2R_i}{N} \leq m\sum_i \log a_i + 2d = N\log\kappa + 2d.$$

Use $4d/(3m) \leq 2d/3$ and $N!/2^N = N\nu_N$ to obtain the explicit bound

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} \leq e^{8d/3}(2\pi)^{2d}N^{2d+1}\kappa^N.$$

With regular internal blocks, every sample has the positive integer row sum $mb_i - 1/2$ in each row of type i. For its adjacency matrix A_G, we have $C(G) \leq \operatorname{per} A_G$. Apply Brégman's bound and the preceding estimates to A_G, then average to obtain

$$\frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N} \leq e^{8d/3}(2\pi)^{2d}N^{2d+1}\kappa^N.$$

The regular model uses only admissible odd m. For a nonregular base, $\kappa < 1$. Take the logarithm of the right-hand side and divide by N:

$$\frac{\log(e^{8d/3}(2\pi)^{2d}N^{2d+1}\kappa^N)}{N} = \log\kappa + \frac{8d/3 + 2d\log(2\pi) + (2d+1)\log N}{N} \longrightarrow \log\kappa < 0,$$

Here we use $\log N/N \to 0$, which follows from

$$0 \leq \frac{\log N}{N} = \frac{1}{N}\int_1^N \frac{dt}{t} \leq \frac{1}{N}\int_1^N \frac{dt}{\sqrt{t}} = \frac{2(\sqrt{N}-1)}{N} \longrightarrow 0.$$

Denote the upper bound by B_N and put $c = -\frac{1}{2}\log\kappa > 0$. Since $\log B_N/N \to -2c$, we have $B_N \le e^{-cN}$ for all sufficiently large N. The same argument applies to κ^N times any fixed-degree polynomial, as will be needed for paths in the next section.

Both normalized cycle expectations therefore tend to zero.

11.6 The added-vertex construction and path expectations

Lemma 11.5 (Random augmentation and path counts).

Fix a tournament G on $N \ge 2$ vertices. Add a vertex v whose edges to the old vertices are independently and fairly oriented. Each path $x_1 \to \cdots \to x_N$ corresponds to the unique candidate cycle $v \to x_1 \to \cdots \to x_N \to v$. Since $x_1 \ne x_N$, the two required new edges are distinct and have joint probability $(1/2)^2 = 1/4$. Represent each cycle with v as its starting vertex; deleting v reverses the correspondence. The condition $N \ge 2$ is necessary: for $N = 1$, $H(G) = 1$ but the augmented tournament has no directed 2-cycle. It is automatic in the blow-up applications here, where $d \ge 3$ is fixed. Therefore

$$\mathbb{E}[C_{\text{aug}} \mid G] = H(G)/4, \qquad 4\nu_{N+1} = \frac{4N!}{2^{N+1}} = \frac{N!}{2^{N-1}} = \mu_N,$$

$$\frac{\mathbb{E}H}{\mu_N} = \frac{\mathbb{E}C_{\text{aug}}}{\nu_{N+1}}.$$

Write G_{aug} for the augmented random tournament and $C_{\text{aug}} = C(G_{\text{aug}})$. For $\mathcal{T}$, let W_{aug} be the entrywise expected augmented adjacency matrix. For $\mathcal{R}$, first fix an augmented sample and let A_{aug} be its adjacency matrix. Apply the row completion and rounding of Section 11.4 to each matrix, obtaining a zero-one matrix $W^{\sharp}_{\text{aug}}$ of order $N+1$. Then

$$\mathbb{E}C_{\text{aug},\mathcal{T}} = C(W_{\text{aug}}) \le \operatorname{per} W_{\text{aug}} \le \operatorname{per} W^{\sharp}_{\text{aug}},$$

$$C(G_{\text{aug}}) \le \operatorname{per} A_{\text{aug}} \le \operatorname{per} W^{\sharp}_{\text{aug}} \qquad (\text{each sample of } \mathcal{R}).$$

The first line uses independence of the internal edges. The second holds for each sample with regular internal blocks.

For the augmented expectation matrix of $\mathcal{T}$, the old row sum is mb_i. For each actual augmented sample of $\mathcal{R}$, the old row sum is at most $mb_i + 1/2$. Use the common choice

$$R'_i = \lceil mb_i + 1/2 \rceil = mb_i + \xi'_i, \qquad \frac{1}{2} \le \xi'_i < \frac{3}{2}.$$

Take the target sum of the new row to be $R_0 = N$. Since $b_i \le d - 1/2$ and $m \ge 2$, we have $mb_i + 1/2 \le N - 1/2$, and hence $m/2 \le R'_i \le N$. All target sums are at most the number of columns $N+1$, so the rounding in Section 11.4 applies. Brégman's bound gives

$$\log \operatorname{per} W^{\sharp}_{\text{aug}} \le m \sum_i \frac{\log(R'_i!)}{R'_i} + \frac{\log(N!)}{N}.$$

Use $R'_i \ge m/4$ and $R'_i \le N+1$ to estimate the remainders for the old rows, keeping the new row separate:

$$\begin{aligned}\log \operatorname{per} W^{\sharp}_{\mathrm{aug}} &\le m\sum_i \log R'_i - N + \log N - 1 \\ &\quad + 2d\log(2\pi(N+1)) + \frac{4d}{3m} + \frac{\log(2\pi N)}{2N} + \frac{1}{12N^2} \\ &\le m\sum_i \log R'_i - N + \log N - 1 \\ &\quad + (2d+1)\log(2\pi(N+1)) + \frac{4d}{3m} + \frac{1}{12}.\end{aligned}$$

Subtract the Stirling lower bound $\log((N+1)!/2^{N+1}) \ge (N+1)\log((N+1)/2) - (N+1)$. The leading terms satisfy

$$\begin{aligned}&m\sum_i \log R'_i + \log N - (N+1)\log((N+1)/2) \\ &= m\sum_i \log(2R'_i/N) + N\log(N/(N+1)) + \log(2N/(N+1)).\end{aligned}$$

The second term is at most zero, and the third is at most $\log 2$. The inequality $\log(1+y) \le y$, valid for $y \ge 0$, follows from the integral $\int_0^y (1+t)^{-1}dt \le y$. Thus

$$\begin{aligned}m\sum_i \log(2R'_i/N) &= N\log\kappa + m\sum_i \log(1+\xi'_i/(mb_i)) \\ &\le N\log\kappa + \sum_i \frac{\xi'_i}{b_i} \le N\log\kappa + 3d.\end{aligned}$$

Using $4d/(3m) \le 2d/3$ now gives

$$\begin{aligned}\log\frac{\operatorname{per} W^{\sharp}_{\mathrm{aug}}}{(N+1)!/2^{N+1}} &\le N\log\kappa + \frac{11d}{3} + \frac{1}{12} + \log 2 \\ &\quad + (2d+1)\log(2\pi(N+1)).\end{aligned}$$

Exponentiate and substitute $(N+1)!/2^{N+1} = (N+1)\nu_{N+1}$. The two models satisfy, respectively,

$$\frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} \le 2e^{11d/3+1/12}(2\pi)^{2d+1}(N+1)^{2d+2}\kappa^N,$$

$$\frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} \le 2e^{11d/3+1/12}(2\pi)^{2d+1}(N+1)^{2d+2}\kappa^N.$$

The second estimate follows by averaging the bound for each augmented sample. For a fixed nonregular base, the logarithm of either right-hand side, divided by N, tends to $\log\kappa < 0$. Thus both normalized path expectations tend to zero.

11.7 The expected extremum and a common threshold

Proof.

For fixed d, there are at most $2^{\binom{d}{2}}$ labeled bases and hence finitely many isomorphism classes. The carousel limit is r_d or $e^{1/d}r_d$. By Section 6.5, every other regular base has a strictly smaller limit, while the normalized expectation for a nonregular base tends to zero.

Thus, for each noncarousel base, its normalized expectation is strictly smaller than the carousel's for all sufficiently large m. Taking the maximum of these finitely many thresholds gives M_d.

Finally, the expectations are invariant under base isomorphism. If π is an isomorphism of bases and the blow-up vertices are written as (i,a), then the map

$$(i, a) \longmapsto (\pi(i), a)$$

preserves all directions between blocks and permutes the independent identically distributed internal blocks, preserving the sample probabilities. It also induces bijections on paths and cycles, so the corresponding expectations agree. Together with the strict inequalities above, this proves uniqueness up to isomorphism. □

12. Proof of Theorem 1.1

We combine the uniform path estimates of Sections 7 and 8 with the spectral comparison of Section 6 to prove Theorem 1.1.

12.1 The determinant ratio

In the path argument the factor is written as

$$\rho_n(S) = \det(I - S^T S/n^2)^{-1/2} \det(I + S/n).$$

Let the nonzero eigenvalues of S occur in pairs $\pm i\lambda_j$, and put $x_j = \lambda_j^2/n^2$. Each $x_j < 1/2$, so all the following factors are positive and

$$\det(I - S^T S/n^2)^{1/2} = \prod_j (1 - x_j) = \det(I + iS/n).$$

Thus the factor is exactly $\det(I + S/n)/\det(I + iS/n)$ from the Hessian argument. The square root is the positive one; the sign of $\det(I + iS/n)$ has not been suppressed.

12.2 Applying the spectral comparison at both parities

The regular equality classification in Section 6 concerns odd orders, but the preceding spectral inequality does not require regularity. To make its use at every order explicit, let $n \geq 2$ and write the permutation in Section 6.3 as all odd indices in increasing order followed by all even indices in decreasing order. At odd order this is the original permutation. At even order it is $(1, 3, \ldots, n-1, n, n-2, \ldots, 2)$. The same entrywise computation gives

$$T_n = \Lambda^T D_0 K_n R_0 \Lambda, \qquad T_n^2 = -U^T K_n^2 U, \qquad U = D_0 \Lambda, \qquad U e_1 = e_1.$$

The signed wrap-around shift satisfies $Z^T T_n Z = T_n$ at either parity. Consequently, all diagonal entries of T_n^{2p} coincide. For every integer $p \geq 1$, the columnwise Fang–Huang comparison and these identities give

$$\|S^p\|_F^2 \leq n \|K_n^p e_1\|_2^2 = \|T_n^p\|_F^2.$$

Substitution into the absolutely convergent logarithmic series of Section 6.4 yields, at every order,

$$\log \rho_n(S) = \sum_{r \geq 0} \frac{\|S^{2r+1}\|_F^2}{(2r+1) n^{4r+2}} \leq \log \rho_n(T_n).$$

The determinant polynomial in Appendix A.1 also holds at both parities. Hence

$$\rho_n(S) \leq r_n = \frac{(1 + 1/n)^n + (1 - 1/n)^n}{(1 + i/n)^n + (1 - i/n)^n} = L + O(n^{-1}).$$

The last step follows directly from the complex-exponential estimates of Section 8.2: for $z \in \{1, -1, i, -i\}$, we have $n \log(1 + z/n) = z + O(n^{-1})$, while the denominator tends to $2 \cos 1 > 0$. The quotient therefore has an $O(n^{-1})$ error. These estimates depend on the order alone, not on the tournament.

12.3 The upper bound

Section 7 divides all tournaments into three classes. The large-variance class has $H(T)/\mu_n = o(n^{-1})$. The low-variance class with exceptional vertices eventually has normalized count below one. In the remaining class, $\tau = \|S\mathbf{1}/(n-1)\|_2^2 \le K_0 \log n$ and

$$\frac{H(T)}{\mu_n} \le \rho_n(S) \exp\left\{-\tau + K\left[\frac{1+\tau+\tau^2}{n} + \frac{\tau^{3/2}+\tau}{\sqrt{n}} + \sqrt{\tau/n}\right]\right\} + o(n^{-1}).$$

For all sufficiently large n, the polynomial error terms containing τ are at most $\tau/4$, while

$$K\sqrt{\tau/n} \le \tau/4 + K^2/n.$$

Thus the total exponent is at most $-\tau/2 + K_1/n$ for an absolute constant K_1. The spectral comparison gives

$$\frac{H(T)}{\mu_n} \le r_n e^{K_1/n} + o(n^{-1}) = L + O(n^{-1}).$$

Since $L > 1$, the first two classes satisfy this final upper bound as well. Taking the largest of the three absolute thresholds and then maximizing over T yields $P(n) \le (L + K_2/n)\mu_n$. The odd carousels and even deletion constructions in Section 8 give $P(n) \ge (L - K_3/n)\mu_n$. Taking $K' = \max(K_2, K_3)$ proves Theorem 1.1.

The constants are independent of the tournament, as required. □

A. Auxiliary algebra and model calculations

A.1 The determinant polynomial of the transitive sign matrix

Let T_n be the transitive tournament sign matrix and set $F_n(t) = \det(I + tT_n)$. We compute F_n by row operations and then recover its principal determinants.

For $n = 1$, $F_1(t) = 1 = ((1+t) + (1-t))/2$, proving the formula in that case. Now assume $n \ge 2$. For the first $n-1$ rows simultaneously perform $R_i \leftarrow R_i - R_{i+1}$. The operation is implemented by an upper-triangular matrix with diagonal 1, so the determinant is preserved. In the resulting matrix the only potentially nonzero entries in the first $n-1$ rows are

$$M_{ii} = 1 + t, \qquad M_{i,i+1} = t - 1$$

The last row is $(-t, \dots, -t, 1)$. Temporarily assume $t \ne -1, 0$ and put

$$r = \frac{1-t}{1+t}.$$

Denote the top-left block of order $(n-1)$ by B. Its determinant is $(1+t)^{n-1}$. The top-right column b has only its last entry nonzero, equal to $t - 1$. Solve $Bx = b$ from the last row upwards:

$$x_{n-1} = \frac{t-1}{1+t} = -r,$$

$$(1+t)x_i + (t-1)x_{i+1} = 0 \quad \Longrightarrow \quad x_i = r x_{i+1} = -r^{n-i}.$$

The Schur complement gives

$$F_n(t) = (1+t)^{n-1}\left(1 - t\sum_{k=1}^{n-1} r^k\right).$$

First multiply the finite geometric sum by $1-r$; adjacent terms cancel:

$$(1-r)\sum_{k=1}^{n-1} r^k = (r+r^2+\cdots+r^{n-1}) - (r^2+\cdots+r^n) = r - r^n.$$

Thus $\sum_{k=1}^{n-1} r^k = r(1-r^{n-1})/(1-r)$. Using $1-r = 2t/(1+t)$, the expression in parentheses becomes

$$\begin{aligned} 1 - t\sum_{k=1}^{n-1} r^k &= 1 - \frac{tr(1-r^{n-1})}{1-r} \\ &= 1 - \frac{1-t}{2}(1-r^{n-1}) \\ &= \frac{1+t}{2} + \frac{1-t}{2} r^{n-1}. \end{aligned}$$

Multiply the bracket by $(1+t)^{n-1}$ and substitute $r = (1-t)/(1+t)$:

$$\begin{aligned} F_n(t) &= \frac{(1+t)^n}{2} + \frac{(1+t)^{n-1}(1-t)}{2}\left(\frac{1-t}{1+t}\right)^{n-1} \\ &= \frac{(1+t)^n}{2} + \frac{(1-t)^n}{2}. \end{aligned}$$

Both sides are polynomials and agree for every t except two values. They therefore also agree at $t = -1, 0$. Substitute $t = 1/d$ and $t = i/d$ to obtain the determinant ratio for the carousel model.

For nonempty U, put $k = |U|$. In the original vertex order, $T_d[U] = T_k$. The coefficient of t^k in $F_k(t) = \det(I + tT_k)$ is $\det T_k$, and hence

$$\det T_d[U] = \det T_k = [t^k]F_k(t) = \frac{1+(-1)^k}{2} = \begin{cases} 1, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}$$

The empty principal determinant is 1 by convention.

A.2 Alternating switching and ordinary carousel isomorphism

Let $d = 2q+1$. Switch the transitive matrix by alternating signs $\varepsilon_i = (-1)^{i-1}$ and $D_\varepsilon = \operatorname{diag}(\varepsilon_i)$. For $i < j$ and $\Delta = j - i$,

$$(D_\varepsilon T_d D_\varepsilon)_{ij} = (-1)^\Delta.$$

Define the carousel labeling map

$$\phi(i) = (q+1)i \pmod d.$$

Since $2(q+1) = d+1$, the numbers $q+1$ and d are coprime, making ϕ a permutation.

If $\Delta = 2r$, then $1 \le r \le q$ and

$$\phi(j) - \phi(i) \equiv (q+1)2r \equiv r \pmod d.$$

This difference belongs to $\{1, \ldots, q\}$, precisely the condition for $\phi(i) \to \phi(j)$ in the carousel. The switched matrix entry is also 1.

If $\Delta = 2r+1$, then $0 \le r \le q-1$ and

$$\phi(j) - \phi(i) \equiv q + 1 + r \pmod d.$$

This difference belongs to $\{q+1,\dots,2q\}$, so the direction is reversed; the switched matrix entry is -1. Skew symmetry handles $i>j$, and all diagonal entries are zero. Thus the map preserves each directed edge and is an ordinary isomorphism.

Sections 6 and 11 also use equality of the diagonal entries of powers of the transitive matrix. Define $Ze_j = e_{j+1}$ for $j<d$ and $Ze_d = -e_1$. When $i,j<d$,

$$(Z^T T_d Z)_{ij} = (T_d)_{i+1,j+1} = (T_d)_{ij}.$$

When $i<d, j=d$,

$$(Z^T T_d Z)_{i,d} = -(T_d)_{i+1,1} = 1 = (T_d)_{i,d}.$$

When $i=d, j<d$,

$$(Z^T T_d Z)_{d,j} = -(T_d)_{1,j+1} = -1 = (T_d)_{d,j}.$$

The remaining diagonal entry is zero in both matrices. Therefore $Z^T T_d Z = T_d$, and for every integer $p \geq 0$,

$$Z^T T_d^{2p} Z = T_d^{2p}.$$

Taking diagonal entries makes adjacent diagonal entries equal, with the two wraparound minus signs canceling.

A.3 Fourth-order determinants and fourth-moment equality

Let $a,b,c,g,e,f \in \mathbb{R}$ and set

$$M = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & g & e \\ -b & -g & 0 & f \\ -c & -e & -f & 0 \end{pmatrix}.$$

Write M_{1j} for the order-three determinant obtained by deleting row 1 and column j, without the cofactor sign. Expanding along the first row gives the following separate calculations:

$$\begin{aligned} M_{12} &= \det \begin{pmatrix} -a & g & e \\ -b & 0 & f \\ -c & -f & 0 \end{pmatrix} \\ &= (-a)(0-f(-f)) - g((-b)0 - f(-c)) + e((-b)(-f) - 0(-c)) \\ &= -af^2 - cgf + bef, \\ M_{13} &= \det \begin{pmatrix} -a & 0 & e \\ -b & -g & f \\ -c & -e & 0 \end{pmatrix} \\ &= (-a)((-g)0 - f(-e)) - 0 + e((-b)(-e) - (-g)(-c)) \\ &= -aef + be^2 - cge, \\ M_{14} &= \det \begin{pmatrix} -a & 0 & g \\ -b & -g & 0 \\ -c & -e & -f \end{pmatrix} \\ &= (-a)((-g)(-f) - 0(-e)) - 0 + g((-b)(-e) - (-g)(-c)) \\ &= -agf + bge - cg^2. \end{aligned}$$

Substituting these minors into the first-row expansion gives

$$\begin{aligned}\det M &= -aM_{12} + bM_{13} - cM_{14}\\ &= (a^2f^2 + acgf - abef) + (-abef + b^2e^2 - bcge) + (acgf - bcge + c^2g^2)\\ &= a^2f^2 + b^2e^2 + c^2g^2 - 2abef + 2acgf - 2bcge\\ &= (af - be + cg)^2.\end{aligned}$$

When all six variables are ± 1, the expression $af - be + cg$ is a sum of three numbers in ± 1, taking values ± 1 or ± 3. Thus the determinant is 1 or 9. Let $D_4(S)$ count four-vertex principal determinants equal to 9. Expand by column multilinearity, choosing in each column either the identity column or the corresponding column of tS:

$$\det(I + tS) = \sum_{U \subseteq [d]} t^{|U|} \det S[U].$$

An odd-order skew-symmetric matrix B satisfies $\det B = \det B^T = \det(-B) = -\det B$, so its determinant vanishes. Every second-order principal determinant is 1. Among fourth-order determinants, D_4 are 9 and all others are 1. Therefore

$$\det(I + tS) = 1 + e_2t^2 + e_4t^4 + O(t^6),$$

$$e_2 = \binom{d}{2}, \qquad e_4 = (\binom{d}{4} - D_4) + 9D_4 = \binom{d}{4} + 8D_4.$$

Use the following expansion only near $t = 0$. Choose $|t|\,\|S\|_{\text{op}} < 1/2$. Every eigenvalue μ then satisfies $|t\mu| < 1/2$. Define $\log(1 + t\mu)$ by the logarithmic series with value 0 at 1. Interchanging the power series with a finite eigenvalue sum is valid, and

$$\det(I + tS) = \prod_{\mu}(1 + t\mu), \qquad \operatorname{tr} S^k = \sum_{\mu} \mu^k.$$

Eigenvalues are counted with multiplicity. The real skew-symmetric canonical form gives diagonalizability over the complex numbers; zero eigenvalues are included.

The logarithmic series for each eigenvalue gives

$$\log\det(I + tS) = \sum_{k \geq 1} \frac{(-1)^{k+1}t^k}{k} \operatorname{tr} S^k = -\frac{t^2}{2}\operatorname{tr} S^2 - \frac{t^4}{4}\operatorname{tr} S^4 + O(t^6).$$

Odd traces vanish because $\operatorname{tr} S^{2j+1} = \operatorname{tr}(S^{2j+1})^T = -\operatorname{tr} S^{2j+1}$. On the other hand, substitute $z = e_2t^2 + e_4t^4 + O(t^6)$ into $\log(1 + z) = z - z^2/2 + O(z^3)$. Since $z^2 = e_2^2t^4 + O(t^6)$ and $z^3 = O(t^6)$, this gives

$$\log\det(I + tS) = e_2t^2 + (e_4 - e_2^2/2)t^4 + O(t^6).$$

Comparing the coefficients of t^4 gives

$$-\operatorname{tr} S^4/4 = e_4 - e_2^2/2, \qquad \operatorname{tr} S^4 = 2e_2^2 - 4e_4 = 2\left(\binom{d}{2}\right)^2 - 4\binom{d}{4} - 32D_4(S).$$

In the transitive matrix, any four vertices in increasing order have $a = b = c = g = e = f = 1$, giving determinant $(1 - 1 + 1)^2 = 1$. Thus $D_4(T_d) = 0$. Subtracting yields

$$\operatorname{tr} T_d^4 - \operatorname{tr} S^4 = 32 D_4(S).$$

If the fourth moments agree, $D_4(S) = 0$. Choose $\varepsilon_1 = 1, \varepsilon_j = s_{1j}$ for $j > 1$, and define $D_\varepsilon = \operatorname{diag}(\varepsilon_1, \ldots, \varepsilon_d)$ and $S' = D_\varepsilon S D_\varepsilon$. Then $s'_{1j} = \varepsilon_1 s_{1j} \varepsilon_j = s_{1j}^2 = 1$. Each principal determinant is multiplied by $(\det D_\varepsilon[U])^2 = 1$, so D_4 is unchanged. Choose any $i < j < k$ all different from 1:

$$\det S'[\{1, i, j, k\}] = (s'_{jk} - s'_{ik} + s'_{ij})^2.$$

If these three vertices form a directed triangle, $s'_{ij} = s'_{jk} = -s'_{ik}$, making the expression in parentheses $3s'_{ij}$ and its square 9, a contradiction. Thus the remaining vertices contain no directed triangle. If $i \to j$ and $j \to k$ but $i \not\to k$, then $k \to i$, producing a triangle. The direction relation is therefore a strict total order, and we can label vertices in this order. Vertex 1 dominates all remaining vertices, making the entire switched tournament transitive. Conversely, signed permutation similarity preserves traces, so it gives fourth-moment equality. For $d = 3$, two vertices remain after switching; order them according to their unique edge.

A.4 Termination of the row-rounding procedure

Definition A.1 (A termination count for rounding).

To prove termination of the two-coordinate rounding in Section 11.4, define

$$q(w) = \#\{j : 0 < w_j < 1\}, \qquad q(W) = \sum_{\ell=1}^{L} q(W_{\ell,\cdot}).$$

Each two-coordinate movement preserves the integer row sum, makes at least one nonintegral coordinate integral, and leaves other integral coordinates unchanged. Thus $q(w)$ decreases by at least 1. Initially $q(w) \le L$, so it cannot decrease indefinitely. If it stopped at $q(w) = 1$, the unique nonintegral coordinate would satisfy

$$w_a = R - \sum_{j \ne a} w_j \in \mathbb{Z},$$

This contradicts nonintegrality, so termination occurs at $q(w) = 0$. Each row requires at most $L - 1$ movements and no other row is changed; the total is at most $L(L - 1)$. By Section 11.4, each movement increases the permanent by $\eta(c_a - c_b) \ge 0$.

A.5 Normalization in the added-vertex bound

We give the normalization calculation used in Section 11.6. Since $md = N$,

$$m \sum_i \log R'_i = m \sum_i \log(2R'_i/N) + N \log(N/2).$$

We obtain

$$\begin{aligned}
&N \log(N/2) + \log N - (N+1) \log((N+1)/2) \\
&= N \log N - N \log 2 + \log N - (N+1) \log(N+1) + (N+1) \log 2 \\
&= N[\log N - \log(N+1)] + [\log N + \log 2 - \log(N+1)] \\
&= N \log(N/(N+1)) + \log(2N/(N+1)).
\end{aligned}$$

The denominator in the ensuing exponential bound is initially $(N+1)!/2^{N+1}$. On changing to the cycle-expectation denominator,

$$\frac{(N+1)!/2^{N+1}}{\nu_{N+1}} = \frac{(N+1)!}{N!} = N + 1.$$

The polynomial degree changes from $2d+1$ to $2d+2$. The added-vertex identity also gives

$$\frac{\mathbb{E}H}{\mu_N} = \frac{4\mathbb{E}C_{\text{aug}}}{4\nu_{N+1}} = \frac{\mathbb{E}C_{\text{aug}}}{\nu_{N+1}}.$$

Thus the normalized path expectation equals the normalized cycle expectation after augmentation.

A.6 A common threshold for finitely many bases

Definition A.2 (Normalized functions on isomorphism classes).

Let T range over the isomorphism classes of bases, choosing a representative for each random blow-up. By Section 11.7, the expectation is independent of the representative. Fix a model $\mathcal{M} \in \{\mathcal{T}, \mathcal{R}\}$ and one of the two counts, and define

$$f_T(m) = \begin{cases} \mathbb{E}C_{\mathcal{M}}(T,m)/\nu_{dm}, & \text{cycles,} \\ \mathbb{E}H_{\mathcal{M}}(T,m)/\mu_{dm}, & \text{paths,} \end{cases} \qquad f_*(m) = f_{\text{Car}_d}(m).$$

Write $\ell_T = \lim_{m\to\infty} f_T(m)$ and $\ell_* = \lim_{m\to\infty} f_*(m)$. In model $\mathcal{R}$, both limits are restricted to odd m.

For this model and count, the carousel limit is $\ell_* > 0$, and every other class satisfies $\ell_T < \ell_*$. The set of noncarousel classes is finite and nonempty: for $d \geq 3$, the transitive base is nonregular whereas the carousel is regular. Hence

$$\delta = \min_{T \ncong \text{Car}_d} (\ell_* - \ell_T) > 0.$$

Nonregular classes have limit zero; regular noncarousel classes are handled by the strict spectral gap. For each class choose M_T so that whenever $m \geq M_T$ its normalized expectation $f_T(m)$ satisfies

$$|f_T(m) - \ell_T| < \delta/3.$$

Impose the same error requirement on the carousel. Let M be the maximum of these finitely many thresholds. Then

$$f_*(m) - f_T(m) > (\ell_* - \delta/3) - (\ell_T + \delta/3) \geq \delta/3 > 0.$$

Taking the maximum of the four thresholds, one for each model and count, gives a common M_d. In the regular internal-block model, m remains restricted to odd integers.

A.7 A calculus proof of the trigonometric inequality used in monotonicity

Section 10.3 uses $\tan y \geq y + y^3/3$ for $0 \leq y < \pi/2$. Set $f(y) = \tan y - y$. Then

$$f'(y) = \tan^2 y \geq 0, \qquad f(0) = 0.$$

Thus $\tan y \geq y$. Next set $g(y) = \tan y - y - y^3/3$. We have

$$g'(y) = \tan^2 y - y^2 \geq 0, \qquad g(0) = 0.$$

The desired inequality follows. Moreover,

$$\alpha(x) = \int_0^1 \frac{dt}{1+x^2t^2} \geq \int_0^1 (1 - x^2t^2)\, dt = 1 - x^2/3,$$

Here we used $(1+z)^{-1} - (1-z) = z^2/(1+z) \geq 0$.

B. Expanded proofs of standard tools

This appendix gives proofs of the standard estimates used in the paper.

B.1 Determinant tools

Principal-minor expansion. For $I + tM$, in column j choose either the identity column e_j or the matrix column $tM_{\cdot j}$. Multilinearity of the determinant gives 2^n choices. If U is the set of columns for which the matrix column is selected, expansion along the remaining identity columns leaves $M[U]$ with a positive sign. Therefore

$$\det(I + tM) = \sum_{U \subseteq [n]} t^{|U|} \det M[U].$$

Rank-one lemma. For vectors a, b, any two columns of ab^{T} are linearly dependent. Thus, when $M = ab^{\mathsf{T}}$, only the terms of orders zero and one remain in the preceding expansion:

$$\det(I + ab^{\mathsf{T}}) = 1 + \sum_i a_i b_i = 1 + b^{\mathsf{T}} a.$$

If M is invertible, then

$$\det(M + ab^{\mathsf{T}}) = \det M \det(I + M^{-1} ab^{\mathsf{T}}) = \det M (1 + b^{\mathsf{T}} M^{-1} a).$$

Hadamard's inequality. Apply Gram-Schmidt orthogonalization to the rows $r_1, \ldots, r_n$ of the matrix to obtain orthogonal vectors $q_1, \ldots, q_n$. Subtracting linear combinations of earlier rows leaves the determinant unchanged, so

$$|\det M|^2 = \det(MM^*) = \prod_i \|q_i\|_2^2 \le \prod_i \|r_i\|_2^2.$$

The inequality also holds when the rows are dependent, since its left-hand side is then zero. Taking square roots yields $|\det M| \le \prod_i \|r_i\|_2$.

We also use differentiation of the determinant. First apply the principal-minor expansion to a small matrix perturbation K:

$$\det(I + K) = 1 + \operatorname{tr} K + \sum_{\substack{U \subseteq [p] \\ |U| \ge 2}} \det K[U] = 1 + \operatorname{tr} K + O(\|K\|_F^2), \qquad K \to 0.$$

The last remainder can be bounded directly, term by term. Write $\delta = \|K\|_F \le 1$; every entry then satisfies $|K_{ij}| \le \delta$. If $|U| = r \ge 2$, the permutation definition of the determinant gives

$$|\det K[U]| \le \sum_{\pi \in \mathfrak{S}_r} \prod_{i=1}^{r} |K[U]_{i,\pi(i)}| \le r! \delta^r \le r! \delta^2.$$

For a fixed order p, set $C_p = \sum_{r=2}^{p} \binom{p}{r} r!$ (the empty sum is zero when $p = 1$). Then

$$|\det(I + K) - 1 - \operatorname{tr} K| \le \sum_{r=2}^{p} \binom{p}{r} r! \delta^2 = C_p \|K\|_F^2.$$

Here C_p may depend on the dimension. Suppose that M is differentiable at t and $M(t)$ is invertible. Then

$$M(t+h) = M(t) + hM'(t) + R(h), \qquad \|R(h)\|_F = o(|h|).$$

The estimate $R(h) = o(h)$ is sufficient. Set

$$K(h) = hM(t)^{-1}M'(t) + M(t)^{-1}R(h).$$

Since $M(t)^{-1}$ is a fixed matrix, we obtain, term by term,

$$\|M(t)^{-1}R(h)\|_F \le \|M(t)^{-1}\|_{\mathrm{op}}\|R(h)\|_F = o(|h|), \qquad \|K(h)\|_F = O(|h|).$$

Factor out $M(t)$ and expand the determinant:

$$\begin{aligned}\det M(t+h) &= \det M(t)\det(I + K(h)) \\ &= \det M(t)\left[1 + \operatorname{tr} K(h) + O(\|K(h)\|_F^2)\right] \\ &= \det M(t)\left[1 + h\operatorname{tr}(M(t)^{-1}M'(t)) + o(|h|)\right].\end{aligned}$$

The remainder terms in the third line satisfy

$$\begin{aligned}\left|\operatorname{tr}(M(t)^{-1}R(h))\right| &\le \sqrt{p}\,\|M(t)^{-1}R(h)\|_F = o(|h|), \\ C_p\|K(h)\|_F^2 &= O(h^2) = o(|h|).\end{aligned}$$

The first inequality applies Cauchy-Schwarz to the p diagonal entries; the second line uses $h^2/|h| = |h| \to 0$.

Subtract $\det M(t)$, divide by $h \neq 0$, and let $h \to 0$. This gives

$$\frac{d}{dt}\det M(t) = \det M(t)\operatorname{tr}[M(t)^{-1}M'(t)].$$

A real logarithm may be taken directly only when the real matrix path satisfies $\det M(t) > 0$ throughout the interval in question. Under this condition, the chain rule gives

$$\frac{d}{dt}\log\det M(t) = \frac{(\det M(t))'}{\det M(t)} = \operatorname{tr}[M(t)^{-1}M'(t)].$$

The Gram comparison in the main text uses $M(t) = I - H(t) > 0$, so this condition is satisfied. The formal-series argument instead uses determinants with constant term one and their formal logarithms; it assumes no common logarithm branch for arbitrary complex numbers.

A scalar counterexample distinguishes differentiability from a second-order remainder. Let $M(t)$ be the one-by-one matrix $[1 + |t|^{3/2}]$. At $t = 0$,

$$M'(0) = \lim_{h\to 0}\frac{|h|^{3/2}}{h} = 0, \qquad \frac{|M(h) - M(0) - hM'(0)|}{h^2} = |h|^{-1/2} \longrightarrow \infty.$$

Thus differentiability alone does not guarantee a second-order remainder. The linear matrix paths used in the main text do satisfy the hypotheses of the differential formula above.

B.2 Matrix norm tools

The Euclidean vector norm is $\|x\|_2 = (\sum_i |x_i|^2)^{1/2}$. When $y = 0$, the inequality $|x^*y| \le \|x\|_2\|y\|_2$ is immediate. When $y \neq 0$, choose the complex scalar $c = (y^*x)/\|y\|_2^2$ and expand term by term:

$$0 \le \|x - cy\|_2^2 = \|x\|_2^2 - cx^*y - \overline{c}\,y^*x + |c|^2\|y\|_2^2 = \|x\|_2^2 - \frac{|y^*x|^2}{\|y\|_2^2}.$$

Multiply by $\|y\|_2^2$ and take the nonnegative square root to obtain the complex Cauchy-Schwarz inequality: $|x^*y| \le \|x\|_2\|y\|_2$.

The operator norm measures the largest stretching factor of a matrix:

$$\|M\|_{\mathrm{op}} = \max_{\|x\|_2=1} \|Mx\|_2.$$

The Frobenius norm treats the entries of a matrix as one long vector:

$$\|M\|_F^2 = \sum_{ij} |M_{ij}|^2 = \mathrm{tr}(M^*M).$$

The trace $\mathrm{tr}\, M$ is the sum of the diagonal entries. Interchanging two compatible factors leaves the trace of their product unchanged, because

$$\mathrm{tr}(AB) = \sum_i \sum_j A_{ij}B_{ji} = \sum_j \sum_i B_{ji}A_{ij} = \mathrm{tr}(BA).$$

This elementary rule for finite sums is used when passing between row-side and column-side Gram matrices.

Since M^*M is Hermitian positive semidefinite, it admits a unitary diagonalization with nonnegative eigenvalues. Write these eigenvalues as s_i^2; the numbers $s_i \ge 0$ are the singular values. Then

$$\|M\|_{\mathrm{op}} = \max_i s_i, \qquad \|M\|_F^2 = \sum_i s_i^2.$$

If $M^*Mv_i = s_i^2 v_i$ and $s_i > 0$, set $u_i = Mv_i/s_i$. Then

$$u_i^* u_j = \frac{v_i^* M^* M v_j}{s_i s_j} = \mathbf{1}\{i = j\}.$$

Completing the orthonormal bases gives the singular value decomposition $M = U\,\mathrm{diag}(s_i)V^*$. This defines the trace norm $\|M\|_* = \sum_i s_i$, and yields

$$\|M\|_{\mathrm{op}} \le \|M\|_F \le \|M\|_*, \qquad \sum_i s_i \le \sqrt{n}\left(\sum_i s_i^2\right)^{1/2}.$$

When both a, b are nonzero, the only nonzero singular value of the rank-one matrix ab^* is $\|a\|\|b\|$, because

$$(ab^*)^*(ab^*) = \|a\|^2 bb^*.$$

If $a = 0$ or $b = 0$, the matrix and all its singular values are zero. Hence $\|ab^*\|_* = \|a\|\|b\|$ in every case. For arbitrary matrices A, B, expansion over the intermediate index gives

$$AB = \sum_k A_{\cdot k}B_{k\cdot}, \qquad \|AB\|_* \le \sum_k \|A_{\cdot k}\|_2 \|B_{k\cdot}\|_2 \le \|A\|_F\|B\|_F.$$

The triangle inequality for the trace norm follows from the dual formula

$$\|M\|_* = \max_{\|Q\|_{\mathrm{op}}\le 1} |\,\mathrm{tr}(Q^*M)|.$$

Substituting $M = U\,\mathrm{diag}(s_i)V^*$ bounds the right-hand side by $\sum_i s_i$, and $Q = UV^*$ attains equality. The triangle inequality, $\|AM\|_* \le \|A\|_{\mathrm{op}}\|M\|_*$, and $|\,\mathrm{tr}(AM)| \le \|A\|_{\mathrm{op}}\|M\|_*$ all follow from the same formula.

The induced one-norm and infinity-norm used in this paper are, respectively,

$$\|M\|_{1\to 1} = \max_j \sum_i |M_{ij}|, \qquad \|M\|_{\infty\to\infty} = \max_i \sum_j |M_{ij}|.$$

Their relation to the Euclidean operator norm has a direct proof:

$$\begin{aligned}\|Mx\|_2^2 &= \sum_i \left|\sum_j M_{ij}x_j\right|^2 \\ &\le \sum_i \left(\sum_j |M_{ij}|\right)\left(\sum_j |M_{ij}||x_j|^2\right) \\ &\le \|M\|_{\infty\to\infty}\|M\|_{1\to 1}\|x\|_2^2.\end{aligned}$$

Thus $\|M\|_{\text{op}} \le \sqrt{\|M\|_{1\to 1}\|M\|_{\infty\to\infty}}$.
Set $u = \mathbf{1}/\sqrt{n}$, $P = uu^{\mathsf{T}} = J/n$, and $\Pi = I - P$. Then

$$P^2 = P, \quad Px = \frac{\sum_i x_i}{n}\mathbf{1}, \quad \Pi x = x - \frac{\sum_i x_i}{n}\mathbf{1}.$$

The projection P extracts the mean direction, while Π removes it. Both are orthogonal projections and therefore have norm at most one. Double centering, $B\mathbf{1} = B^{\mathsf{T}}\mathbf{1} = 0$, is equivalent to $PB = BP = 0$.

B.3 Remainders and Stirling order estimates

The notation $f_n = O(g_n)$ means that fixed K, n_0 exist such that, for $n \ge n_0$, we have $|f_n| \le K|g_n|$; $f_n = o(g_n)$ means $f_n/g_n \to 0$. If there is also a graph parameter T, uniformity requires the same K, n_0 to work for every admissible T.

For example, $\log n/n = o(1)$ but is not $O(1/n)$. Summability of the deletion weights is therefore needed to retain an $O(1/n)$ error.

Calculus gives the exact remainder formulas

$$\log(1+x) - x = -\int_0^x \frac{u}{1+u}\,du, \qquad |\log(1+x) - x| \le \frac{x^2}{2(1-|x|)} \quad (|x|<1),$$

$$e^x - 1 = x\int_0^1 e^{tx}\,dt, \qquad |e^x - 1| \le |x|e^{|x|}.$$

Consequently, $|x_n| \le K/n$ implies $e^{x_n} = 1 + O(1/n)$. For complex exponentials, $e^z - 1 = z\int_0^1 e^{tz}\,dt$ holds as well, with the same absolute-value bound.

The paper only needs Stirling's estimate in the form

$$\log(n!) = (n+1/2)\log n - n + c + O(1/n),$$

The value of c is immaterial for the applications. To prove its existence, set $f(x) = \log x$ and define the trapezoidal error on $[k, k+1]$ by

$$E_k = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} f(x)\,dx.$$

Put $w(x) = (x-k)(k+1-x)$. Then $w(k) = w(k+1) = 0$, $w'(k) = 1$, $w'(k+1) = -1$, and $w'' = -2$. Integrating by parts twice gives

$$\begin{aligned}\int_k^{k+1} wf'' &= [wf']_k^{k+1} - \int_k^{k+1} w'f' \\ &= -[w'f]_k^{k+1} + \int_k^{k+1} w''f \\ &= f(k) + f(k+1) - 2\int_k^{k+1} f.\end{aligned}$$

Hence

$$E_k = \frac{1}{2}\int_k^{k+1} (x-k)(k+1-x)f''(x)\,dx, \quad |E_k| \le \frac{1}{12k^2}.$$

Thus $\sum_{k\ge 1} E_k$ converges absolutely; define the fixed real number $c = 1 + \sum_{k=1}^{\infty} E_k$. For $n \ge 2$,

$$\left|\sum_{k=n}^{\infty} E_k\right| \le \frac{1}{12}\sum_{k=n}^{\infty} k^{-2} \le \frac{1}{12}\int_{n-1}^{\infty} x^{-2}\,dx = \frac{1}{12(n-1)} \le \frac{1}{6n}.$$

This gives an explicit K/n bound on the tail sum. Summing over $k = 1, \dots, n-1$,

$$\sum_{k=1}^{n-1} \frac{\log k + \log(k+1)}{2} = \log(n!) - \frac{1}{2}\log n,$$

$$\int_1^n \log x\,dx = n\log n - n + 1.$$

Therefore

$$\log(n!) = (n+1/2)\log n - n + 1 + \sum_{k=1}^{n-1} E_k = (n+1/2)\log n - n + c + O(1/n).$$

Exponentiation gives $n! = e^c\sqrt{n}(n/e)^n[1 + O(1/n)]$. This suffices to track the $\sqrt{n}$ factors and fixed constants in the main text.

B.4 Convexity tools

A function f is convex on an interval if $f((1-t)x + ty) \le (1-t)f(x) + tf(y)$ for $0 \le t \le 1$. If its second derivative exists and $f'' \ge 0$, then f' is increasing and each tangent lies below the graph:

$$f(y) - f(x) - f'(x)(y-x) = \int_x^y [f'(u) - f'(x)]\,du \ge 0.$$

For weights $p_i \ge 0$ with $\sum_i p_i = 1$, set $\bar{x} = \sum_i p_i x_i$. Multiply the tangent bound by the weights and sum:

$$\sum_i p_i f(x_i) \ge \sum_i p_i[f(\bar{x}) + f'(\bar{x})(x_i - \bar{x})] = f(\bar{x}).$$

This is Jensen's inequality; the direction reverses for concave functions. For an integer $n \ge 1$ and positive real numbers $x_1, \dots, x_n > 0$, the second derivative of $\log x$ is $-1/x^2 < 0$, so

$$\frac{1}{n}\sum_i \log x_i \le \log\left(\frac{1}{n}\sum_i x_i\right), \qquad \prod_i x_i \le \left(\frac{\sum_i x_i}{n}\right)^n.$$

This is AM-GM. The product inequality extends to $x_i \geq 0$: if some $x_i = 0$, the left-hand side is zero and the right-hand side is nonnegative, so no logarithm of that element is needed.

Young's inequality here is simply completion of a square: $0 \leq (u - v)^2$ gives $2uv \leq u^2 + v^2$. With $u = \sqrt{\tau}/2$ and $v = K/\sqrt{n}$, we obtain

$$K\sqrt{\tau/n} \leq \tau/4 + K^2/n.$$

An error such as $\sqrt{\tau/n}$, which appears larger than $1/n$, can therefore be absorbed by the existing negative term $-\tau$. If $-\tau$ is discarded, this inequality no longer yields a uniform $O(1/n)$ bound.

Discrete concavity also requires only slopes. If the values $h(k)$ at integer points satisfy $h(k+1) - h(k) \leq h(k) - h(k-1)$, the piecewise linear interpolation has decreasing slopes and is therefore concave; Jensen's inequality can then be used at noninteger means. The half-integer correction $1/(32m)$ in the paper comes precisely from the quadratic term in this interpolation.

B.5 Brégman's bound

This is the standard permanent inequality needed for the high-variance reduction. Let A be a zero-one matrix whose row i contains r_i ones. If a row is zero, the permanent is zero. Otherwise we must prove

$$\operatorname{per} A \leq \prod_i (r_i!)^{1/r_i}.$$

Let Ω be the set of admissible matching permutations and $N = |\Omega| = \operatorname{per} A$. The result is immediate when $N = 0$. Henceforth assume $N > 0$; choose a permutation uniformly from Ω and denote it σ.

We need one tool from finite probability. The entropy of a variable X is $\mathsf{H}(X) = -\sum_x p_x \log p_x$, with the convention $0 \log 0 = 0$. If it has at most d possible values, concavity of log gives

$$\mathsf{H}(X) = \sum_{p_x > 0} p_x \log(1/p_x) \leq \log \left(\sum_{p_x > 0} p_x / p_x \right) \leq \log d.$$

Equality holds for a uniform distribution, so $\mathsf{H}(\sigma) = \log N$. Only when $p(x) > 0$ do we define the conditional probabilities $p(y \mid x) = p(x, y)/p(x)$. Conditional entropy is defined by

$$\mathsf{H}(Y \mid X) = \sum_{x : p(x) > 0} p(x) \left[- \sum_{y : p(x,y) > 0} p(y \mid x) \log p(y \mid x) \right].$$

In the following logarithmic decomposition, sum over $p(x, y) > 0$. Then $p(x) > 0$ and $p(y \mid x) > 0$, so $p(x, y) = p(x)p(y \mid x)$ gives

$$\begin{aligned} \mathsf{H}(X, Y) &= - \sum_{(x,y) : p(x,y) > 0} p(x, y) [\log p(x) + \log p(y \mid x)] \\ &= \mathsf{H}(X) + \mathsf{H}(Y \mid X). \end{aligned}$$

Iteration gives the chain rule. Independently of σ, choose a reveal order ρ uniformly from all row permutations, and reveal $\sigma(i)$ in that order. When row i is revealed, it can choose only among the admissible columns not already occupied by earlier rows. Let $N_i(\sigma, \rho)$ be their number. The conditional entropy bound gives

$$\log N \leq \sum_i \mathbb{E}_{\sigma,\rho} \log N_i(\sigma, \rho).$$

Fix σ and row i, and set $R_i = \{j : A_{ij} = 1\}$, with $|R_i| = r_i$. Each column $j \in R_i$ is used by the unique row $\sigma^{-1}(j)$ in the matching. These r_i rows are distinct and include i. In the random reveal order, the position of row i among these r_i rows is uniform.

If it is revealed when k of these rows remain, exactly k columns in R_i remain unoccupied. Thus

$$N_i \mid \sigma \text{ is uniform on } \{1, 2, \dots, r_i\}, \qquad \mathbb{E}_\rho \log N_i = \frac{1}{r_i} \sum_{k=1}^{r_i} \log k = \frac{\log(r_i!)}{r_i}.$$

This does not depend on the fixed matching σ. Substitution and exponentiation give

$$\log \operatorname{per} A \le \sum_i \frac{\log(r_i!)}{r_i}, \qquad \operatorname{per} A \le \prod_i (r_i!)^{1/r_i}.$$

In Section 7.2, discrete concavity of $f(k) = \log(k!)/k$ strengthens this estimate by a variance penalty.

B.6 Gram and determinant perturbations

For a real symmetric or Hermitian matrix, $H \ge 0$ means that $x^* H x \ge 0$ for every vector; $H \le K$ means $K - H \ge 0$. If $0 \le H \le q^2 I$ and $0 \le q < 1$, then $I - H > 0$, and the eigenvalues of its inverse are at most $1/(1 - q^2)$.

Two meanings of nonnegativity must be distinguished. In matrix scaling, $X \ge 0$ means that every entry satisfies $X_{ij} \ge 0$. In a Gram comparison, $H \ge 0$ for a symmetric matrix means that the preceding quadratic form is nonnegative, namely positive semidefiniteness. These conditions are not equivalent. In each of the following two examples, denote the displayed matrix by H:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ has nonnegative entries, but for } x = (1, -1)^{\mathsf{T}} \text{ we have } x^{\mathsf{T}} H x = -2 < 0;$$

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \text{ has negative entries, but } x^{\mathsf{T}} H x = (x_1 - x_2)^2 \ge 0.$$

Each subsequent argument states which meaning it uses.

For a matrix satisfying $\|Z\|_{\mathrm{op}} < 1$, define $\mathcal{G}(Z) = \det(I - Z^* Z)^{-1/2}$ using the positive square root. Then $I - Z^* Z$ is positive definite and its real logarithm is defined. This quantity depends only on the singular values, so deleting zero rows or columns, or interchanging $Z^* Z$ and $Z Z^*$, leaves it unchanged. If the current matrix has n rows, let $I \subseteq [n]$ be the deleted row set, $R = [n] \setminus I$ the retained row set, and $t = |I|$. The projection P_R onto the retained row coordinates gives

$$Z^* Z - Z^* P_R Z = Z^* (I - P_R) Z \ge 0.$$

Log-determinant differentiation and the integral formula in the determinant tools bound the change by the sum of the squared norms of the deleted rows. For an order-n dense kernel with a fixed spectral gap and $|Z_{ij}| \le C/n$, we have $\sum_{i \in I, j} |Z_{ij}|^2 \le tn(C/n)^2 = C^2 t/n$, so the loss is $O(t/n)$. Using only the square root of the Frobenius displacement would often give the weaker $O(\sqrt{t/n})$. This explains the separate deletion lemma in the main text.

Finally, we prove $|\det(I + E)| \le e^{\|E\|_*}$. Write $E = UDV^*$, where $D = \operatorname{diag}(s_i)$. A unitary similarity gives

$$\det(I + E) = \det[V^*(I + E)V] = \det(I + QD), \qquad Q = V^* U.$$

The principal-minor expansion gives

$$\det(I+QD)=\sum_{I\subseteq[n]}\det Q[I]\prod_{i\in I}s_i.$$

Each $Q[I]$ is a compression of a unitary matrix, so its operator norm and all its column norms are at most one. Hadamard's inequality gives $|\det Q[I]|\le 1$. Therefore

$$|\det(I+E)|\le\sum_I\prod_{i\in I}s_i=\prod_i(1+s_i)\le e^{\sum_i s_i}=e^{\|E\|_*}.$$

Another useful fact is the following: if a square matrix M satisfies $(M+M^*)/2\ge I$ (for real matrices, its symmetric part is at least I), then $\|M^{-1}\|_{\mathrm{op}}\le 1$. Indeed,

$$\|Mx\|\|x\|\ge|x^*Mx|\ge\mathrm{Re}(x^*Mx)\ge\|x\|^2,$$

Thus $\|Mx\|\ge\|x\|$. The matrix has trivial nullspace and, being square, is invertible. Setting $y=Mx$ gives $\|M^{-1}y\|\le\|y\|$.

Author statement

Xiangyu Ye completed the companion Lean formalization of the earlier path bound, whose upper leading constant is $C_*>L$; its precise statement and revision are recorded in the earlier repository. The present `formal/` directory separately covers the actual second derivatives of Wormald's original angular function, the Hessian matrix formula, positive definiteness, determinant and positive rational square root, together with explicitly listed auxiliary results. The sharp quantitative asymptotic in Theorem 1.1 has not been formalized end to end in Lean; results not listed in the coverage table are also outside this project's certification scope. AI assistance was used in the exposition and parts of the proof checking. These checks do not constitute machine certification of the entire present manuscript; see `formal/COVERAGE.zh.md` and the revision-specific verification records.

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